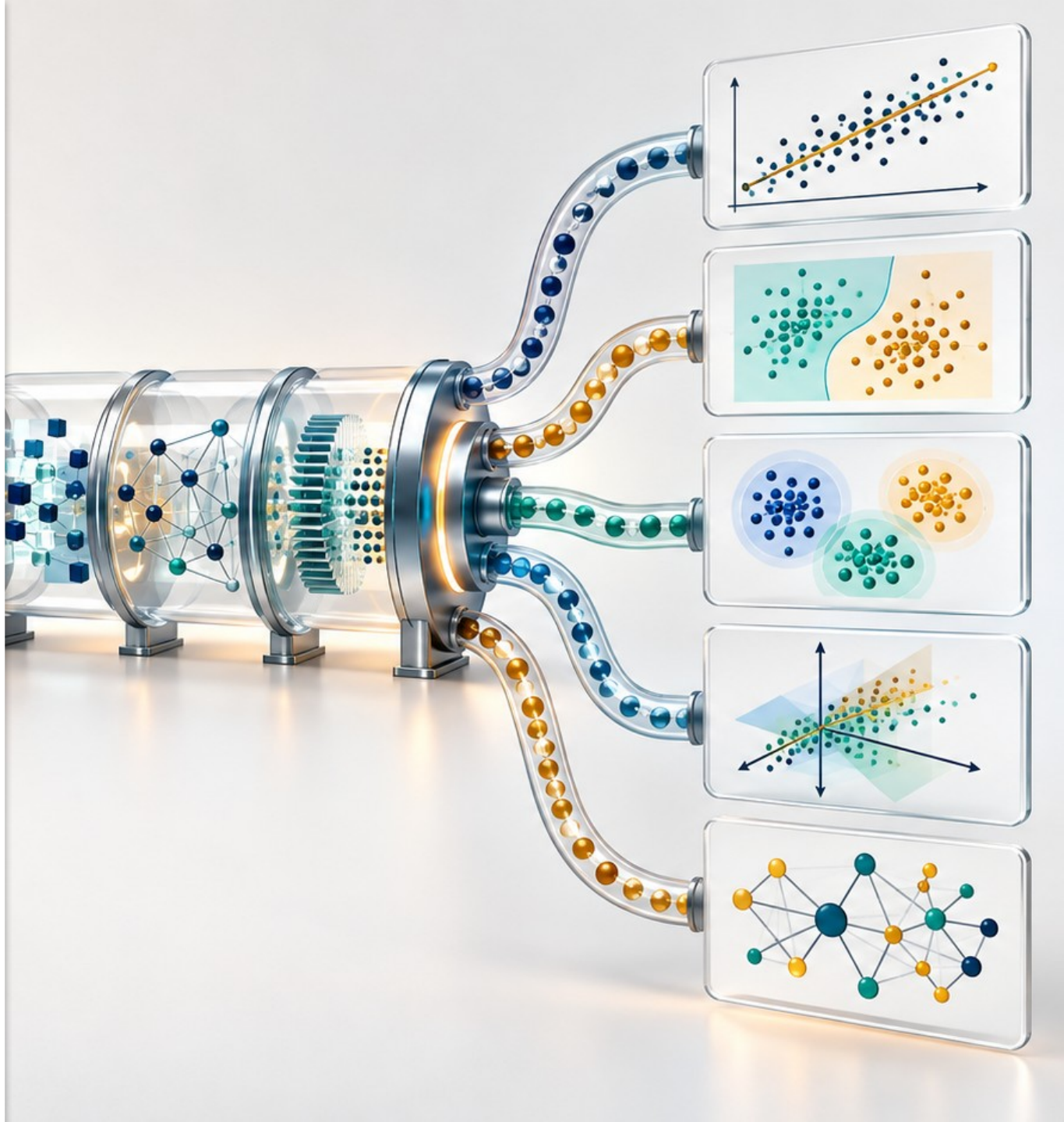
WSQ COURSE SLIDES

Data Mining and Machine Learning Fundamentals for Beginners

Course Code TGS-2020503264
Data Analytics · MED-ACE-3018-1

PYTHON + STREAMLIT · 15 LABS

Version 17.0 · 10 September 2026
Tertiary Infotech Academy Pte Ltd · UEN 201200696W



DATA MODEL METRIC CONTROL

Digital attendance · evidence trail

1

AM check-in

2

PM check-in

3

Assessment check-in

4

Attendance ≥75%

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DATA MODEL METRIC CONTROL

Your trainer



Dr Alfred Ang

Principal trainer · data science, machine learning, optimisation, and applied AI education

Facilitation focus

Mechanisms · reproducible evidence · business interpretation · responsible decisions

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DATA MODEL METRIC CONTROL

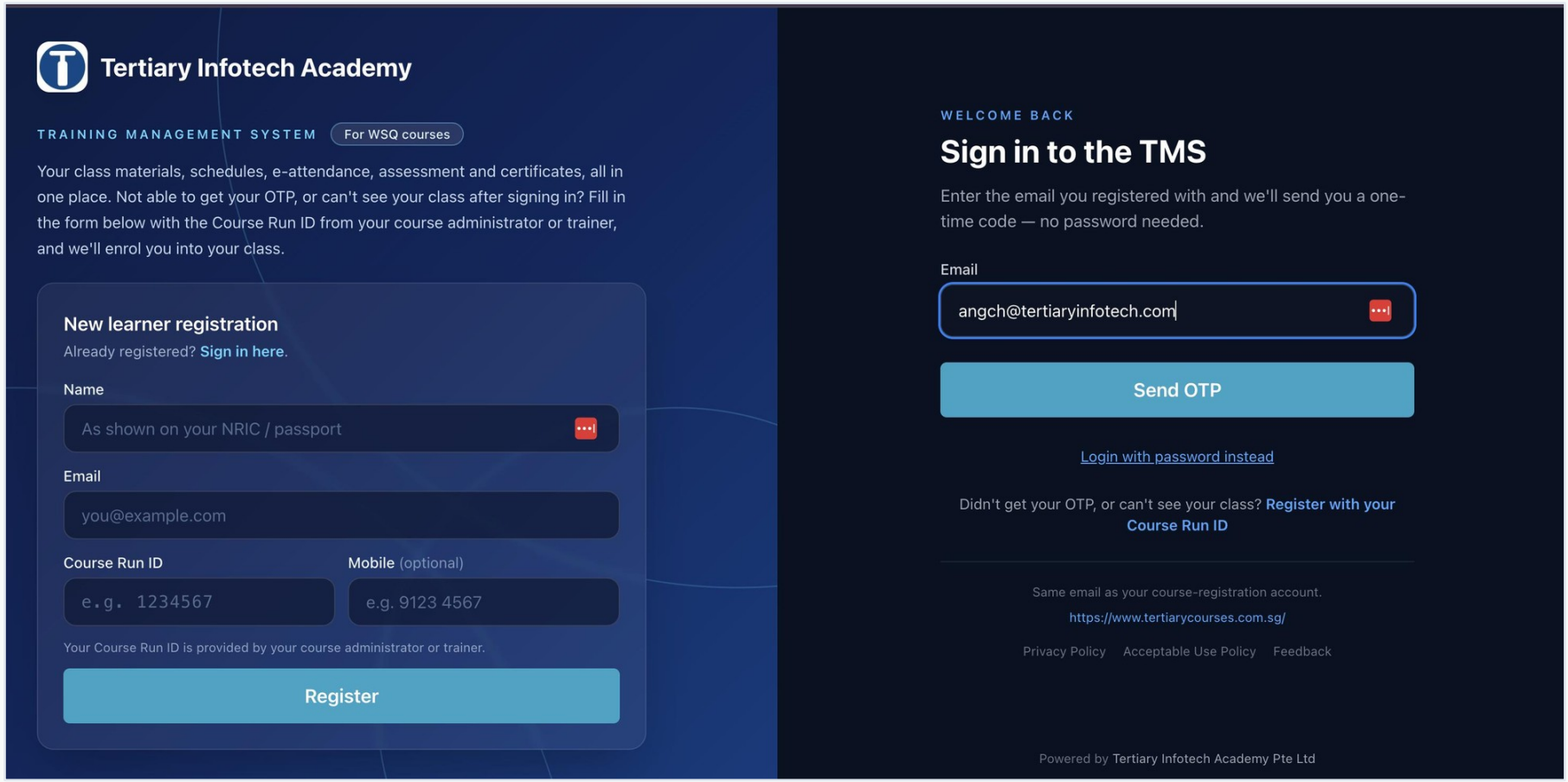
Ground rules for an evidence-led lab

1. **Be punctual**
2. **Participate actively**
3. **Ask questions**
4. **Respect differences**
5. **Protect data**
6. **Verify claims**

Trace the named state at each boundary; a missing state breaks reproducibility.

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Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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DATA MODEL METRIC CONTROL

Day 1 · prepare, predict, and classify

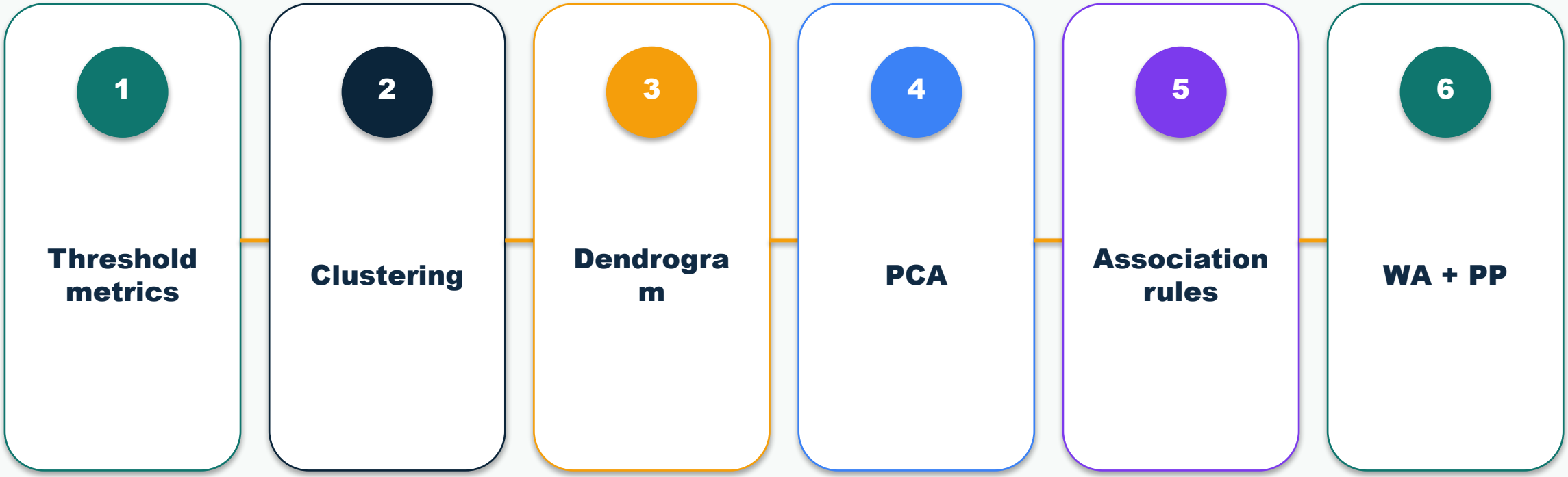


Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Day 2 · discover, compress, and associate



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Learning outcomes mapped to evidence

LO1: Apply data mining and machine learning principles to assess business insights.

LO2: Integrate and aggregate information from multiple datasets to build data models.

LO3: Apply predictive data modelling techniques to identify underlying trends in data.

LO4: Apply machine learning classification techniques to gain new insights from data.

LO5: Apply clustering techniques to discover patterns, visualise evidence, and make decisions.

LO6: Develop prototype algorithms with dimension-reduction and feature-ranking techniques.

LO7: Construct association rules to identify patterns across transactional datasets.

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DATA MODEL METRIC CONTROL

Assessment evidence chain



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

TSC abilities A1–A4

- # **A1 Integrate information from multiple datasets**
- # **A2 Apply predictive modelling**
- # **A3 Identify patterns across datasets**
- # **A4 Develop prototype algorithms**

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DATA MODEL METRIC CONTROL

TSC abilities A5–A8

- **A5 Decide which patterns warrant further analysis**
- **A6 Assemble aggregations into data models**
- **A7 Use machine learning for new insight**
- **A8 Assess organisational impact**

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DATA MODEL METRIC CONTROL

TSC ability A9 · interactive evidence

A9 Support interactive data visualisation	
proof	readable chart + labels
interaction	selected fields + parameters
decision	interpretation + limitation

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DATA MODEL METRIC CONTROL

TSC knowledge K1–K4

- **K1 Data collection, mining, cleaning and analysis**
- **K2 Statistical principles**
- **K3 Communicating data effectively**
- **K4 Organisational domains and processes**

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TSC knowledge K5–K8

- **K5 Exploratory visual analysis and predictive modelling**
- **K6 Machine-learning algorithms**
- **K7 Building a data model**
- **K8 Established and novel analytical techniques**

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DATA MODEL METRIC CONTROL

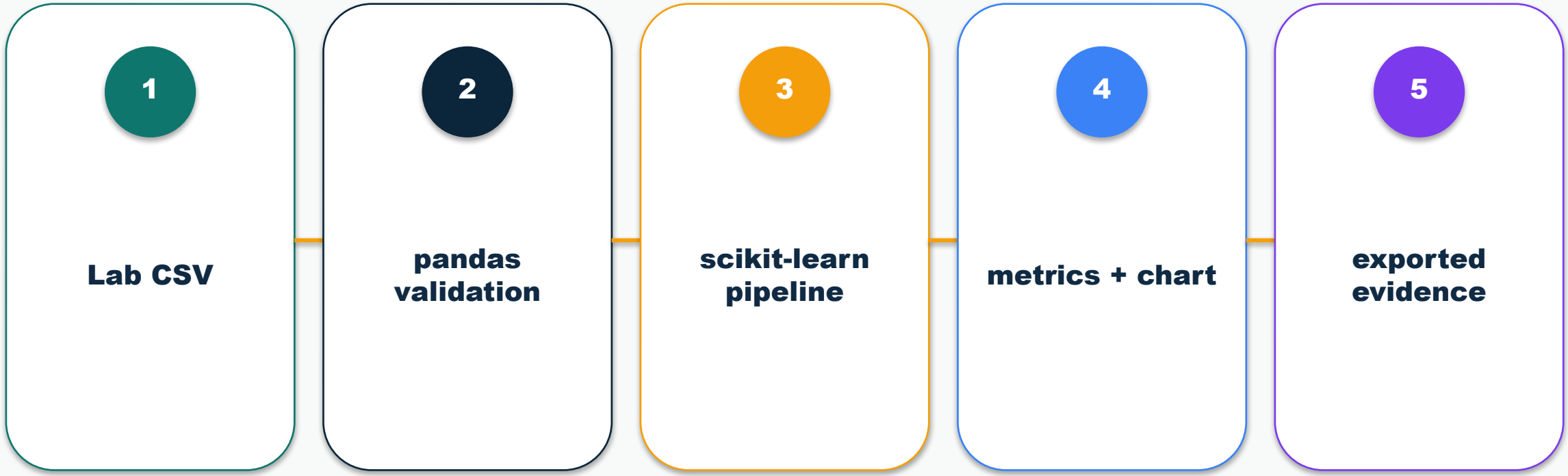
TSC knowledge K9–K10

K9 Applying software tools to analyse data	
K10 Statistical and experimental techniques	
evidence	runnable central app
mapping	WA Q1–Q10

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DATA MODEL METRIC CONTROL

Central app architecture



Trace the named state at each boundary; a missing state breaks reproducibility.

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Model outputs require judgement

FAILURE MODE

A visually convincing chart can hide leakage, bias, or a weak sample.

CONTROL

Pair every result with a baseline, error metric, data scope, and human review.

VERIFICATION

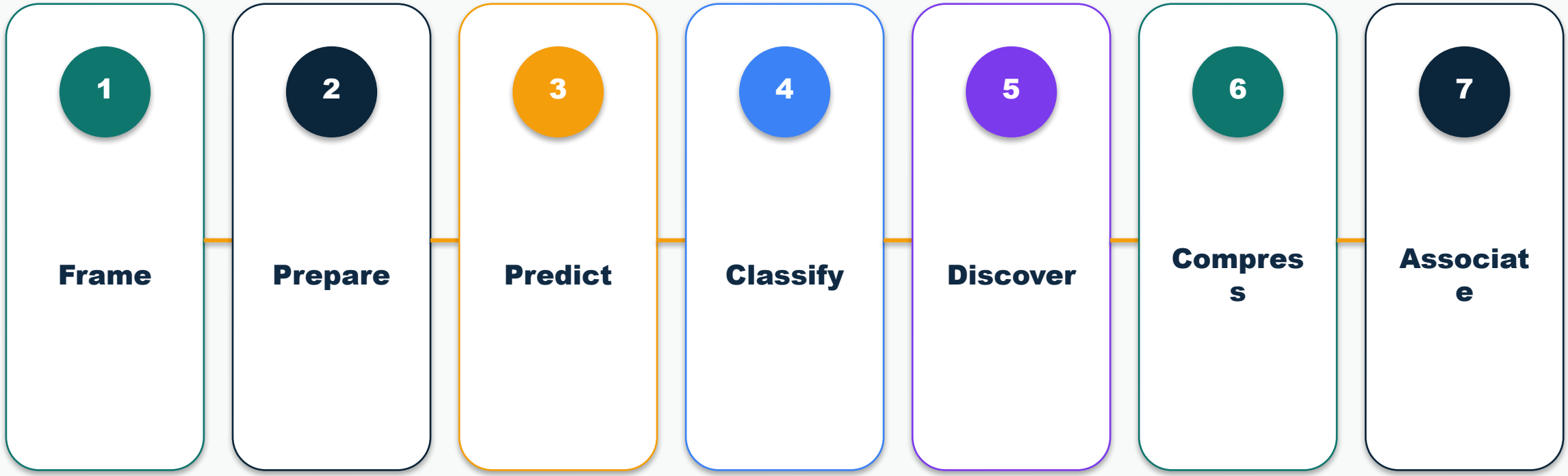
A second learner can reproduce the output from the same file and parameters.

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16 HOURS

DATA MODEL METRIC CONTROL

Two-day learning arc

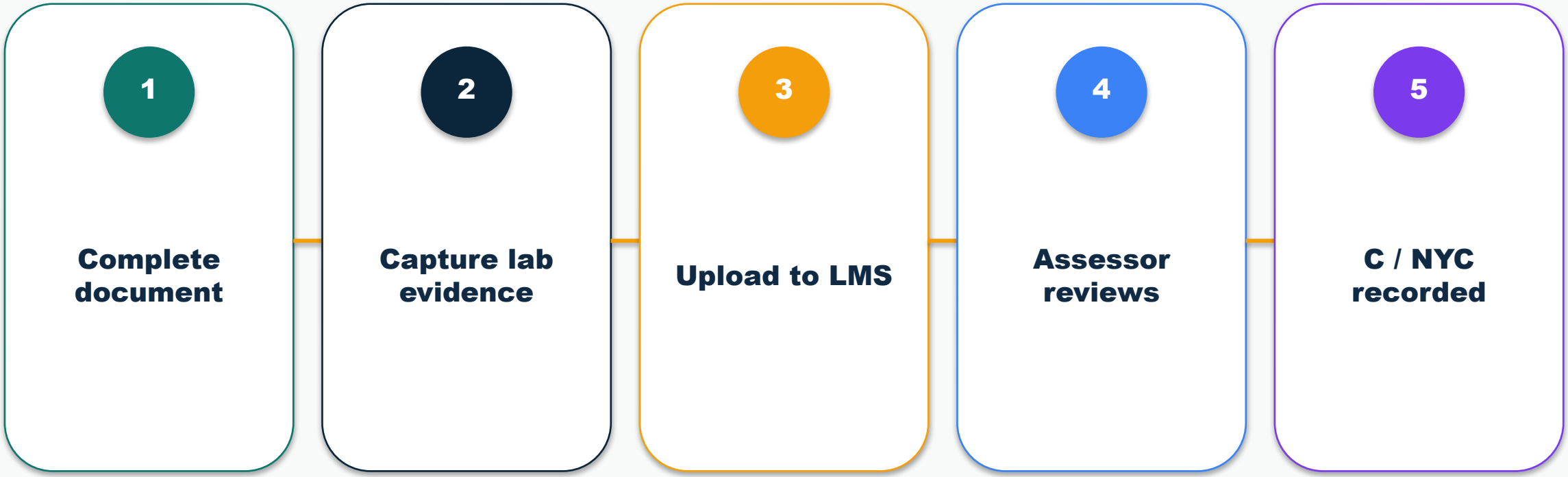


Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

LMS evidence flow



Trace the named state at each boundary; a missing state breaks reproducibility.

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OPEN BOOK · 2 HOURS

DATA MODEL METRIC CONTROL

Assessment briefing

FAILURE MODE

Unapproved collaboration or copied work invalidates individual evidence.

CONTROL

Use only approved course materials; complete every item and keep evidence traceable.

VERIFICATION

WA answers and PP outputs are uploaded to the LMS and signed off by the assessor.

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VISUAL LEGEND

DATA MODEL METRIC CONTROL

How to read the technical slides

1

Mechanism

2

Contract

3

Calculation

4

Evidence

The final control panel on each concept names a failure, a control, and observable proof.

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TOPIC 01

Overview of Data Mining and Machine Learning

LO1 · A8 · K4 · From business question to monitored evidence

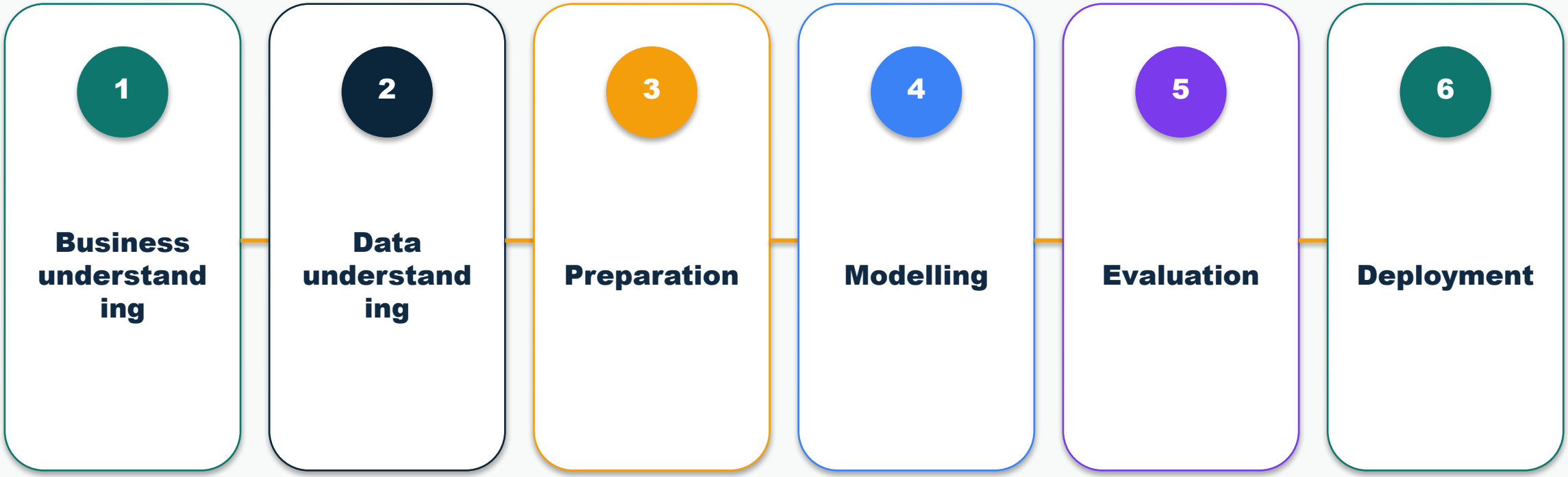
Mechanism → contract → calculation → evidence → control



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DATA MODEL METRIC CONTROL

CRISP-DM evidence loop · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

CRISP-DM evidence loop · concrete contract

objective	measurable decision
unit_of_analysis	one row
target	observed outcome
acceptance	metric + threshold

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DATA MODEL METRIC CONTROL

CRISP-DM evidence loop · worked calculation

decision value = benefit of correct action − cost of errors

WORKED EXAMPLE

$80 benefit − $18 expected error cost = $62 net value

READOUT

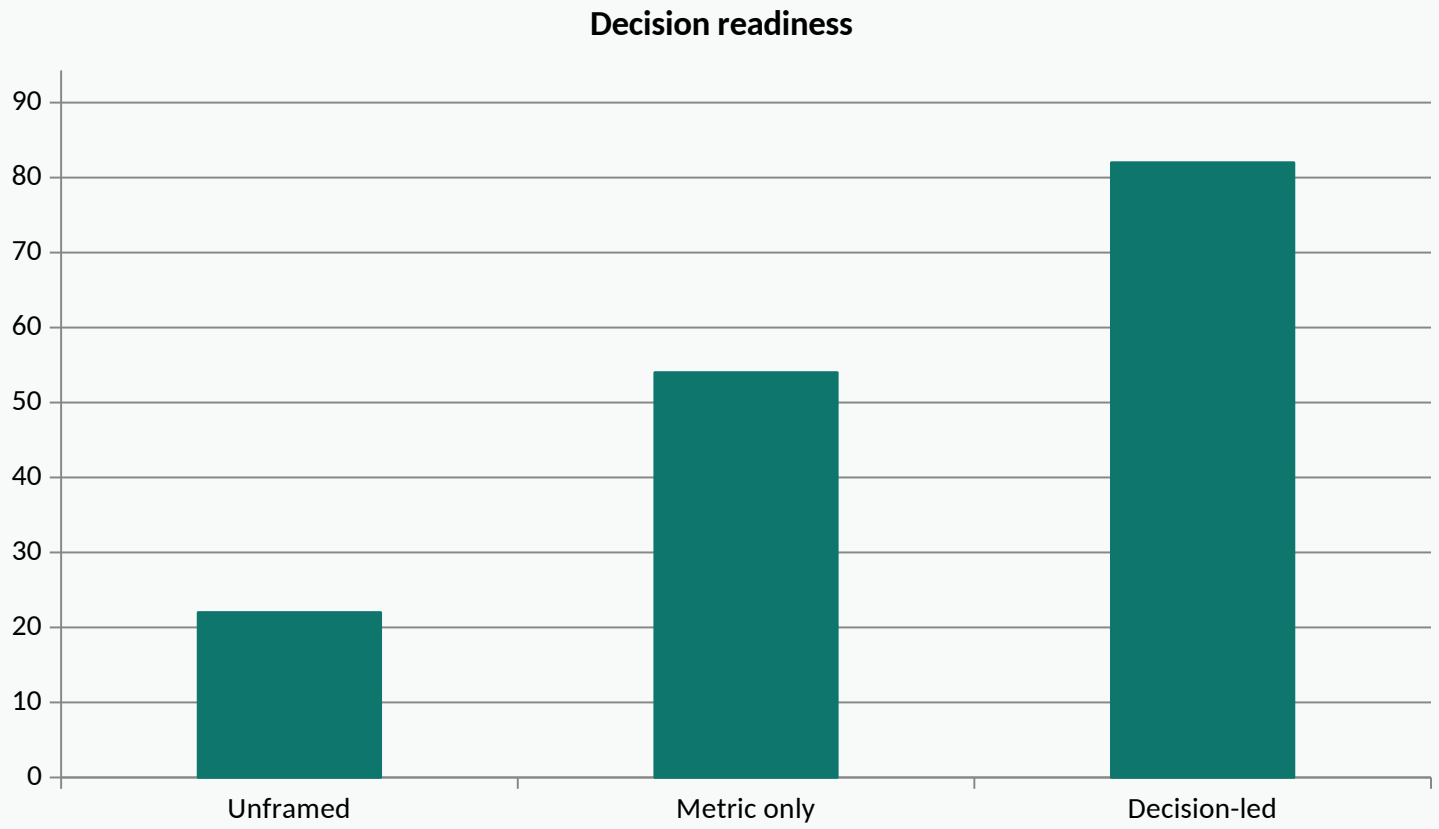
expected decision value

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

CRISP-DM evidence loop · evidence



WHAT THE DATA SHOWS

A reviewer can trace every model output to a decision and acceptance rule.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

CRISP-DM evidence loop · failure and control

FAILURE MODE

Optimising a convenient model metric that does not change the business decision.

CONTROL

Write the decision, owner, action, and cost matrix before modelling.

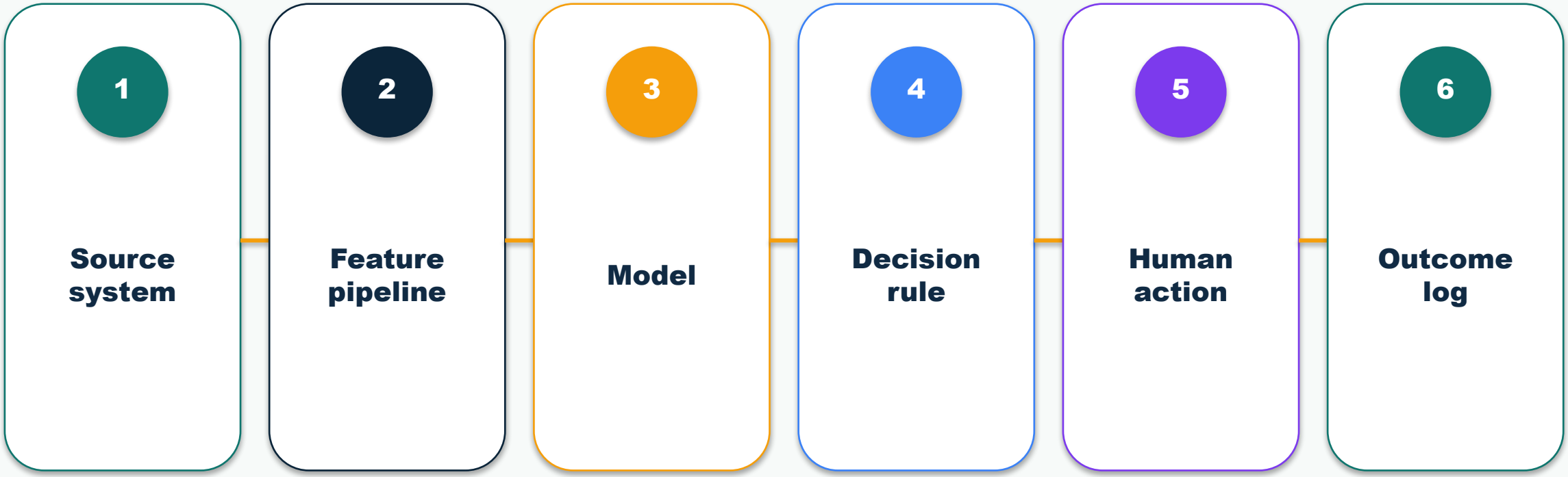
VERIFICATION

A reviewer can trace every model output to a decision and acceptance rule.

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DATA MODEL METRIC CONTROL

Machine-learning system boundary · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Machine-learning system boundary · concrete contract

inputs	versioned fields
model_artifact	fitted parameters
threshold	operating point
outcome	delayed ground truth

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DATA MODEL METRIC CONTROL

Machine-learning system boundary · worked calculation

$$\text{prediction} \neq \text{decision}; \text{decision} = \text{policy}(\text{prediction}, \text{context})$$

WORKED EXAMPLE

0.73 churn probability + threshold 0.65 → retention review

READOUT

decision trace coverage

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Machine-learning system boundary · evidence



WHAT THE DATA SHOWS

Logs contain input version, score, threshold, action, and later outcome.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Machine-learning system boundary · failure and control

FAILURE MODE

Treating a prediction as an instruction and hiding ownership of the final action.

CONTROL

Separate inference, policy, and human approval in the architecture.

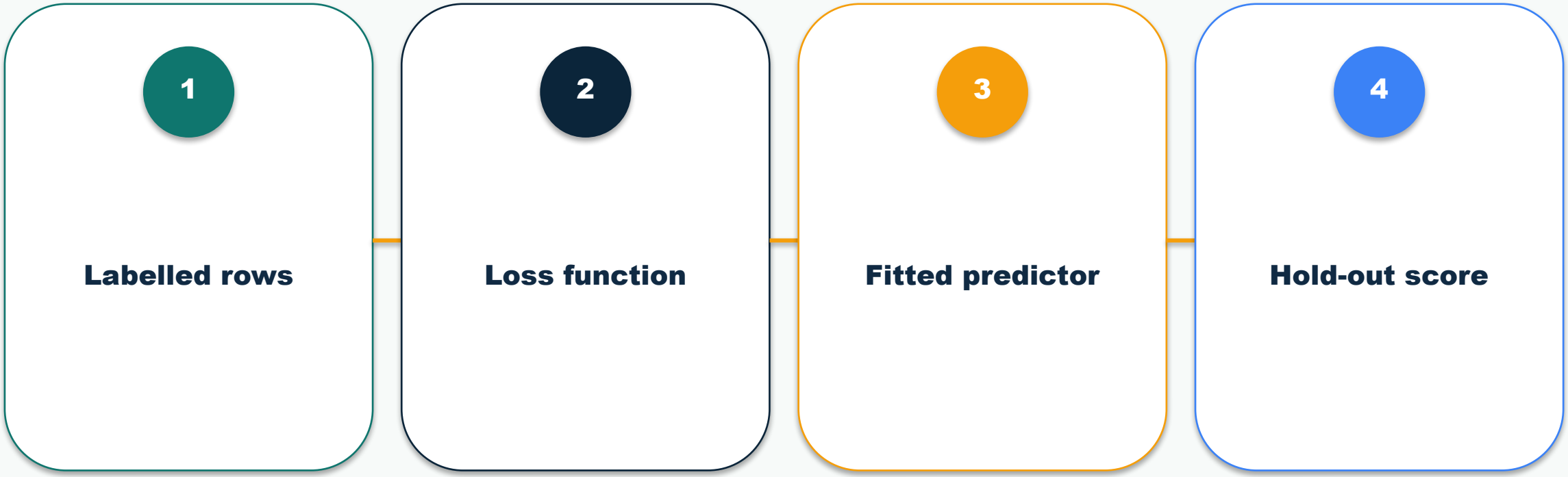
VERIFICATION

Logs contain input version, score, threshold, action, and later outcome.

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DATA MODEL METRIC CONTROL

Supervised versus unsupervised evidence · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Supervised versus unsupervised evidence · concrete contract

supervised	X plus y
unsupervised	X only
classification	discrete y
regression	continuous y

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DATA MODEL METRIC CONTROL

Supervised versus unsupervised evidence · worked calculation

supervised minimises error to y; clustering minimises within-group distance

WORKED EXAMPLE

churn label trains a classifier; spend patterns form clusters without labels

READOUT

external or internal validation

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Supervised versus unsupervised evidence · evidence

SUPERVISED

- Labelled target
- Predict held-out y
- Loss / class metric

UNSUPERVISED

- No target label
- Discover structure
- Stability / utility

The evidence contract changes with the learning question; the scores are not interchangeable.

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DATA MODEL METRIC CONTROL

Supervised versus unsupervised evidence · failure and control

FAILURE MODE

Calling clusters 'true customer types' without external validation.

CONTROL

Use business review, stability tests, and later outcomes to validate segments.

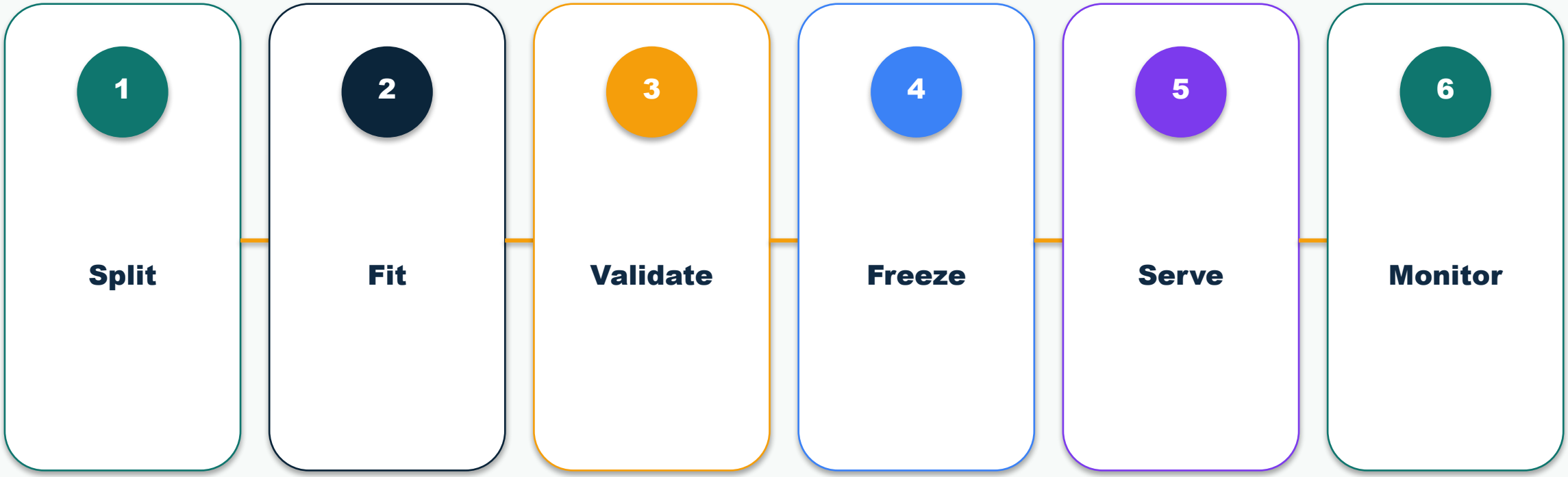
VERIFICATION

The evaluation method matches the presence or absence of ground truth.

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DATA MODEL METRIC CONTROL

Train–evaluate–deploy lifecycle · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Train–evaluate–deploy lifecycle · concrete contract

train	parameter learning
validation	selection
test	final estimate
production	unseen stream

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DATA MODEL METRIC CONTROL

Train–evaluate–deploy lifecycle · worked calculation

generalisation gap = training score − test score

WORKED EXAMPLE

0.96 train − 0.81 test = 0.15 gap

READOUT

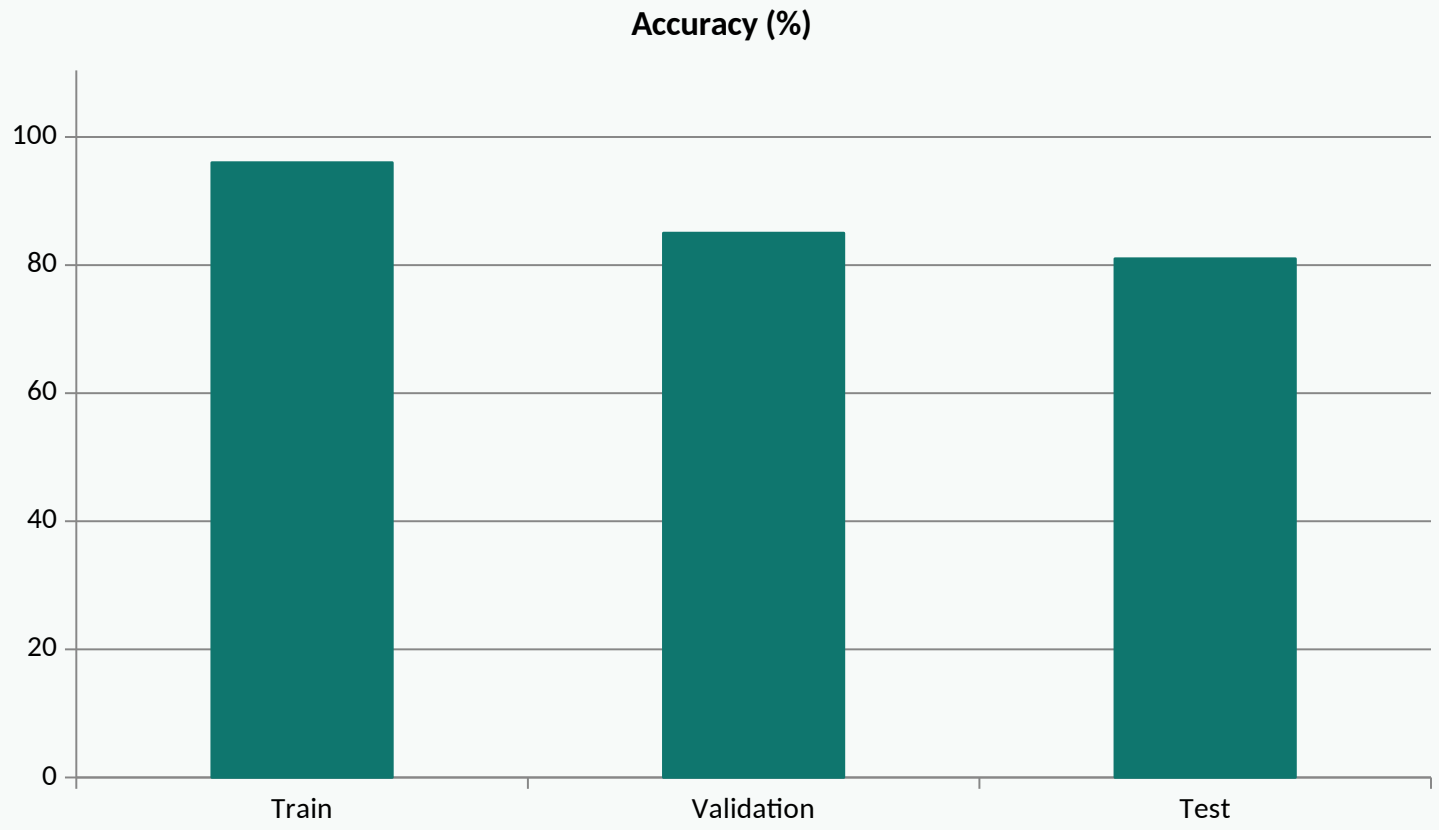
generalisation gap

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Train–evaluate–deploy lifecycle · evidence



WHAT THE DATA SHOWS

Re-running the experiment with the same seed reproduces the split and score.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Train–evaluate–deploy lifecycle · failure and control

FAILURE MODE

Repeatedly tuning against the test set leaks test information into model selection.

CONTROL

Lock a final test set and record experiment parameters separately.

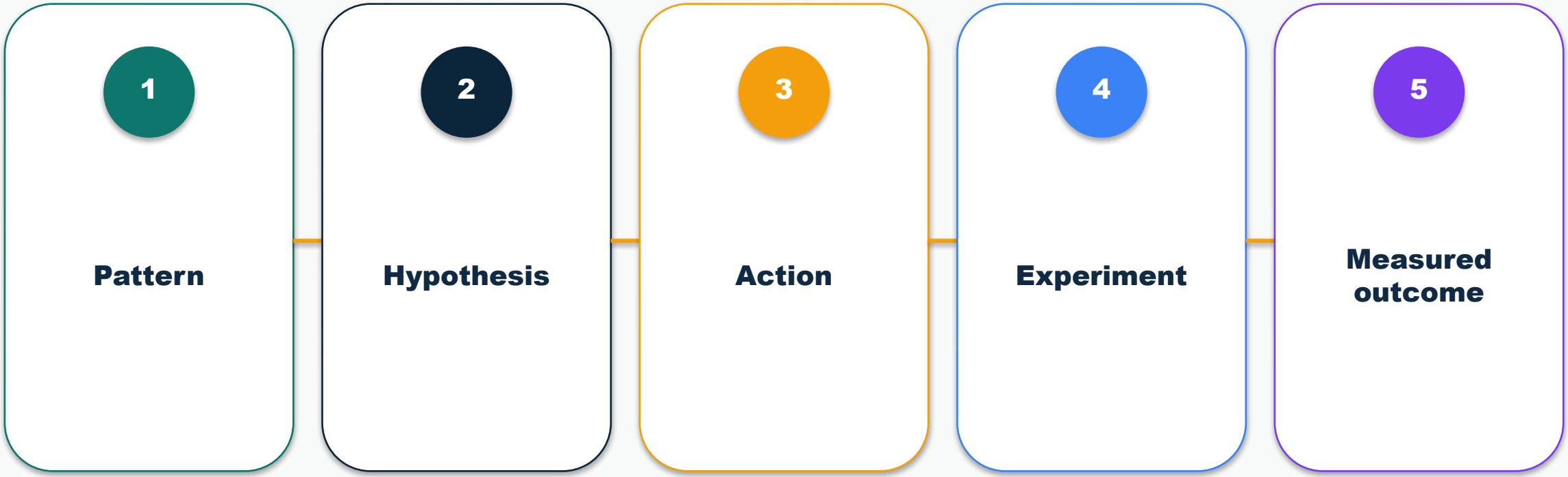
VERIFICATION

Re-running the experiment with the same seed reproduces the split and score.

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DATA MODEL METRIC CONTROL

Business insight impact · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Business insight impact · concrete contract

segment	affected population
action	operational change
guardrail	harm limit
outcome	causal measure

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DATA MODEL METRIC CONTROL

Business insight impact · worked calculation

$$\text{uplift} = \text{outcome_treatment} - \text{outcome_control}$$

WORKED EXAMPLE

18.4% retention − 14.1% control = 4.3 pp uplift

READOUT

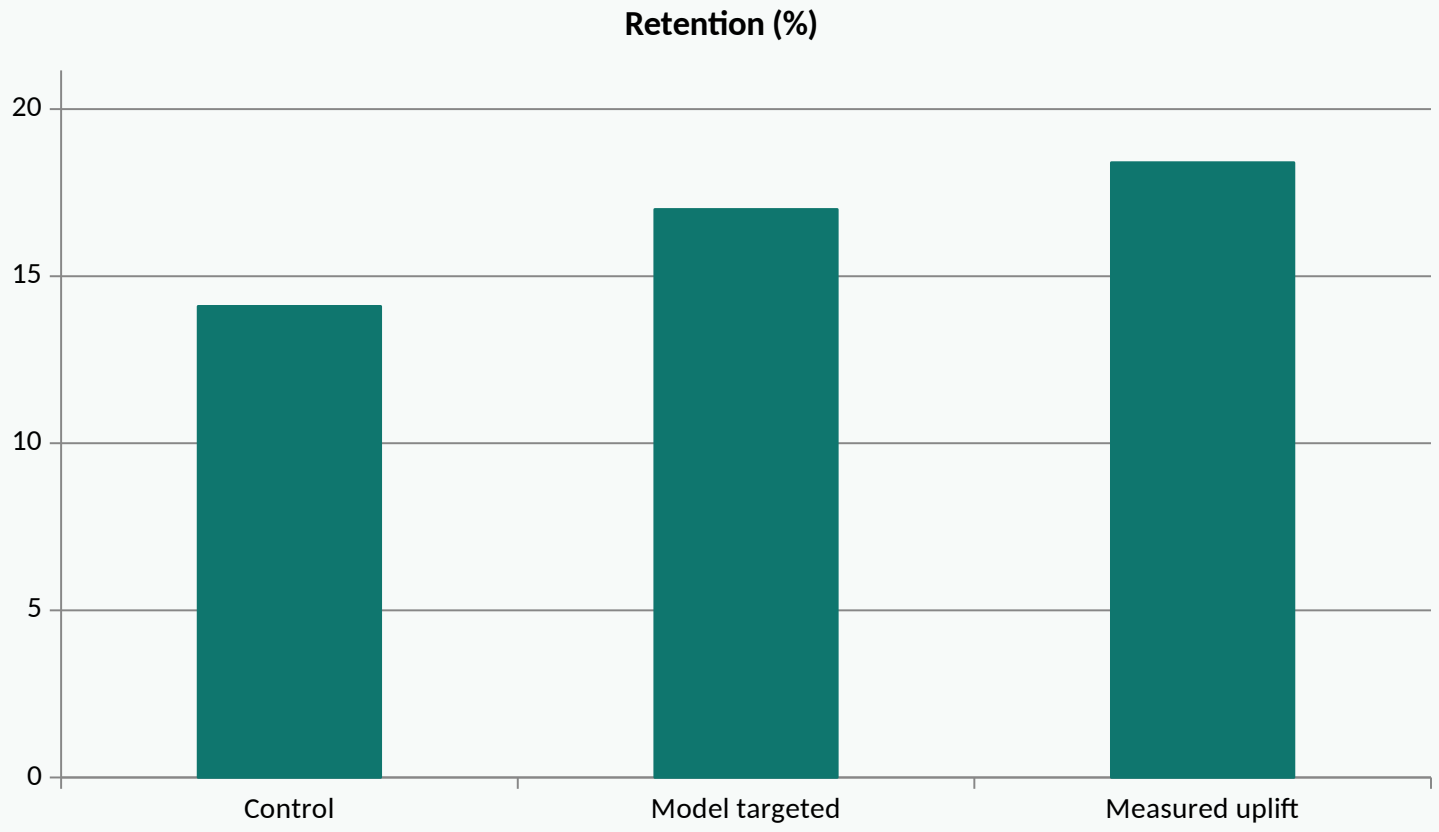
uplift and guardrails

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Business insight impact · evidence



WHAT THE DATA SHOWS

The insight statement includes scope, uncertainty, expected action, and limitation.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Business insight impact · failure and control

FAILURE MODE

Reporting correlation as causal impact.

CONTROL

Use controlled trials or explicitly label observational evidence.

VERIFICATION

The insight statement includes scope, uncertainty, expected action, and limitation.

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DATA MODEL METRIC CONTROL

Lab 01 · Data audit and business question

CENTRAL APP

Data audit & preparation

OPEN CENTRAL LAB PORTAL

INPUT Lab 01 mock CSV

MECHANISM Data audit & preparation

EVIDENCE metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 01 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Data audit and business question produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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DATA MODEL METRIC CONTROL

Lab 15 · Detect and explain anomalies

CENTRAL APP

Anomaly detection

OPEN CENTRAL LAB PORTAL

INPUT	Lab 15 mock CSV
MECHANISM	Anomaly detection
EVIDENCE	metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 15 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Detect and explain anomalies produces a reproducible artifact and a defensible interpretation—not only a screenshot.

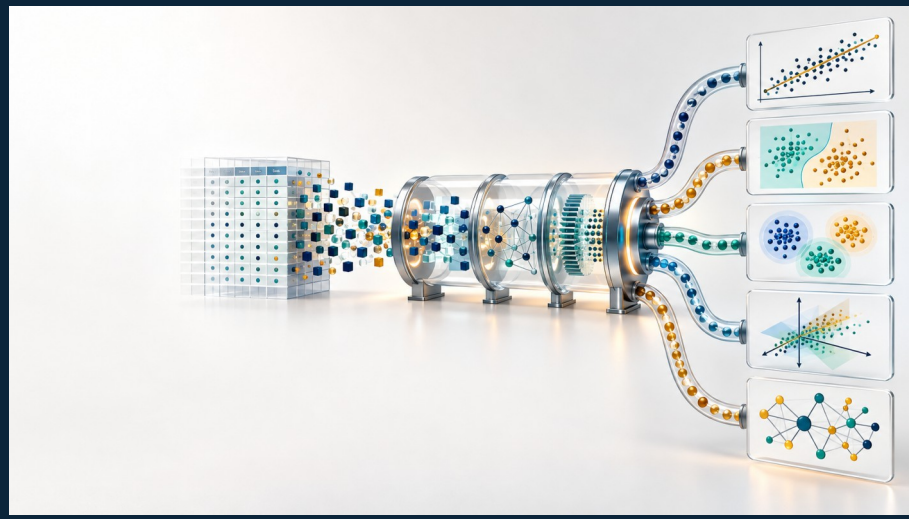
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TOPIC 02

Data Preparation

LO2 · A1/A6 · K1/K7 · Reliable tables before reliable models

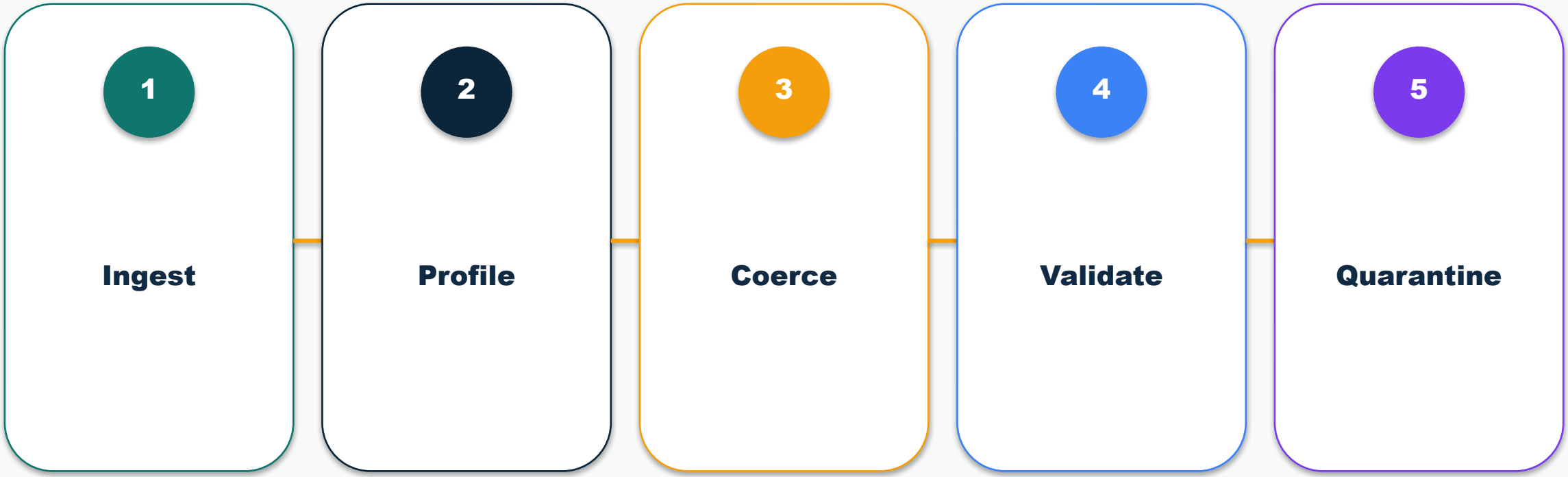
Mechanism → contract → calculation → evidence → control



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DATA MODEL METRIC CONTROL

Schema and data types · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Schema and data types · concrete contract

customer_id	string key
event_time	datetime
amount	float >= 0
target	explicit role

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DATA MODEL METRIC CONTROL

Schema and data types · worked calculation

validity rate = valid rows / received rows

WORKED EXAMPLE

9,720 valid / 10,000 received = 97.2%

READOUT

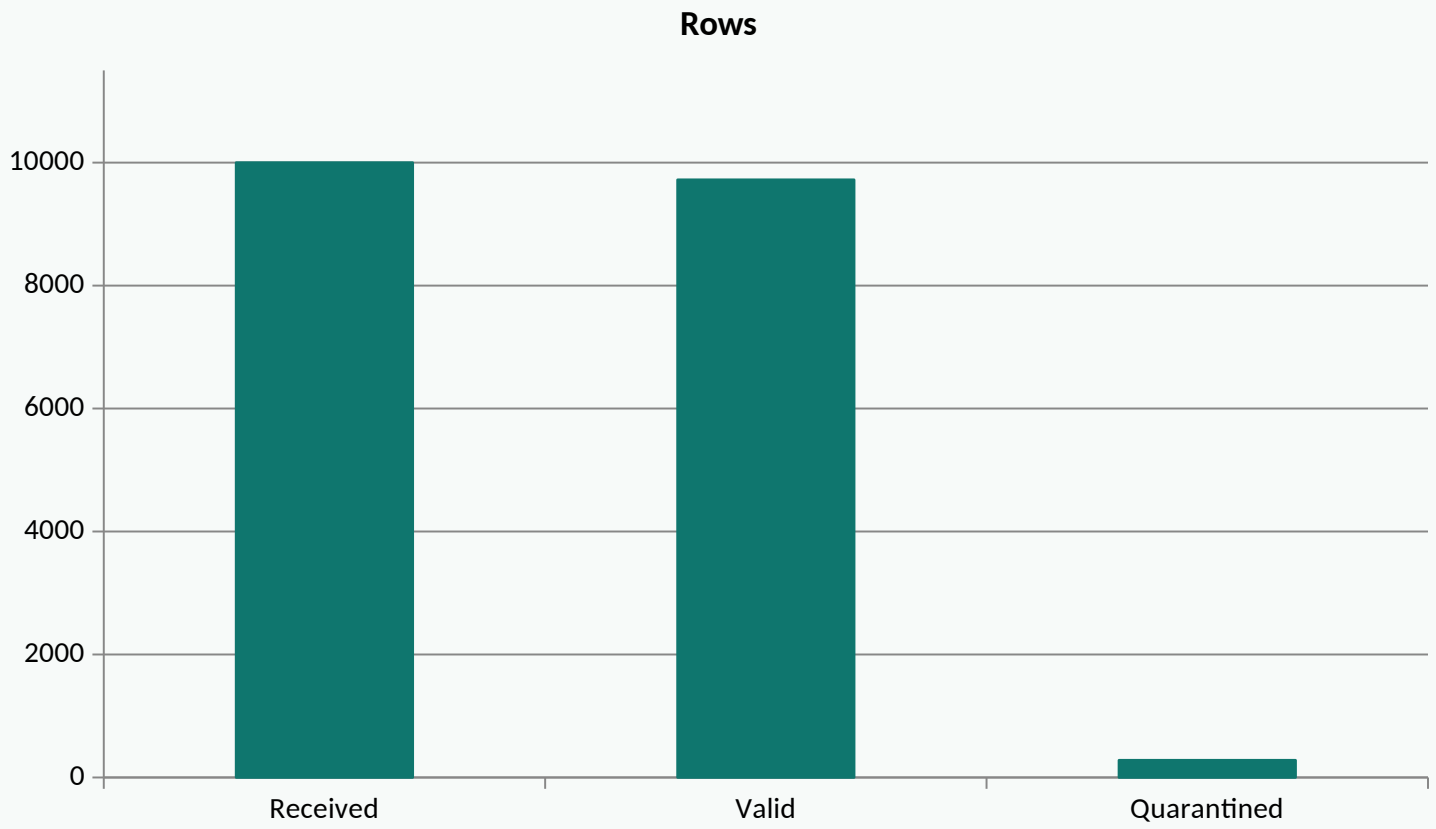
validity rate

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Schema and data types · evidence



WHAT THE DATA SHOWS

Row counts reconcile across received, accepted, and quarantined records.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Schema and data types · failure and control

FAILURE MODE

Silent type coercion turns invalid values into missing values without an audit trail.

CONTROL

Validate ranges and log rejected rows before conversion.

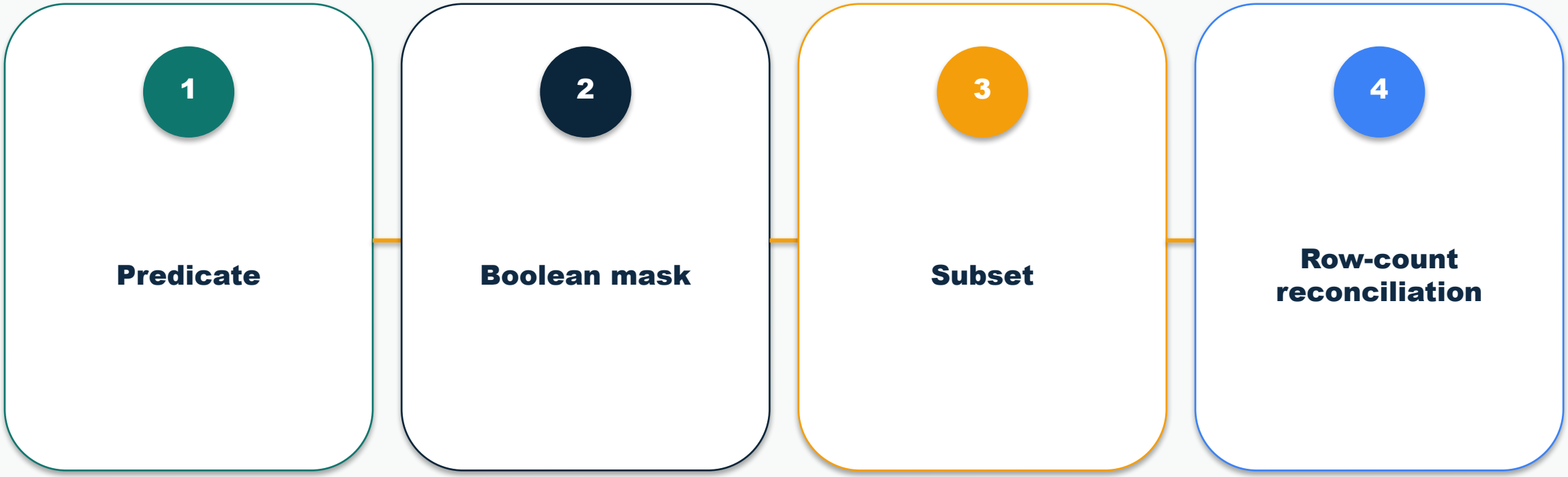
VERIFICATION

Row counts reconcile across received, accepted, and quarantined records.

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DATA MODEL METRIC CONTROL

Filtering and row selection · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Filtering and row selection · concrete contract

predicate	amount > 0
scope	date window
null policy	explicit
index	preserved identifier

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DATA MODEL METRIC CONTROL

Filtering and row selection · worked calculation

$$\text{selected} = \Sigma\, \mathrm{I}(\text{condition is true})$$

WORKED EXAMPLE

1,250 rows satisfy region='North' and amount>0

READOUT

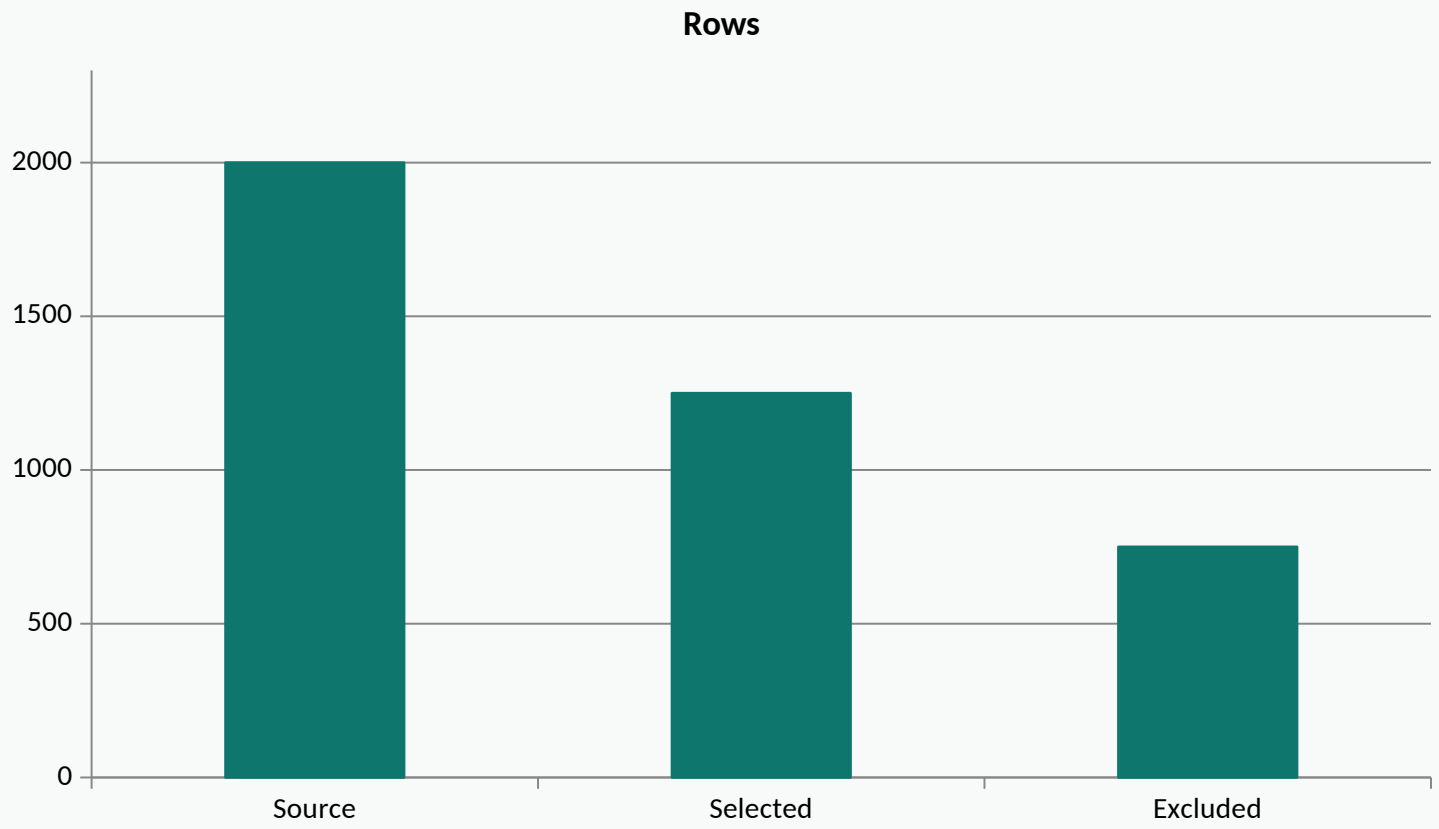
selection rate

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Filtering and row selection · evidence



WHAT THE DATA SHOWS

Selected + excluded + null-policy rows equals the source row count.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Filtering and row selection · failure and control

FAILURE MODE

A filter unintentionally removes nulls because null comparisons evaluate false.

CONTROL

Profile null counts before and after every predicate.

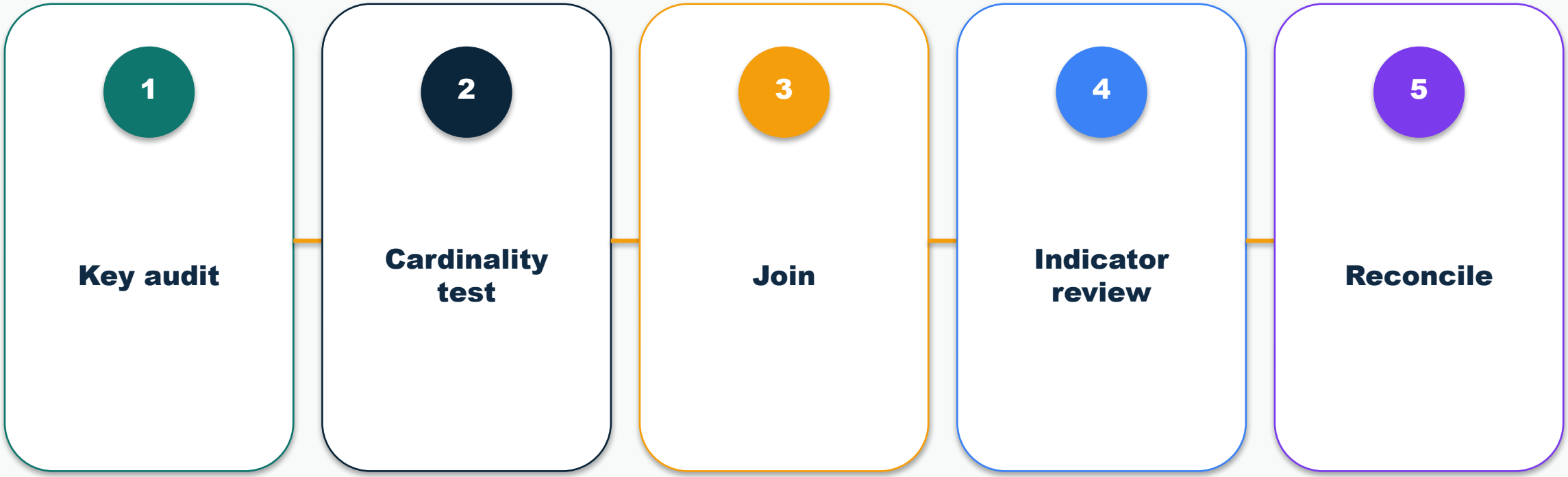
VERIFICATION

Selected + excluded + null-policy rows equals the source row count.

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DATA MODEL METRIC CONTROL

Join and concatenate · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Join and concatenate · concrete contract

key	customer_id
left cardinality	many
right cardinality	one
join type	inner/left

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DATA MODEL METRIC CONTROL

Join and concatenate · worked calculation

join expansion = output rows / left rows

WORKED EXAMPLE

168 output / 160 orders = 1.05× warns of duplicate customer keys

READOUT

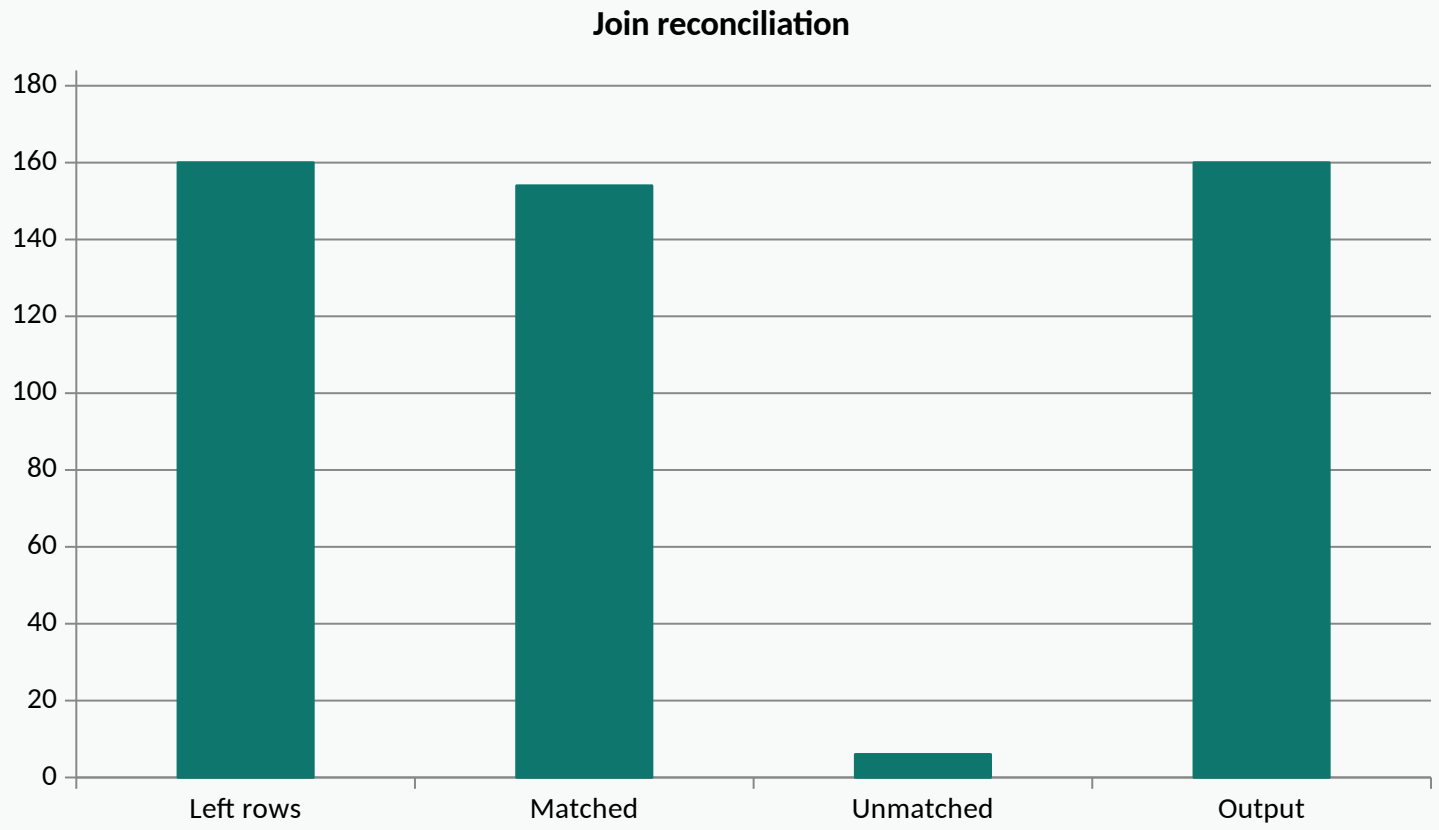
match rate and expansion

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Join and concatenate · evidence



WHAT THE DATA SHOWS

Revenue before and after the intended join reconciles within the documented scope.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Join and concatenate · failure and control

FAILURE MODE

Many-to-many keys duplicate facts and inflate revenue.

CONTROL

Assert key uniqueness on the dimension table and inspect merge indicators.

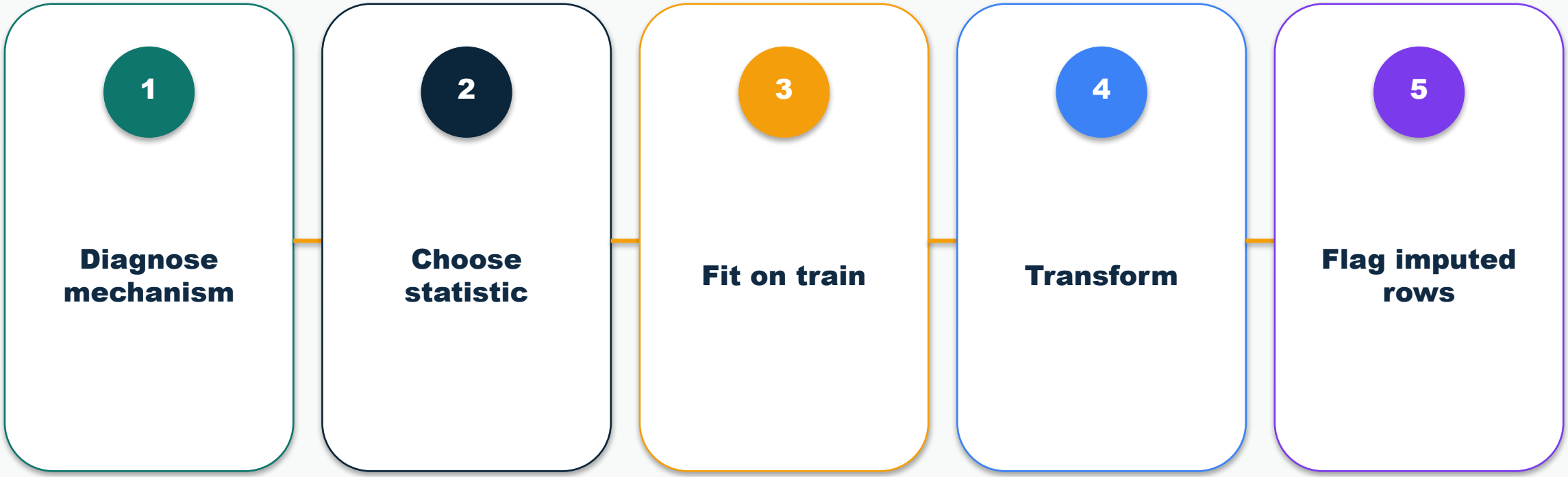
VERIFICATION

Revenue before and after the intended join reconciles within the documented scope.

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DATA MODEL METRIC CONTROL

Missing-value treatment · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Missing-value treatment · concrete contract

MCAR/MAR/MNAR hypothesis	
numeric	median/mean
category	mode/Unknown
indicator	was_missing

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DATA MODEL METRIC CONTROL

Missing-value treatment · worked calculation

median = middle ordered value; robust to extreme values

WORKED EXAMPLE

[20, 22, 24, 95] → median 23, mean 40.25

READOUT

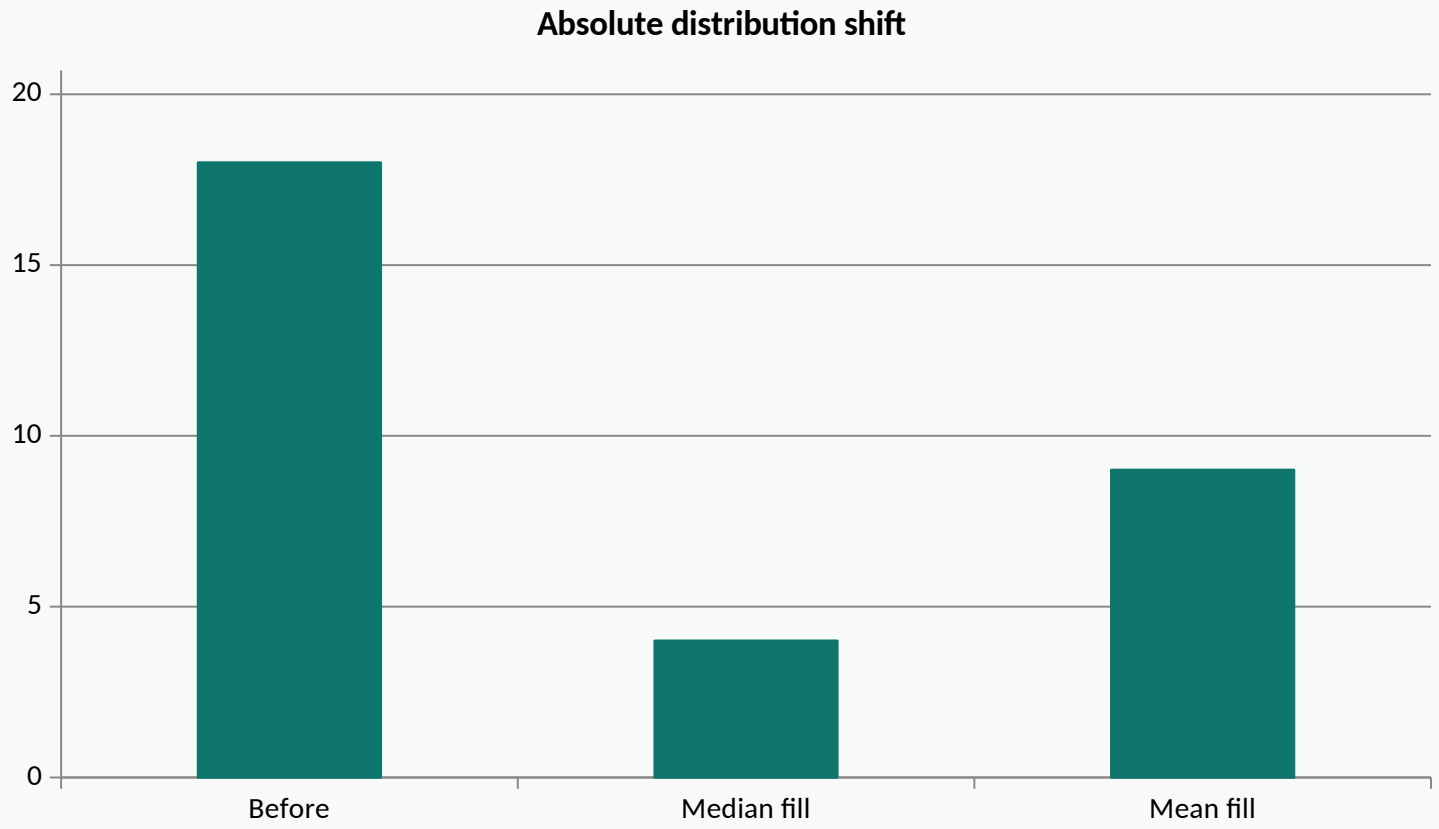
distribution shift

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Missing-value treatment · evidence



WHAT THE DATA SHOWS

No unintended nulls remain and train/test preprocessing is identical.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Missing-value treatment · failure and control

FAILURE MODE

Computing imputation values on the full dataset leaks test distribution.

CONTROL

Fit imputers only on training data inside a pipeline.

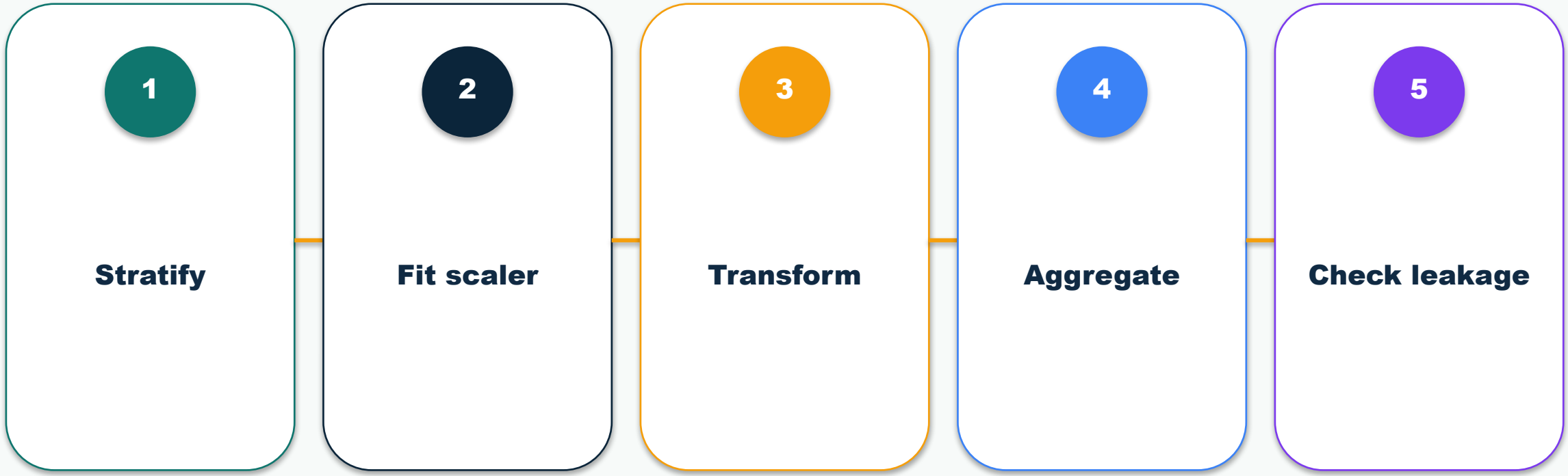
VERIFICATION

No unintended nulls remain and train/test preprocessing is identical.

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DATA MODEL METRIC CONTROL

Scaling, sampling, and aggregation · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Scaling, sampling, and aggregation · concrete contract

#	**z = (x−μ)/σ**	
#	**min-max = (x−min)/(max−min)**	
{}	**group key**	declared
{}	**measure**	count/sum/mean

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DATA MODEL METRIC CONTROL

Scaling, sampling, and aggregation · worked calculation

$$z = (850 - 500)/140 = 2.50$$

WORKED EXAMPLE

Revenue 850 is 2.5 standard deviations above training mean

READOUT

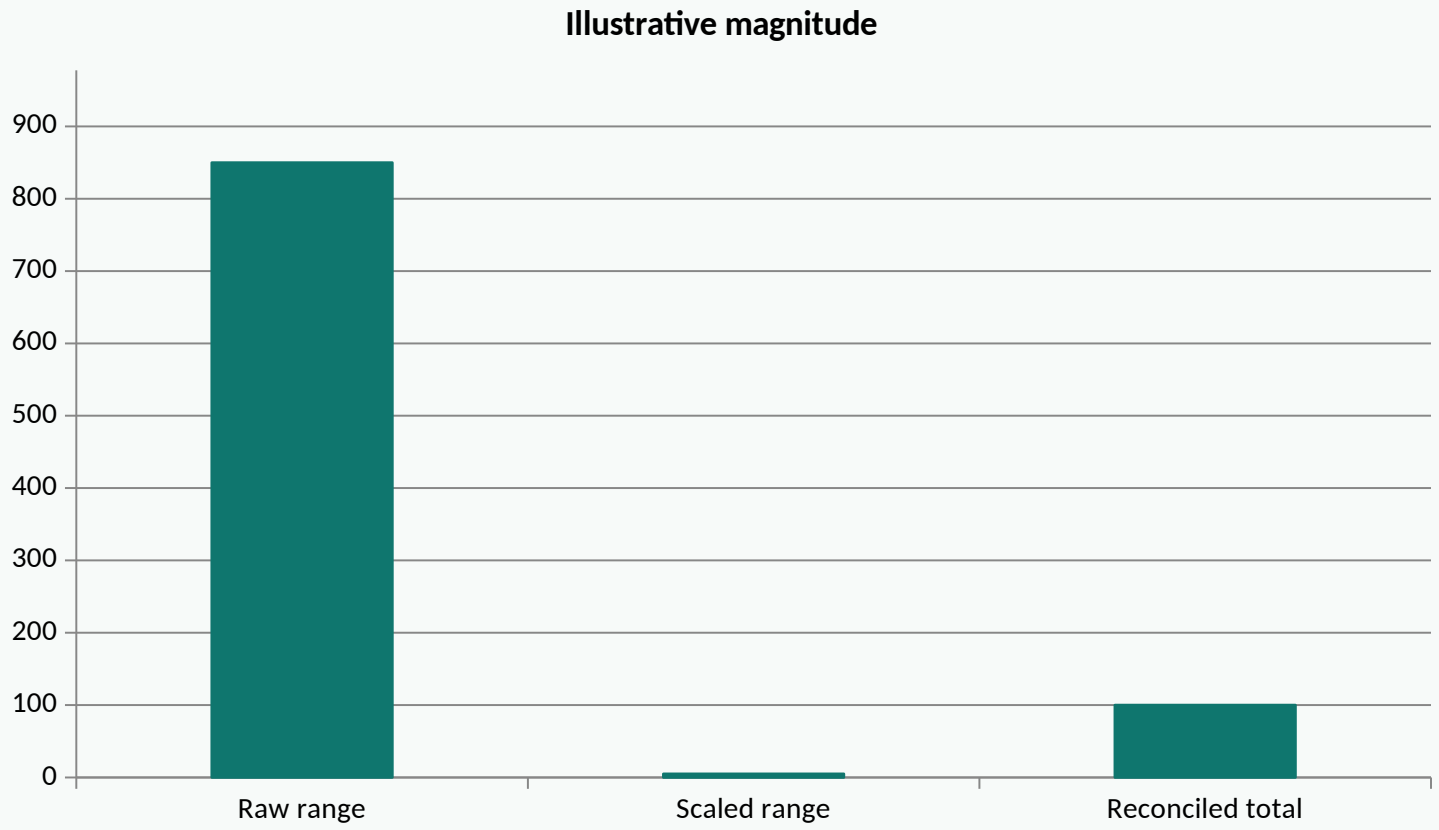
standardised range and group totals

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Scaling, sampling, and aggregation · evidence



WHAT THE DATA SHOWS

Training features centre near zero and group totals reconcile to source totals.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Scaling, sampling, and aggregation · failure and control

FAILURE MODE

Scaling before splitting lets test statistics influence training features.

CONTROL

Split first; fit the scaler within the training pipeline.

VERIFICATION

Training features centre near zero and group totals reconcile to source totals.

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DATA MODEL METRIC CONTROL

Lab 02 · Clean, impute, and export

CENTRAL APP

Data audit & preparation

OPEN CENTRAL LAB PORTAL

INPUT Lab 02 mock CSV

MECHANISM Data audit & preparation

EVIDENCE metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 02 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Clean, impute, and export produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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DATA MODEL METRIC CONTROL

Lab 03 · Filter, join, and aggregate

CENTRAL APP

Join & aggregate

OPEN CENTRAL LAB PORTAL

INPUT Lab 03 mock CSV

MECHANISM Join & aggregate

EVIDENCE metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 03 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Filter, join, and aggregate produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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DATA MODEL METRIC CONTROL

Lab 04 · Scale features before modelling

CENTRAL APP

Regression

OPEN CENTRAL LAB PORTAL

INPUT Lab 04 mock CSV

MECHANISM Regression

EVIDENCE metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 04 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Scale features before modelling produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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TOPIC 03

Regression

LO3 · A2 · K2/K10 · Predicting continuous outcomes with controlled error

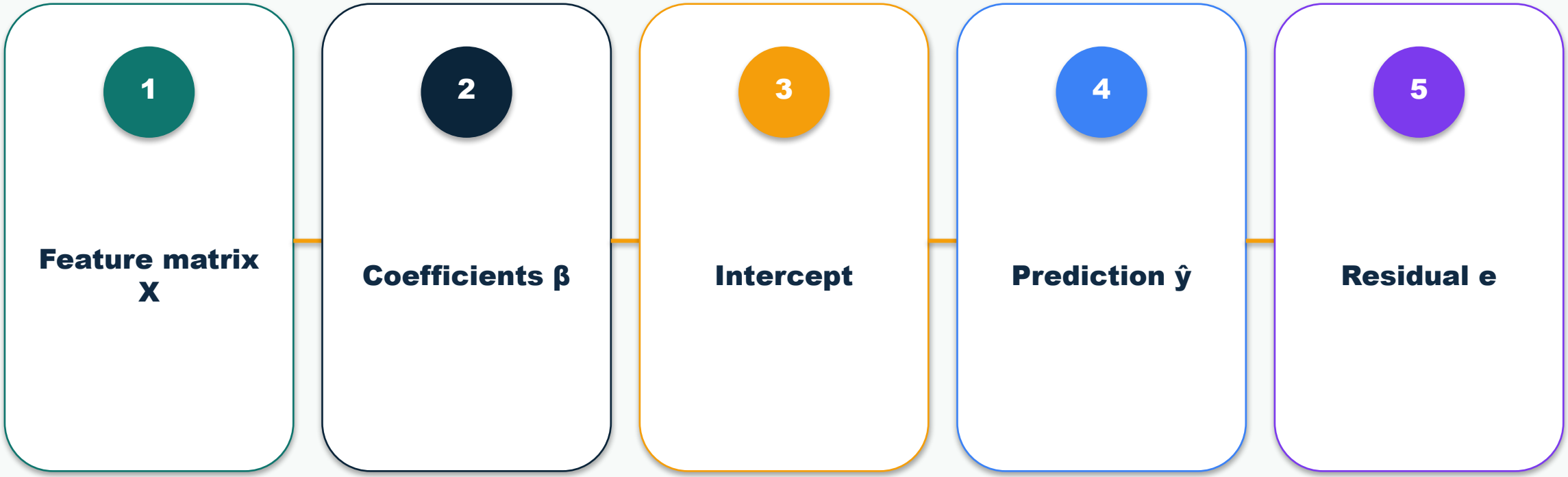
Mechanism → contract → calculation → evidence → control



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DATA MODEL METRIC CONTROL

Linear model mechanism · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Linear model mechanism · concrete contract

row	observation
column	feature
β	marginal slope
ŷ	numeric outcome

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DATA MODEL METRIC CONTROL

Linear model mechanism · worked calculation

$$\hat{y} = \beta_0 + \beta_1 x_1 + \ldots + \beta p x p$$

WORKED EXAMPLE

$$\hat{y} = 120{,}000 + 5{,}200 \times 92 + 7{,}200 \times 10 = \$670{,}400$$

READOUT

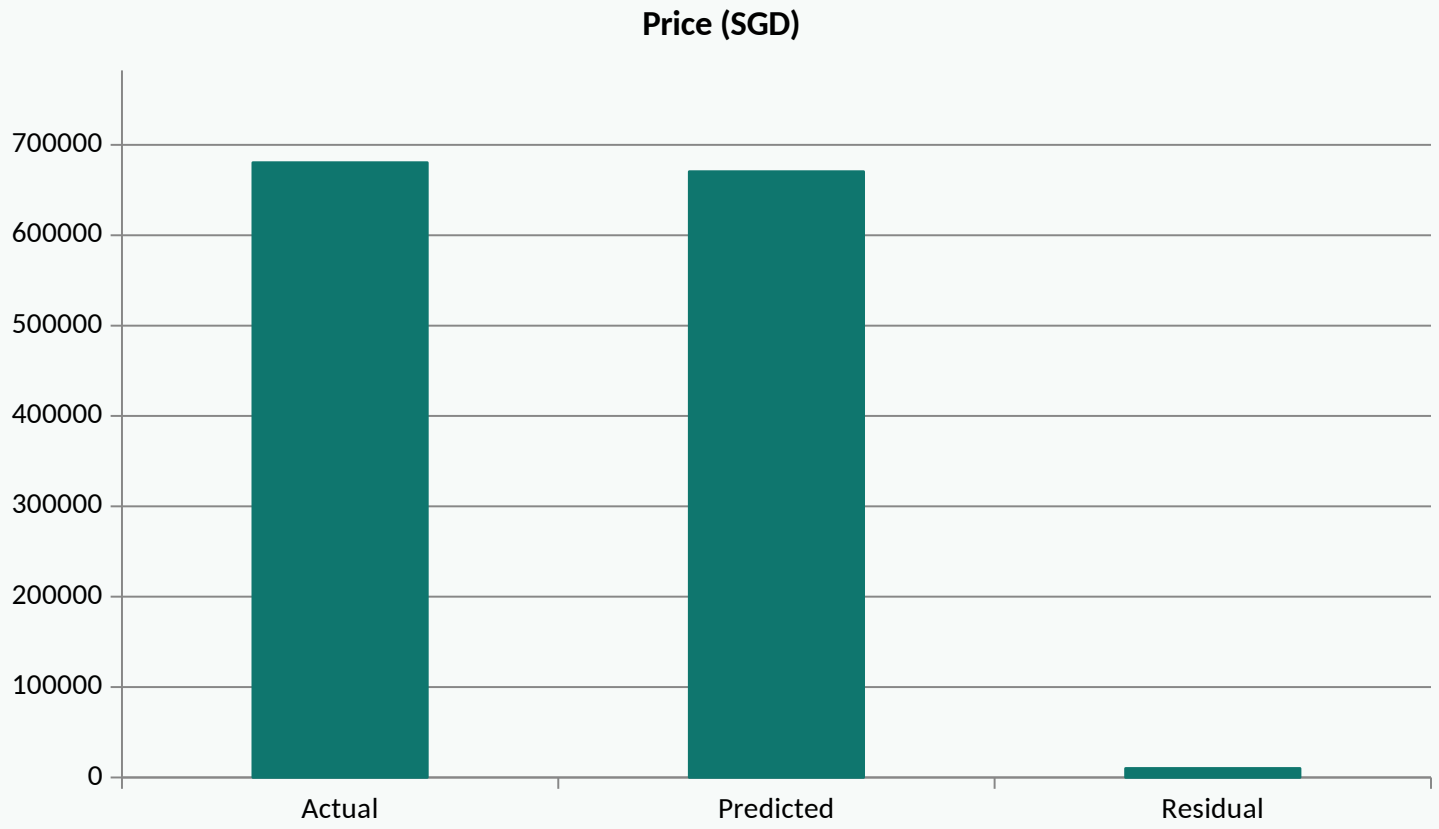
coefficient and residual

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Linear model mechanism · evidence



WHAT THE DATA SHOWS

Predictions, coefficients, and residuals can be reproduced from the frozen pipeline.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Linear model mechanism · failure and control

FAILURE MODE

Interpreting a coefficient causally when correlated variables or selection bias are present.

CONTROL

Use coefficients as conditional associations unless the design supports causality.

VERIFICATION

Predictions, coefficients, and residuals can be reproduced from the frozen pipeline.

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DATA MODEL METRIC CONTROL

Residual diagnostics · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Residual diagnostics · concrete contract

- # residual $e = y - \hat{y}$
- # zero-centred cloud
- # constant variance
- # independent errors

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DATA MODEL METRIC CONTROL

Residual diagnostics · worked calculation

$$SSE = \Sigma(y_i - \hat{y}_i)^2$$

WORKED EXAMPLE

errors [10, −5, 15] → SSE = 100+25+225 = 350

READOUT

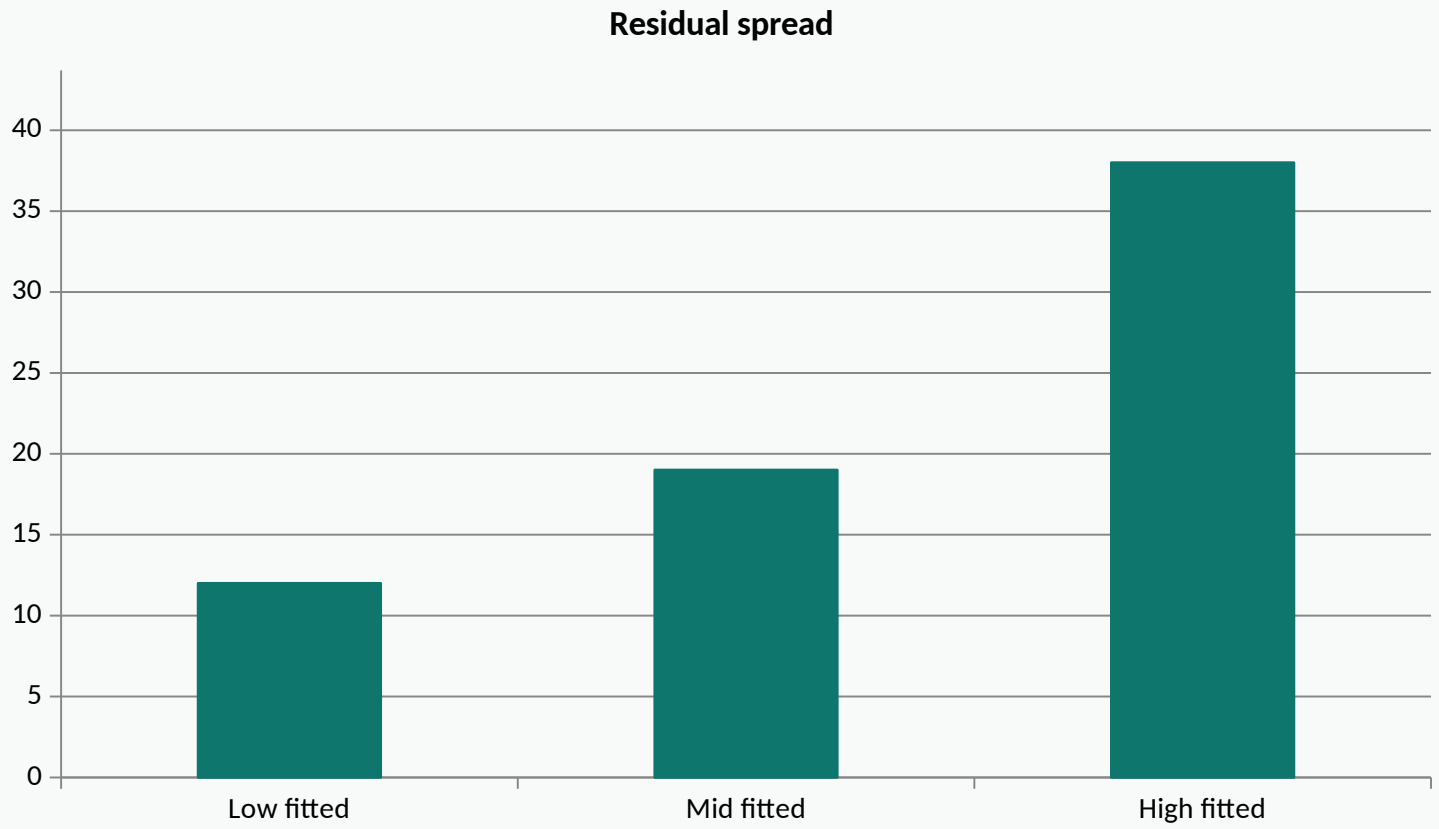
residual pattern

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Residual diagnostics · evidence



WHAT THE DATA SHOWS

Residuals show no systematic curve, drift, or group-specific bias.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Residual diagnostics · failure and control

FAILURE MODE

A funnel shape means error variance grows with the prediction.

CONTROL

Transform the target, add relevant features, or use robust intervals.

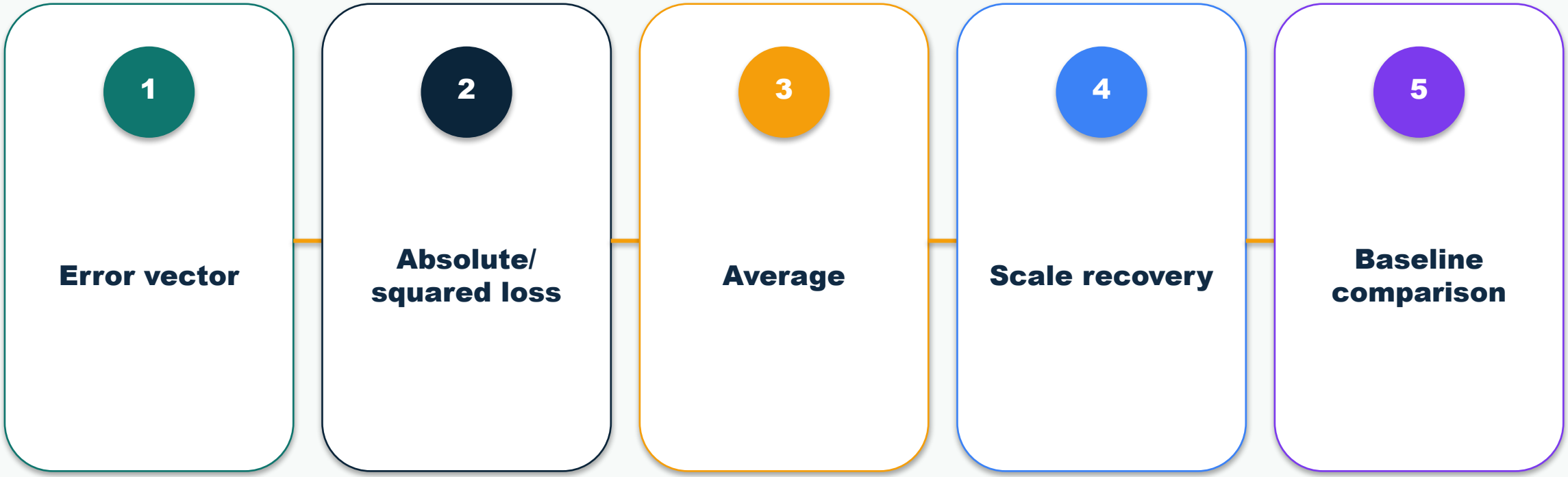
VERIFICATION

Residuals show no systematic curve, drift, or group-specific bias.

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DATA MODEL METRIC CONTROL

MAE, MSE, RMSE, and R² · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

MAE, MSE, RMSE, and R² · concrete contract

MAE	linear penalty
MSE	squared penalty
RMSE	target units
R²	versus mean baseline

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DATA MODEL METRIC CONTROL

MAE, MSE, RMSE, and R² · worked calculation

$$R^2 = 1 - SSE/SST$$

WORKED EXAMPLE

SSE 1,800; SST 9,000 → R² = 0.80

READOUT

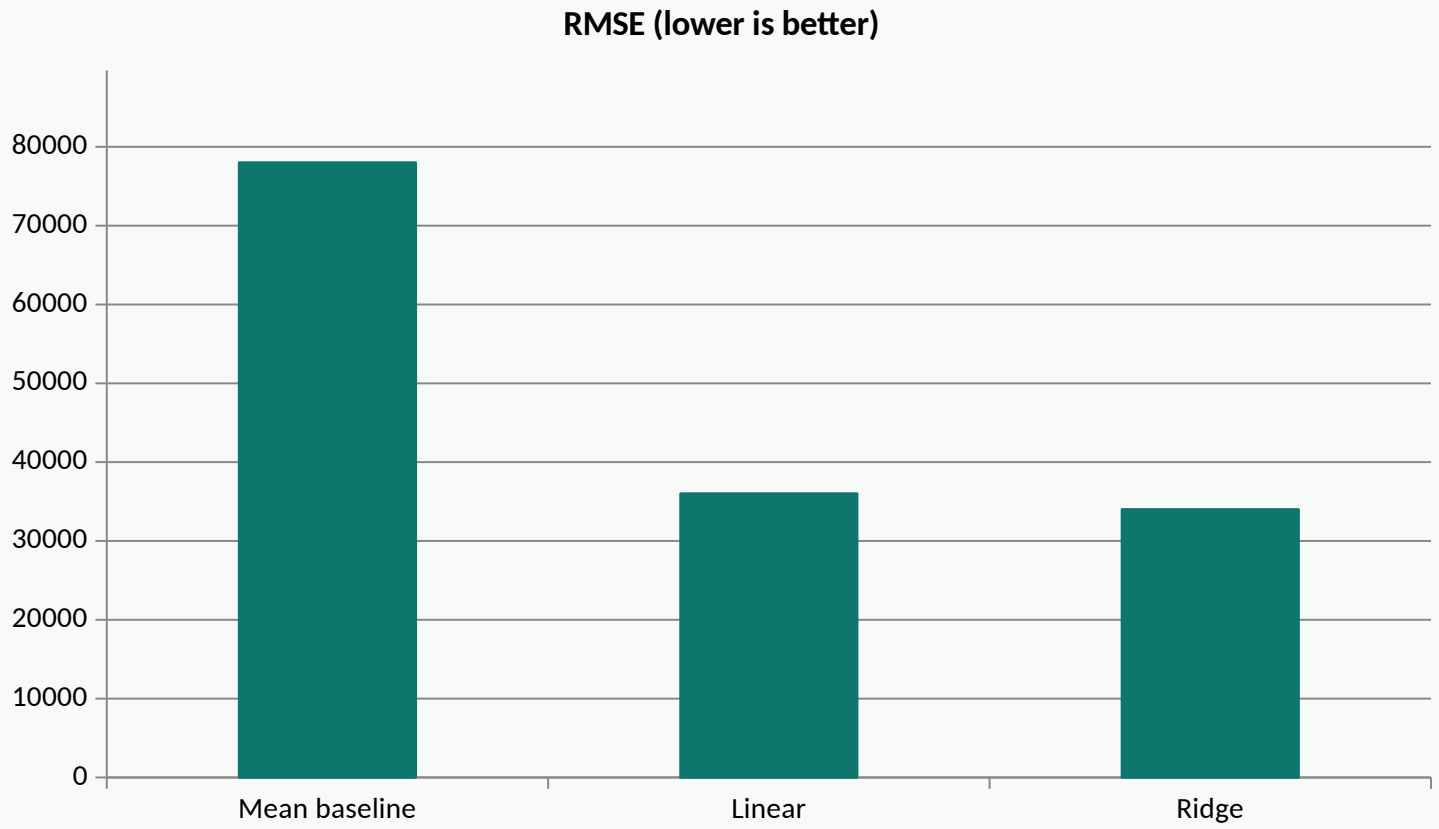
hold-out MAE/RMSE/R²

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

MAE, MSE, RMSE, and R² · evidence



WHAT THE DATA SHOWS

Metrics are computed on unseen rows and compared with a simple baseline.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

MAE, MSE, RMSE, and R^2 · failure and control

FAILURE MODE

High R^2 can coexist with large business-critical errors in a subgroup.

CONTROL

Report absolute error and segment diagnostics alongside R^2.

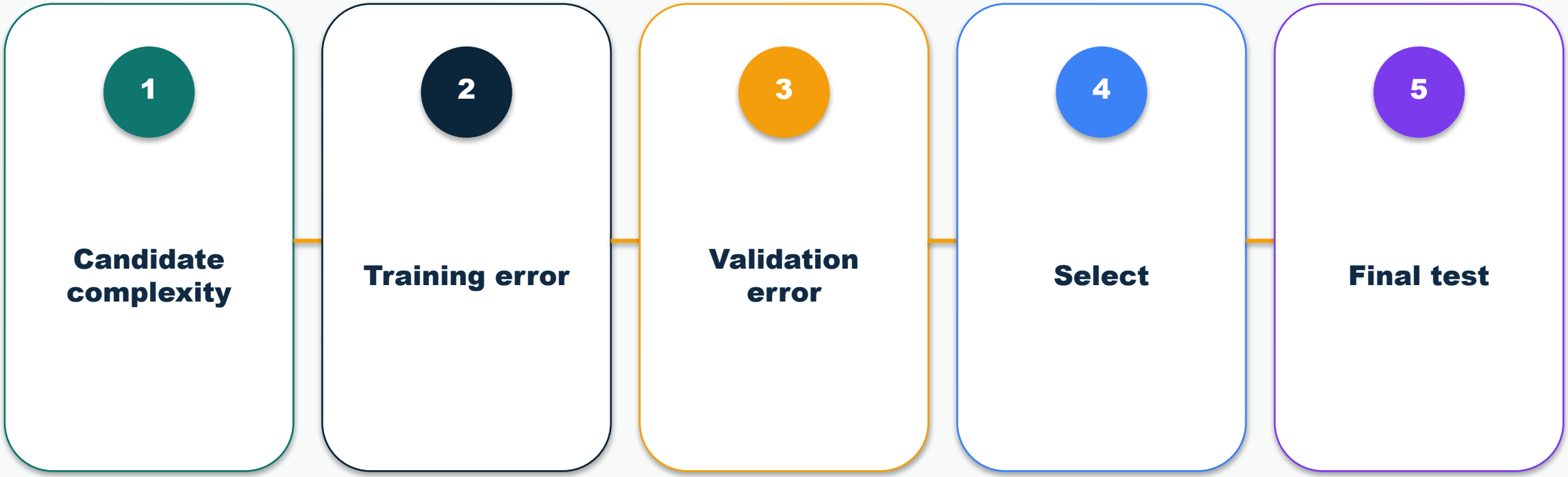
VERIFICATION

Metrics are computed on unseen rows and compared with a simple baseline.

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DATA MODEL METRIC CONTROL

Underfitting, overfitting, and validation · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Underfitting, overfitting, and validation · concrete contract

underfit	both errors high
fit	low validation error
overfit	widening gap
test	used once

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DATA MODEL METRIC CONTROL

Underfitting, overfitting, and validation · worked calculation

$$\text{gap} = \text{train RMSE} - \text{test RMSE}$$

WORKED EXAMPLE

train 12k vs test 47k → −35k gap indicates overfit

READOUT

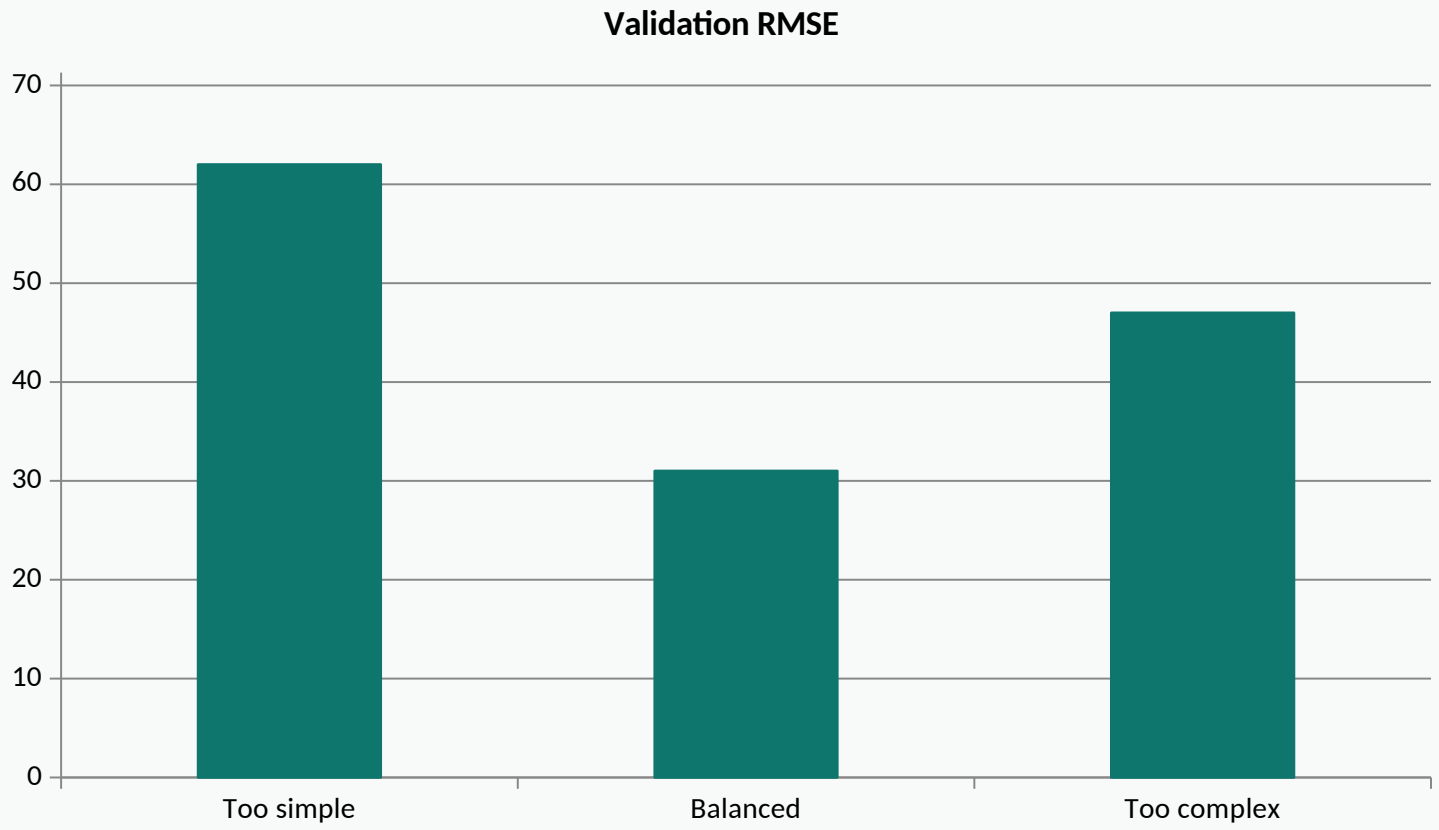
validation curve

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Underfitting, overfitting, and validation · evidence



WHAT THE DATA SHOWS

Learning curves show whether more data or a simpler model is likely to help.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Underfitting, overfitting, and validation · failure and control

FAILURE MODE

Selecting the deepest model because it minimises training error.

CONTROL

Choose complexity at the validation minimum and retain a final test set.

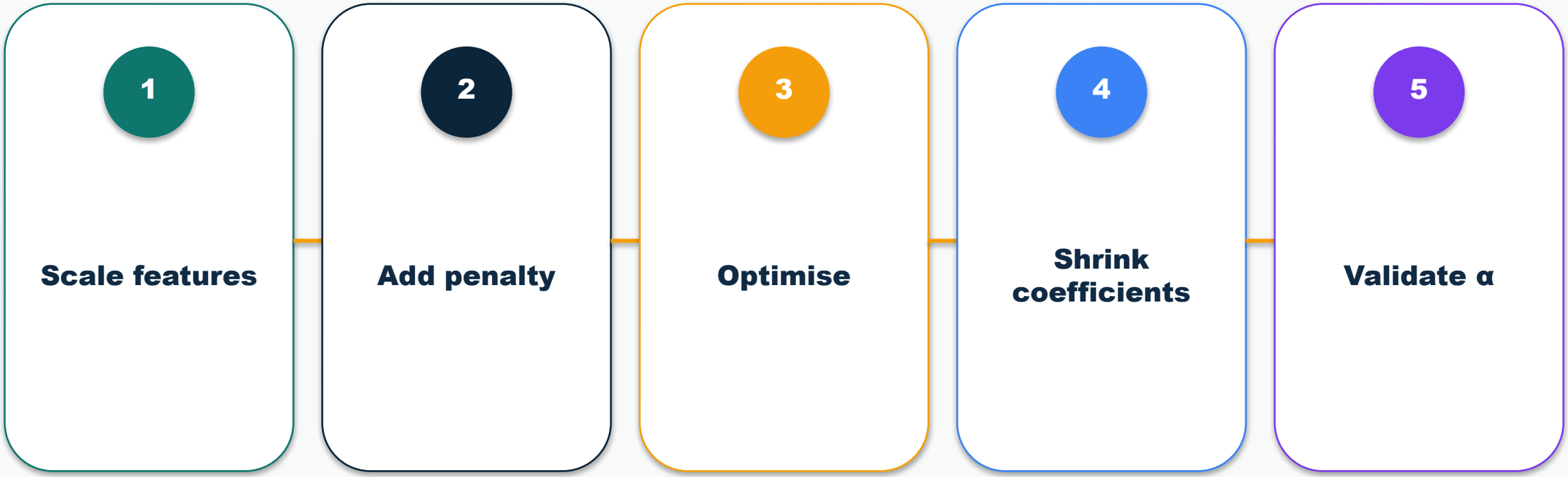
VERIFICATION

Learning curves show whether more data or a simpler model is likely to help.

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DATA MODEL METRIC CONTROL

Ridge and Lasso regularisation · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Ridge and Lasso regularisation · concrete contract

Ridge	L2 penalty
Lasso	L1 penalty
α	penalty strength
intercept	usually unpenalised

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DATA MODEL METRIC CONTROL

Ridge and Lasso regularisation · worked calculation

Ridge: minimise SSE + αΣβ²; Lasso: SSE + αΣ|β|

WORKED EXAMPLE

α=1 shrinks unstable coefficients while retaining all Ridge features

READOUT

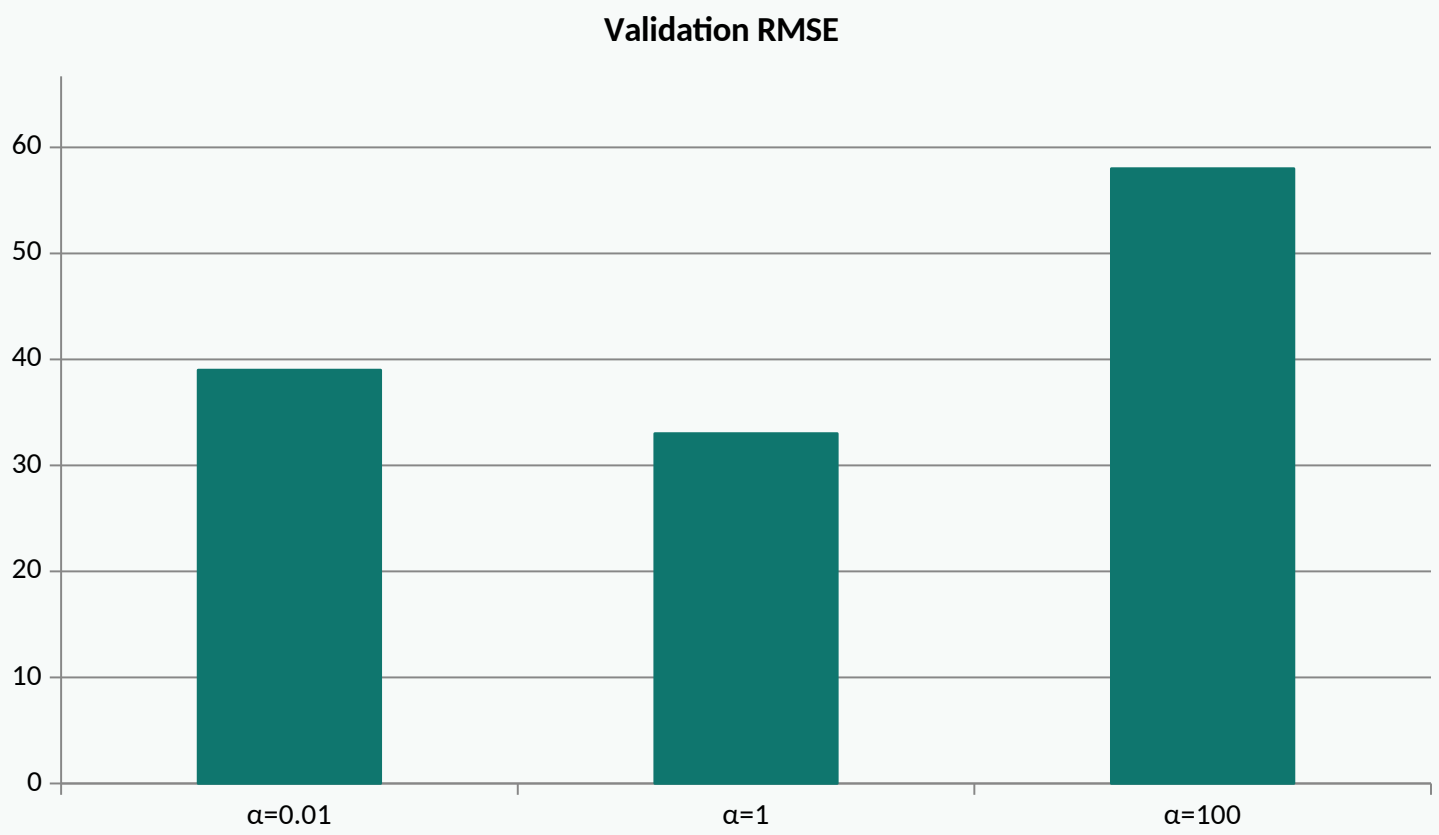
cross-validated error versus α

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Ridge and Lasso regularisation · evidence



WHAT THE DATA SHOWS

Selected α improves or preserves validation error without collapsing useful signal.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Ridge and Lasso regularisation · failure and control

FAILURE MODE

Comparing penalty strengths on unscaled features makes the penalty uneven.

CONTROL

Standardise within the pipeline before Ridge or Lasso.

VERIFICATION

Selected α improves or preserves validation error without collapsing useful signal.

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DATA MODEL METRIC CONTROL

Lab 05 · Build a regression baseline

CENTRAL APP

Regression

OPEN CENTRAL LAB PORTAL

INPUT Lab 05 mock CSV

MECHANISM Regression

EVIDENCE metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 05 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Build a regression baseline produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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DATA MODEL METRIC CONTROL

Lab 06 · Control overfitting with Ridge and Lasso

CENTRAL APP

Regression

OPEN CENTRAL LAB PORTAL

INPUT	Lab 06 mock CSV
MECHANISM	Regression
EVIDENCE	metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 06 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Control overfitting with Ridge and Lasso produces a reproducible artifact and a defensible interpretation—not only a screenshot.

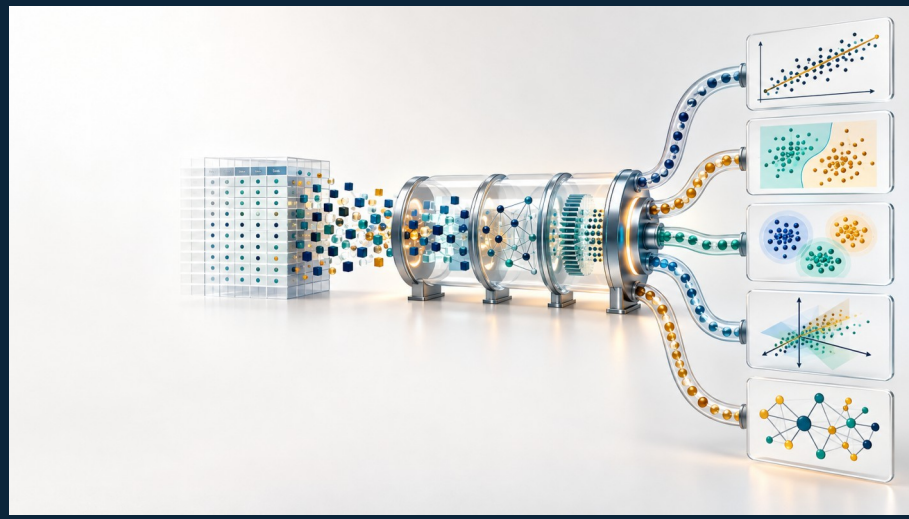
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TOPIC 04

Classification

LO4 · A7 · K6 · Separating classes and choosing an operating threshold

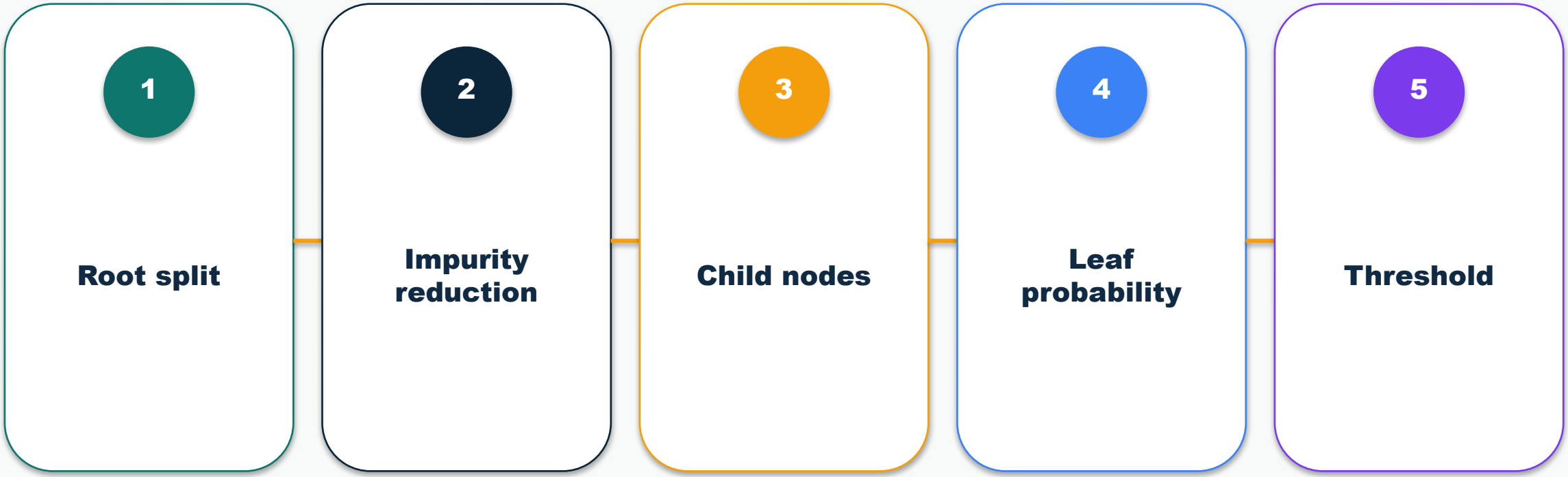
Mechanism → contract → calculation → evidence → control



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DATA MODEL METRIC CONTROL

Decision tree and ensemble path · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Decision tree and ensemble path · concrete contract

- # feature threshold
- # Gini/entropy
- # max_depth
- # min_samples_leaf

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DATA MODEL METRIC CONTROL

Decision tree and ensemble path · worked calculation

$$\text{Gini} = 1 - \Sigma p_k^2$$

WORKED EXAMPLE

$p(\text{churn})=0.7 \rightarrow \text{Gini}=1-0.7^2-0.3^2=0.42$

READOUT

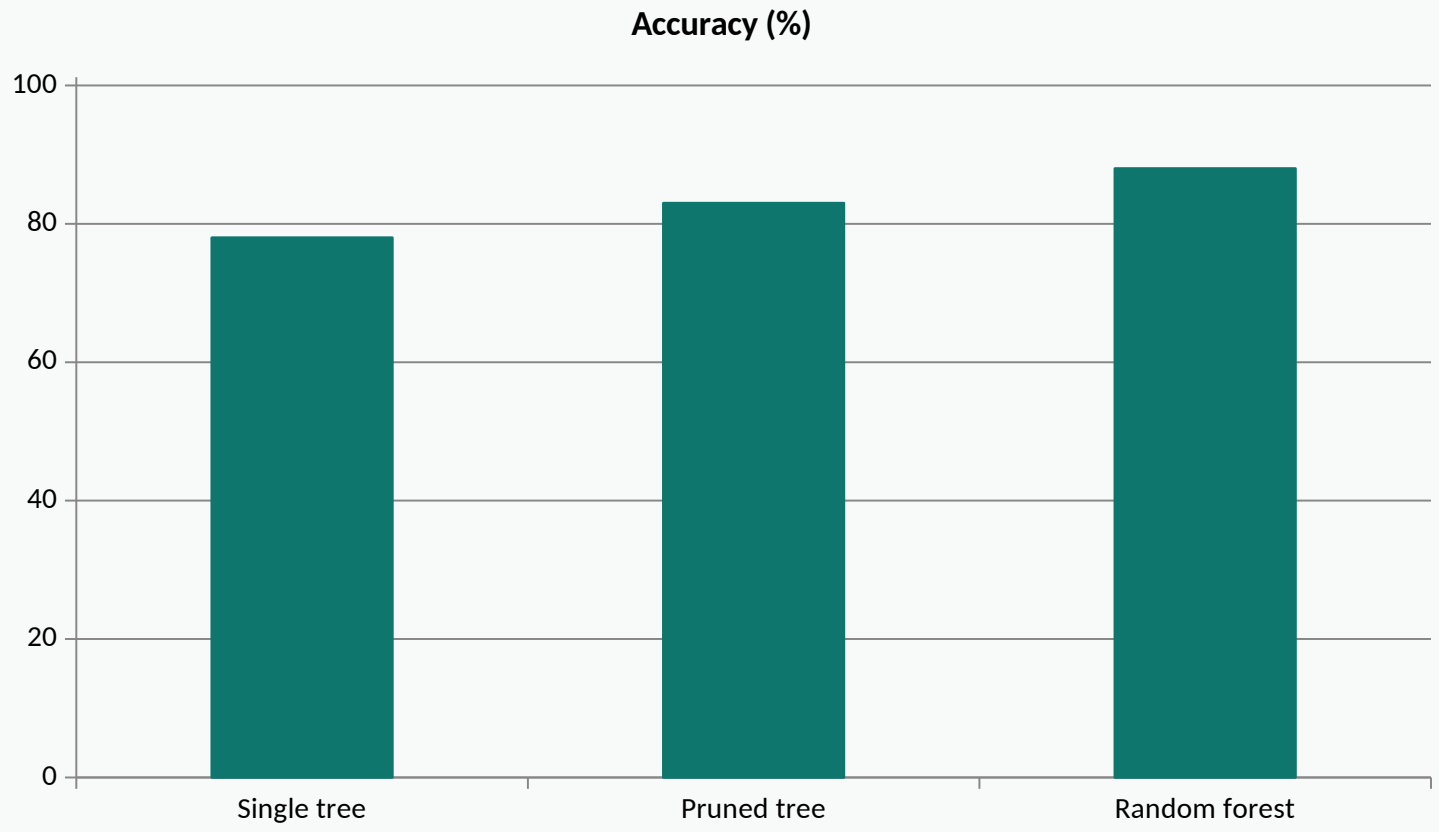
depth, leaf support, validation score

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Decision tree and ensemble path · evidence



WHAT THE DATA SHOWS

Leaf counts, feature importance, and hold-out scores remain plausible.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Decision tree and ensemble path · failure and control

FAILURE MODE

Deep leaves memorise rare combinations and become unstable.

CONTROL

Limit depth/leaf size and compare with bagged Random Forest estimates.

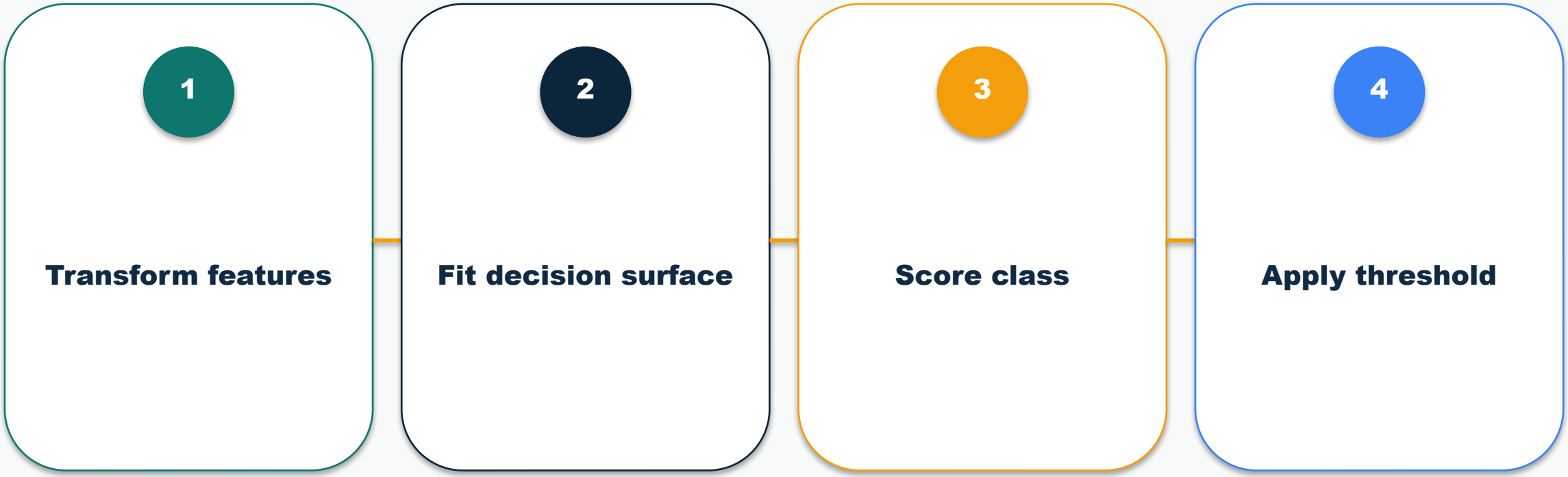
VERIFICATION

Leaf counts, feature importance, and hold-out scores remain plausible.

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DATA MODEL METRIC CONTROL

Logistic, KNN, SVM, and Naive Bayes · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Logistic, KNN, SVM, and Naive Bayes · concrete contract

logistic	log-odds
KNN	neighbourhood vote
SVM	margin/kernel
NB	conditional likelihood

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DATA MODEL METRIC CONTROL

Logistic, KNN, SVM, and Naive Bayes · worked calculation

$$\text{logit}(p)=\beta_0+\beta x;\ p = 1/(1+\exp(-z))$$

WORKED EXAMPLE

$z=1.386 \rightarrow p\approx0.80$

READOUT

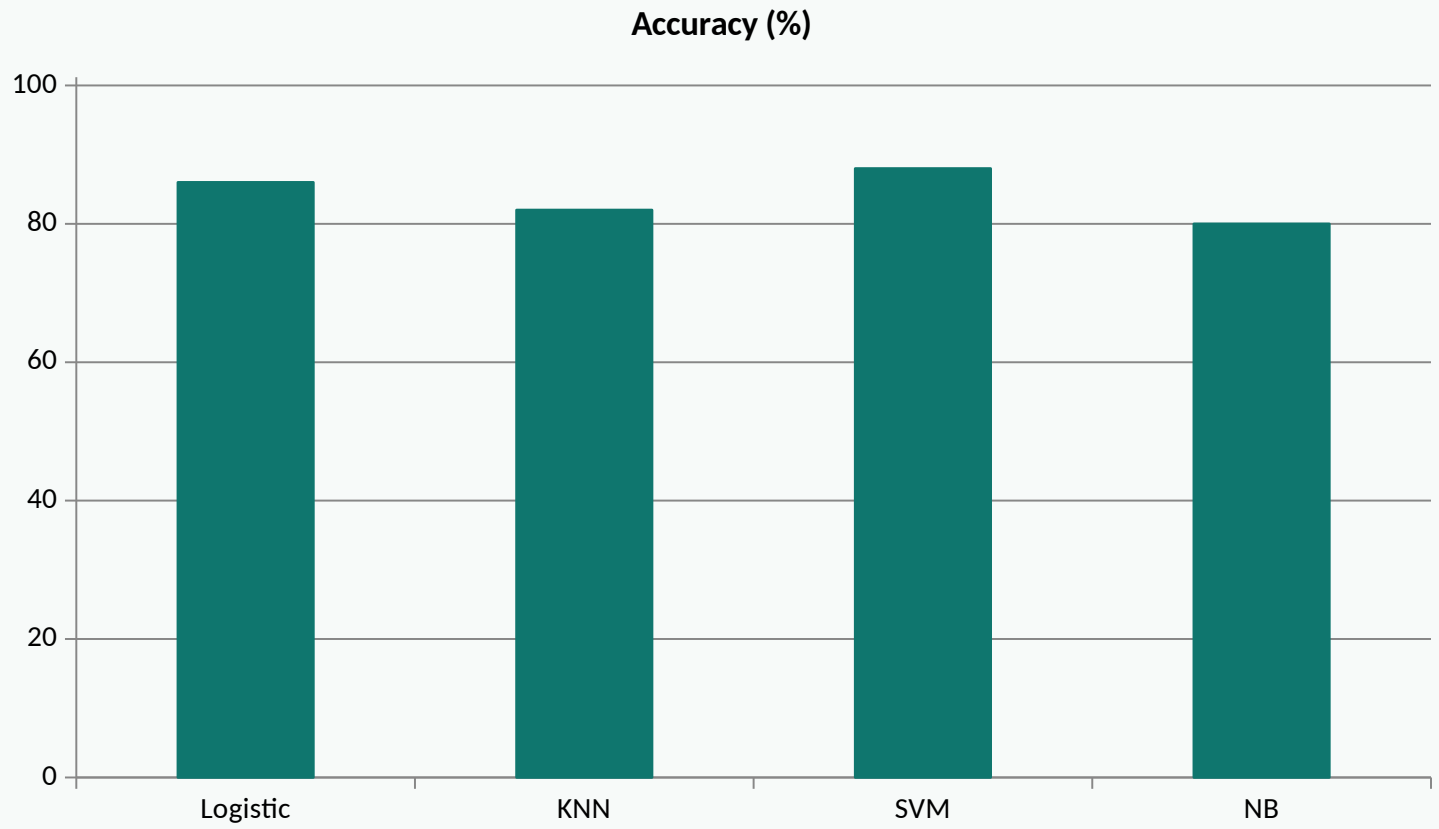
probability calibration and discrimination

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Logistic, KNN, SVM, and Naive Bayes · evidence



WHAT THE DATA SHOWS

Each classifier is evaluated on the same rows, labels, and metrics.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Logistic, KNN, SVM, and Naive Bayes · failure and control

FAILURE MODE

KNN and SVM are distorted when features have incompatible scales.

CONTROL

Use a shared preprocessing pipeline and compare models under identical folds.

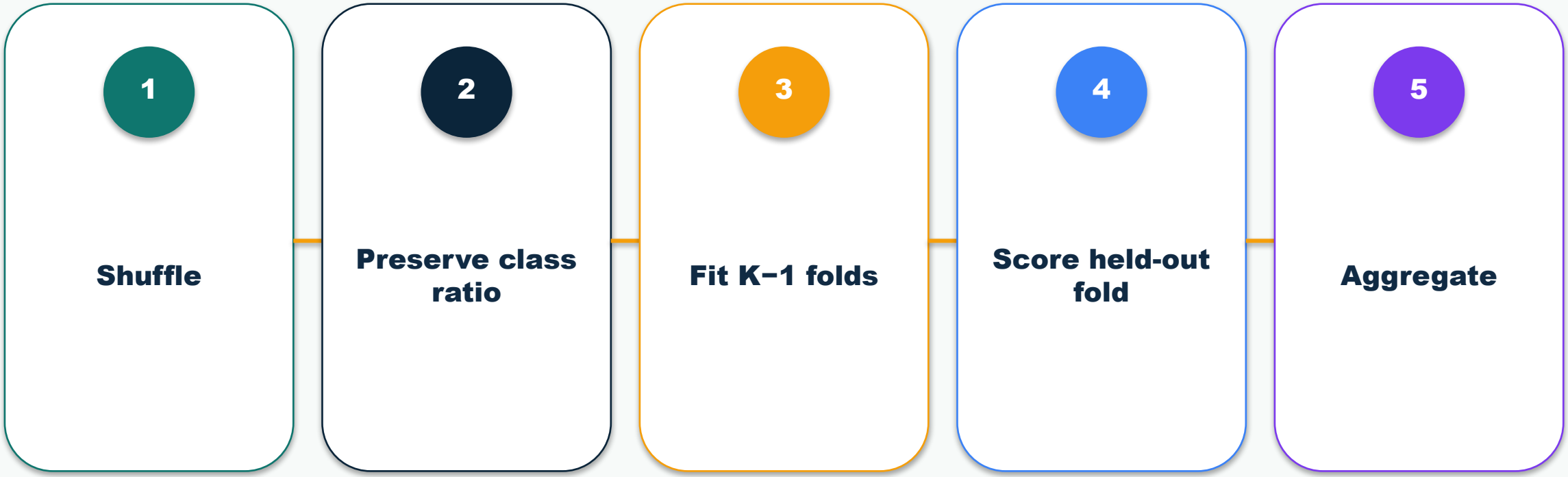
VERIFICATION

Each classifier is evaluated on the same rows, labels, and metrics.

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DATA MODEL METRIC CONTROL

Stratified K-fold cross-validation · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Stratified K-fold cross-validation · concrete contract

k	number of folds
stratify	class balance
seed	reproducibility
mean ± standard deviation	

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DATA MODEL METRIC CONTROL

Stratified K-fold cross-validation · worked calculation

$$\text{CV mean} = \Sigma\ \text{score}_i\ /\ k$$

WORKED EXAMPLE

[.84,.87,.85,.86,.88] → mean .86

READOUT

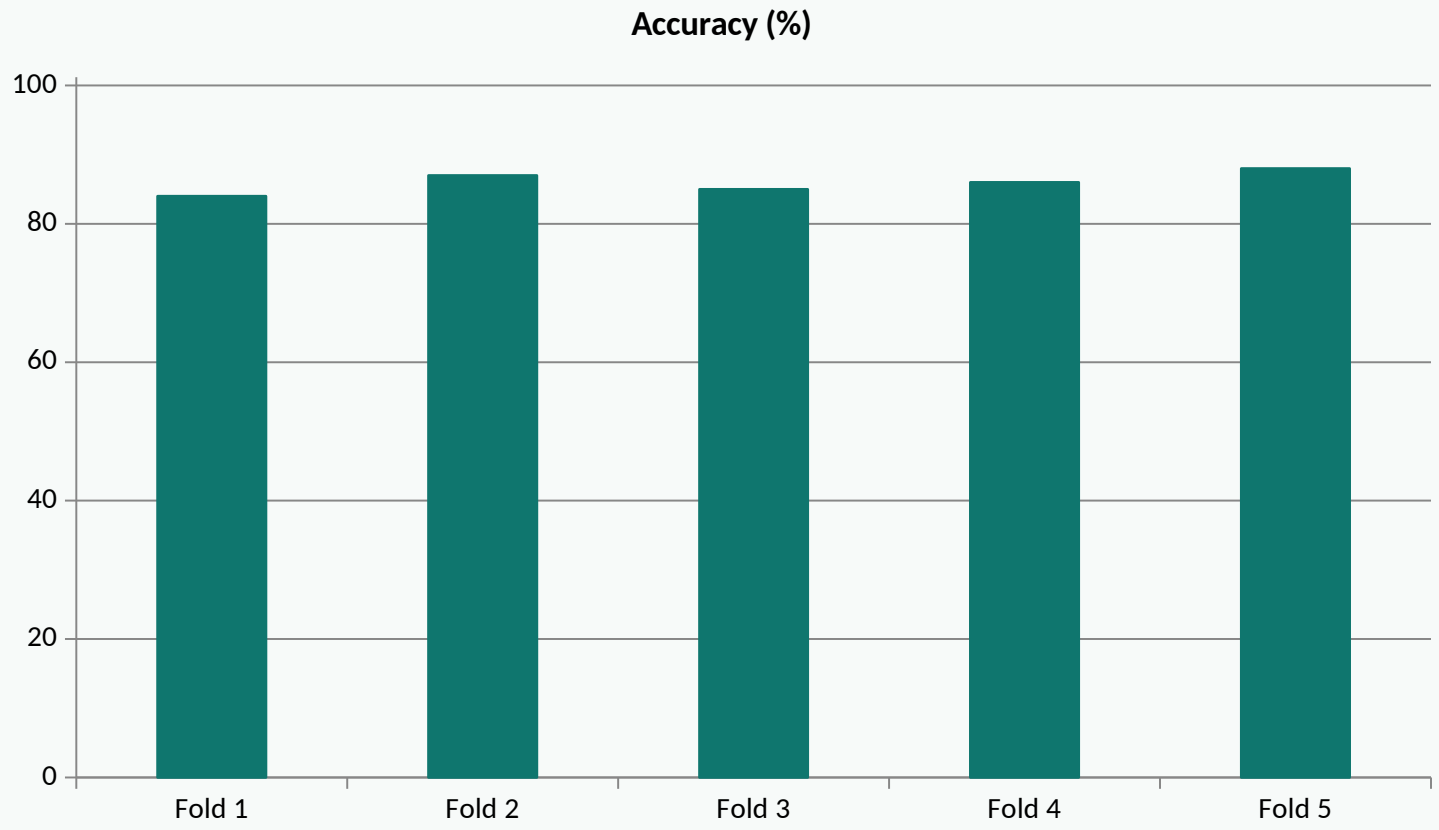
mean and fold variability

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Stratified K-fold cross-validation · evidence



WHAT THE DATA SHOWS

Every row is held out once and class proportions are stable across folds.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Stratified K-fold cross-validation · failure and control

FAILURE MODE

Preprocessing outside the cross-validation loop leaks fold information.

CONTROL

Put imputation, encoding, scaling, and model in one pipeline.

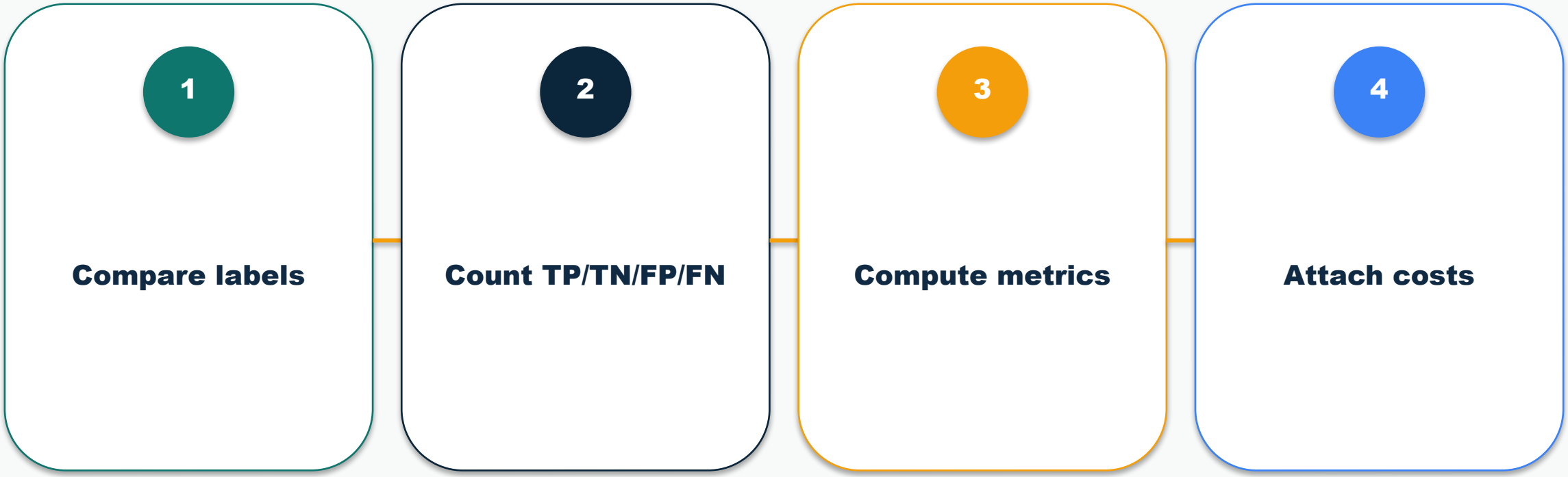
VERIFICATION

Every row is held out once and class proportions are stable across folds.

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DATA MODEL METRIC CONTROL

Confusion matrix and class metrics · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Confusion matrix and class metrics · concrete contract

- # precision=TP/(TP+FP)
- # recall=TP/(TP+FN)
- # specificity=TN/(TN+FP)
- # F1 harmonic mean

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DATA MODEL METRIC CONTROL

Confusion matrix and class metrics · worked calculation

$$F1 = 2PR/(P+R)$$

WORKED EXAMPLE

P=.75, R=.60 → F1=.667

READOUT

precision, recall, F1, specificity

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Confusion matrix and class metrics · evidence

ACTUAL \ PREDICTED	POSITIVE	NEGATIVE
POSITIVE	TP 45	FN 30
NEGATIVE	FP 15	TN 210

METRIC READOUT

Precision 75%
Recall 60%
F1 66.7%
Cases 300

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DATA MODEL METRIC CONTROL

Confusion matrix and class metrics · failure and control

FAILURE MODE

Accuracy hides failure when the positive class is rare.

CONTROL

Report the full matrix and select metrics based on error cost.

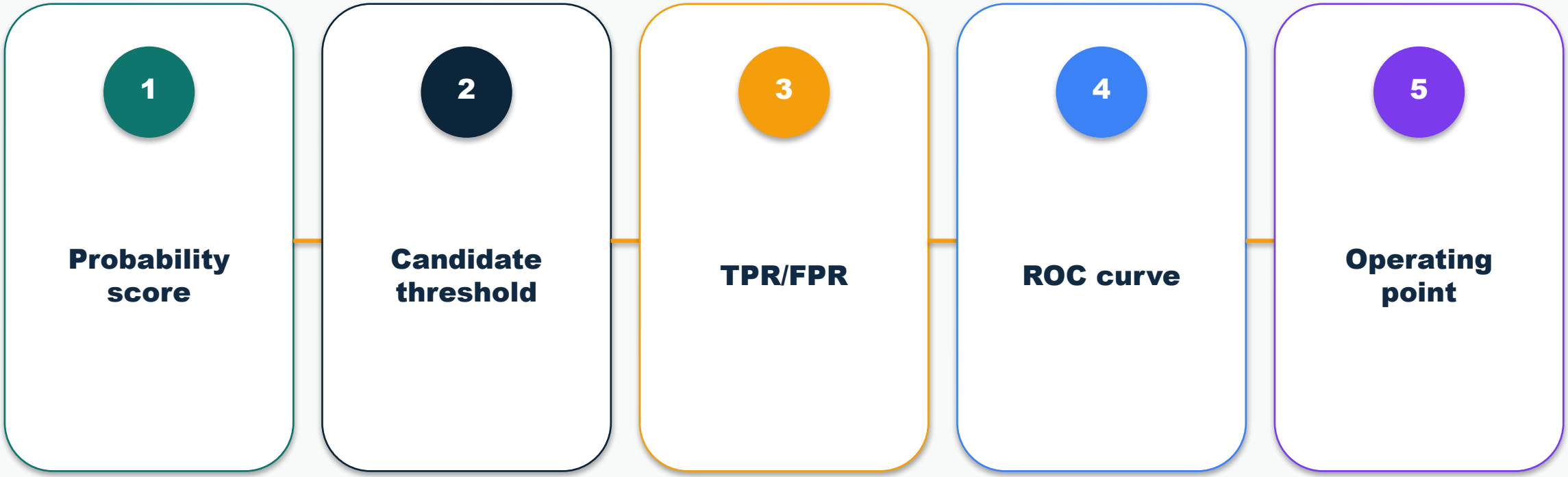
VERIFICATION

Matrix cell counts sum to the evaluated sample size.

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DATA MODEL METRIC CONTROL

ROC, AUC, and threshold policy · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

ROC, AUC, and threshold policy · concrete contract

TPR=recall	
FPR=FP/(FP+TN)	
AUC	ranking probability
threshold	policy choice

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DATA MODEL METRIC CONTROL

ROC, AUC, and threshold policy · worked calculation

$$\text{expected cost} = FP \times C_{FP} + FN \times C_{FN}$$

WORKED EXAMPLE

15 FP×$20 + 30 FN×$120 = $3,900

READOUT

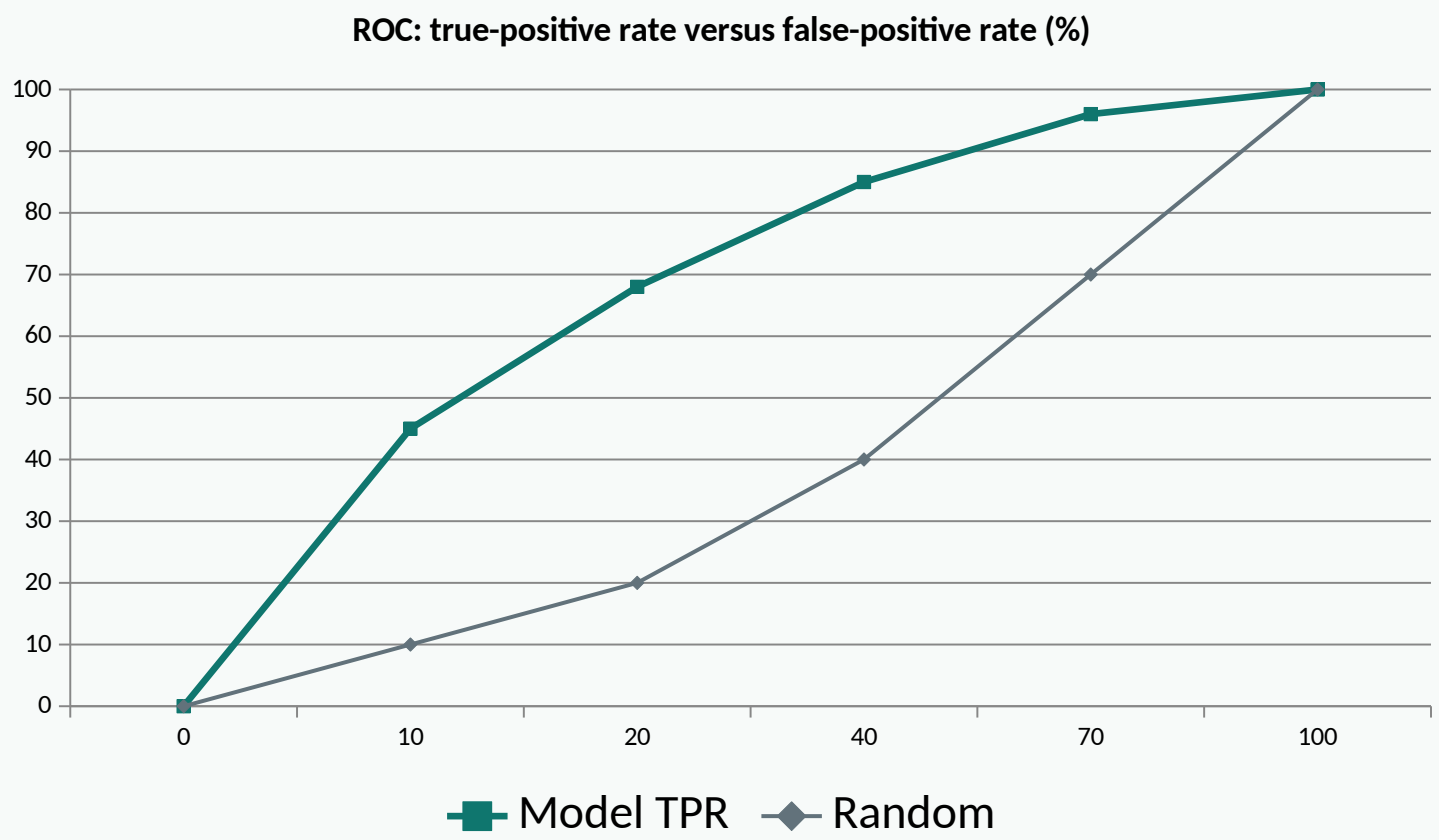
AUC plus operating cost

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

ROC, AUC, and threshold policy · evidence



OPERATING POINT

AUC summarises ranking.

The threshold still controls TPR, FPR, workload, and cost.

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DATA MODEL METRIC CONTROL

ROC, AUC, and threshold policy · failure and control

FAILURE MODE

Choosing 0.5 by convention when false negatives are much costlier.

CONTROL

Select a threshold from business costs and capacity constraints.

VERIFICATION

The chosen point states threshold, TPR, FPR, volume, and expected cost.

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DATA MODEL METRIC CONTROL

Lab 07 · Compare classification algorithms

CENTRAL APP

Classification

OPEN CENTRAL LAB PORTAL

INPUT Lab 07 mock CSV

MECHANISM Classification

EVIDENCE metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 07 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Compare classification algorithms produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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DATA MODEL METRIC CONTROL

Lab 08 · Use stratified K-fold validation

CENTRAL APP

Cross-validation

OPEN CENTRAL LAB PORTAL

INPUT	Lab 08 mock CSV
MECHANISM	Cross-validation
EVIDENCE	metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 08 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Use stratified K-fold validation produces a reproducible artifact and a defensible interpretation —not only a screenshot.

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DATA MODEL METRIC CONTROL

Lab 09 · Interpret confusion metrics and error costs

CENTRAL APP

Classification

OPEN CENTRAL LAB PORTAL

INPUT Lab 09 mock CSV

MECHANISM Classification

EVIDENCE metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 09 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Interpret confusion metrics and error costs produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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TOPIC 05

Clustering

LO5 · A5/A9 · K3/K5 · Discovering structure without labels

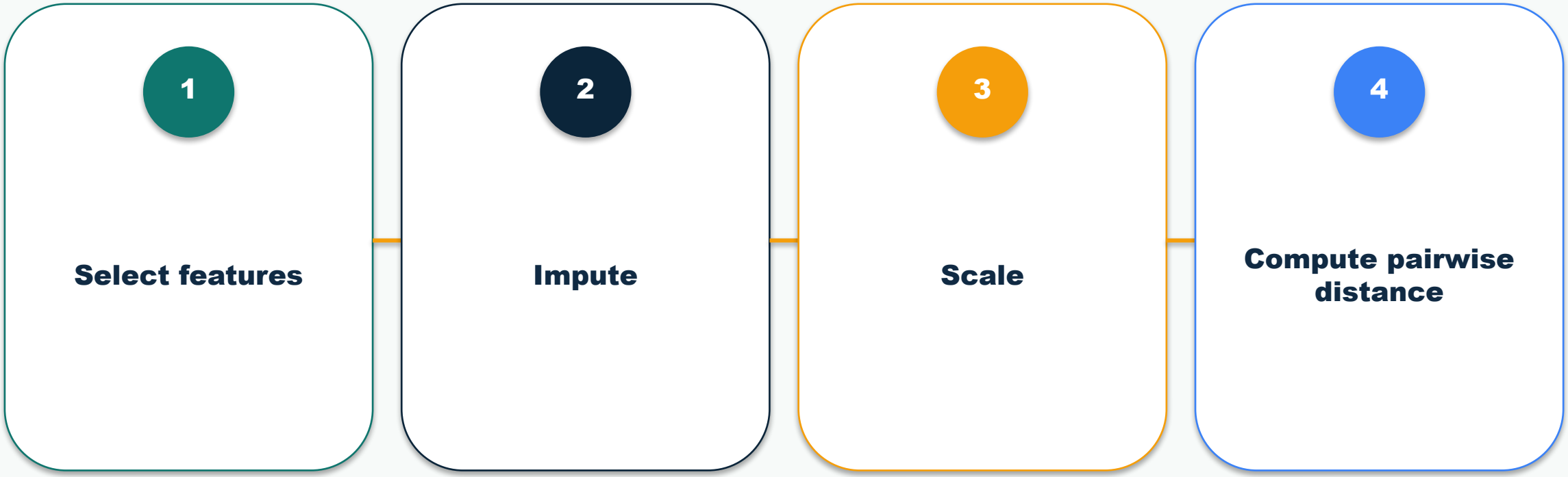
Mechanism → contract → calculation → evidence → control



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DATA MODEL METRIC CONTROL

Distance and scaling · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Distance and scaling · concrete contract

Euclidean	continuous
Manhattan	absolute steps
cosine	direction
categorical	requires encoding

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DATA MODEL METRIC CONTROL

Distance and scaling · worked calculation

$$d(x,y)=\sqrt{\Sigma(x_i-y_i)^2}$$

WORKED EXAMPLE

(2,3) to (5,7) → √(9+16)=5

READOUT

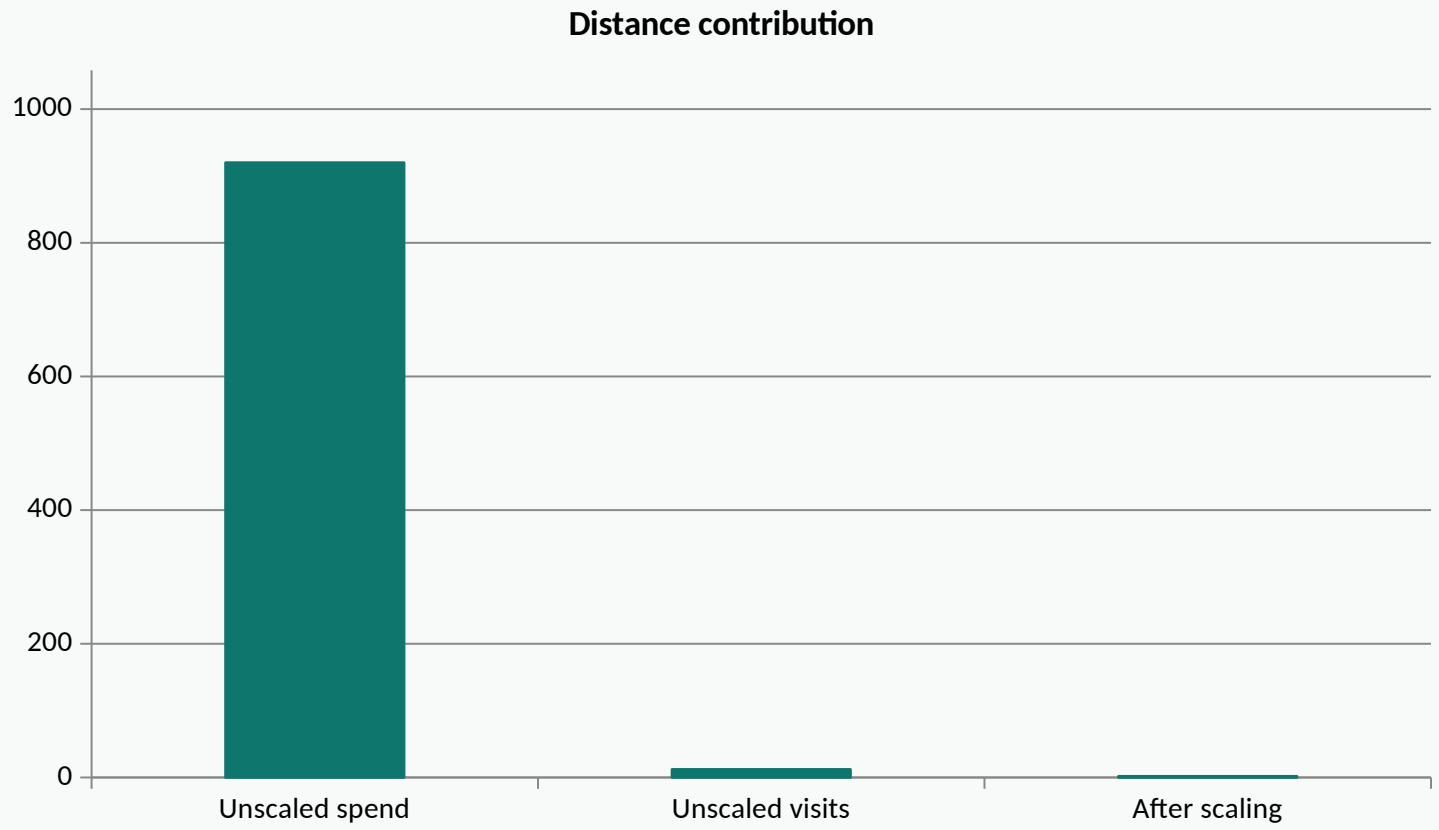
distance distribution

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Distance and scaling · evidence



WHAT THE DATA SHOWS

Nearest neighbours are sensible under domain review and robust to scaling choices.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Distance and scaling · failure and control

FAILURE MODE

A high-magnitude spend field dominates an unscaled visit-frequency field.

CONTROL

Standardise numeric variables and justify the distance metric.

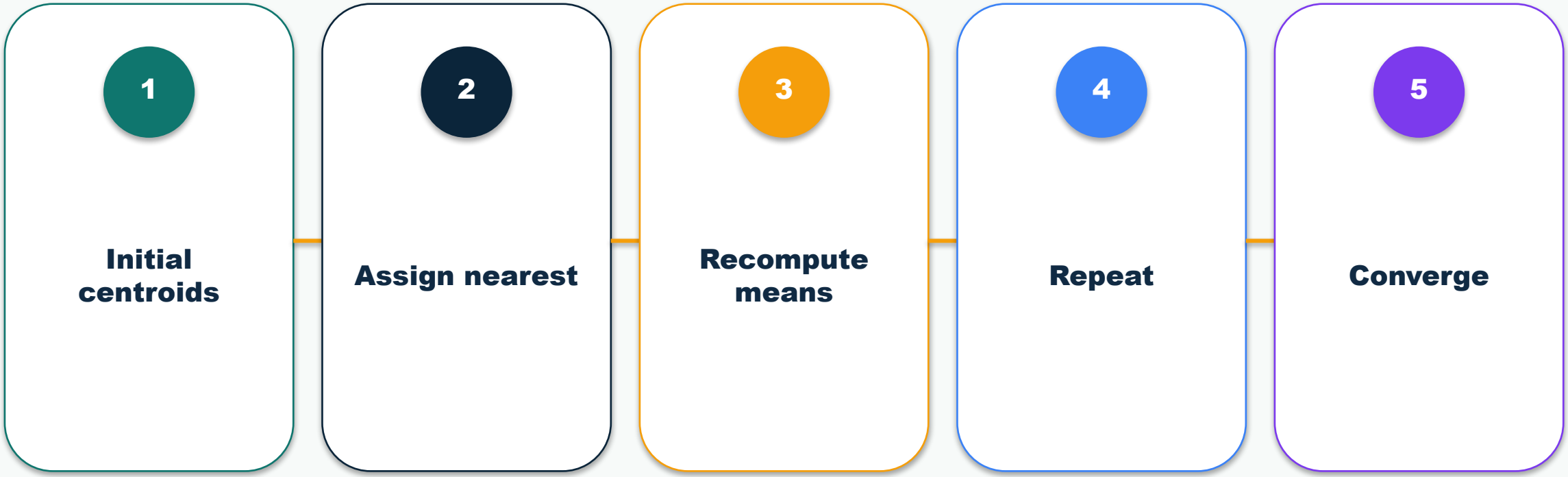
VERIFICATION

Nearest neighbours are sensible under domain review and robust to scaling choices.

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DATA MODEL METRIC CONTROL

K-means assignment/update loop · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

K-means assignment/update loop · concrete contract

k	requested clusters
centroid	feature mean
inertia	within-cluster SSE
n_init	multiple starts

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DATA MODEL METRIC CONTROL

K-means assignment/update loop · worked calculation

$$\text{inertia} = \Sigma ||x_i - \mu\text{cluster}(i)||^2$$

WORKED EXAMPLE

three compact groups reduce inertia from 940 to 218

READOUT

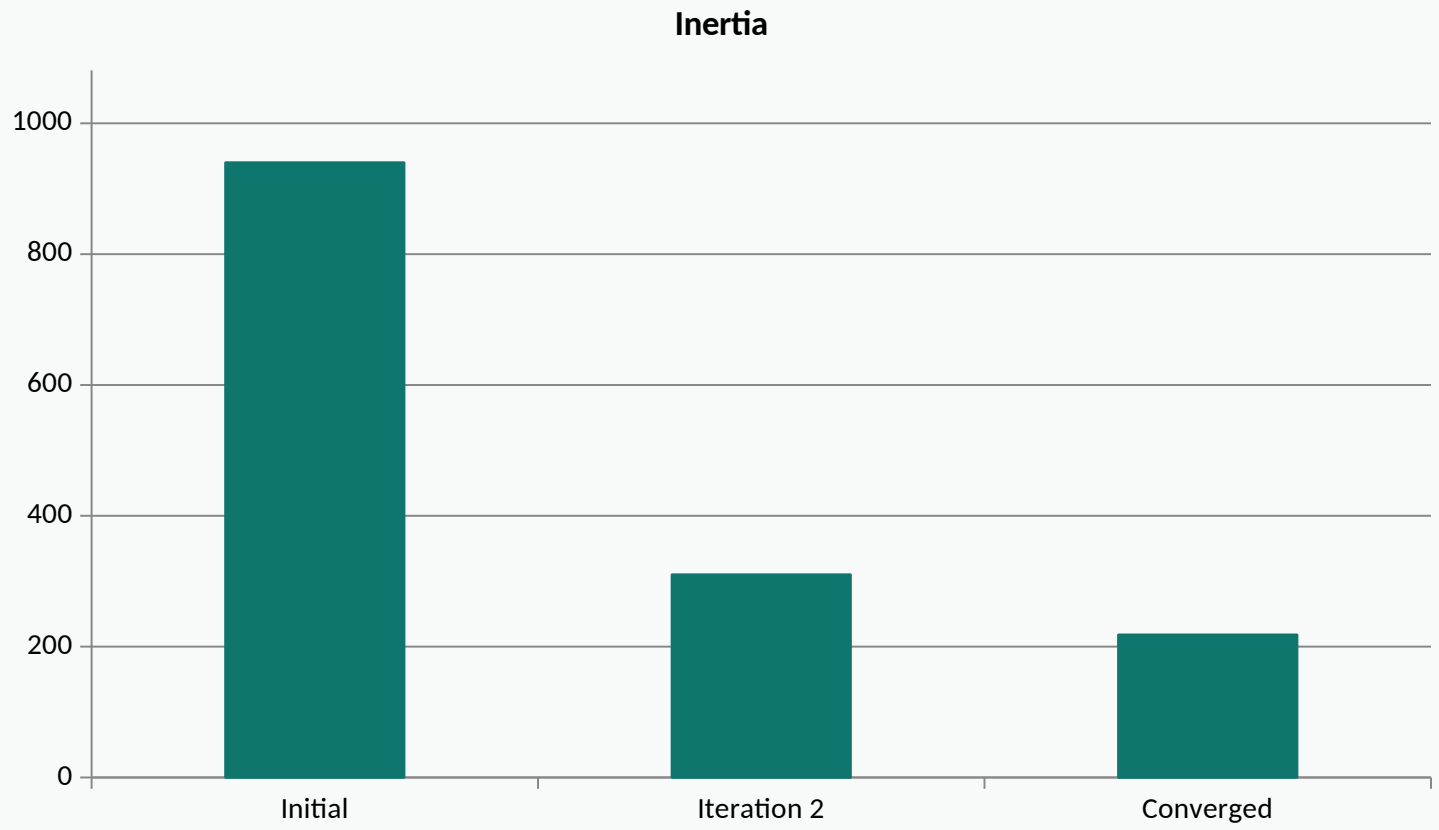
inertia and stability

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

K-means assignment/update loop · evidence



WHAT THE DATA SHOWS

Centroids, inertia, and labels reproduce under the recorded seed.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

K-means assignment/update loop · failure and control

FAILURE MODE

One random start reaches a poor local minimum.

CONTROL

Use multiple initialisations and compare label stability.

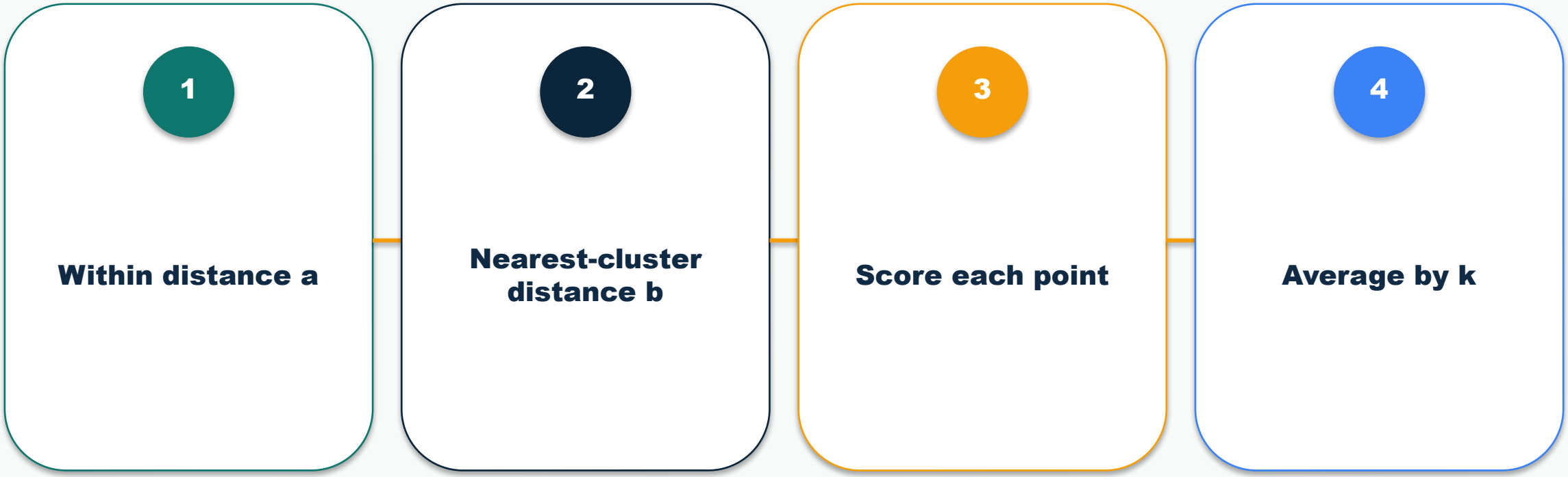
VERIFICATION

Centroids, inertia, and labels reproduce under the recorded seed.

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DATA MODEL METRIC CONTROL

Silhouette analysis · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Silhouette analysis · concrete contract

range	−1 to 1
near 1	separated
near 0	boundary
negative	likely misassigned

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DATA MODEL METRIC CONTROL

Silhouette analysis · worked calculation

$$s=(b-a)/\max(a,b)$$

WORKED EXAMPLE

a=2, b=5 → s=0.60

READOUT

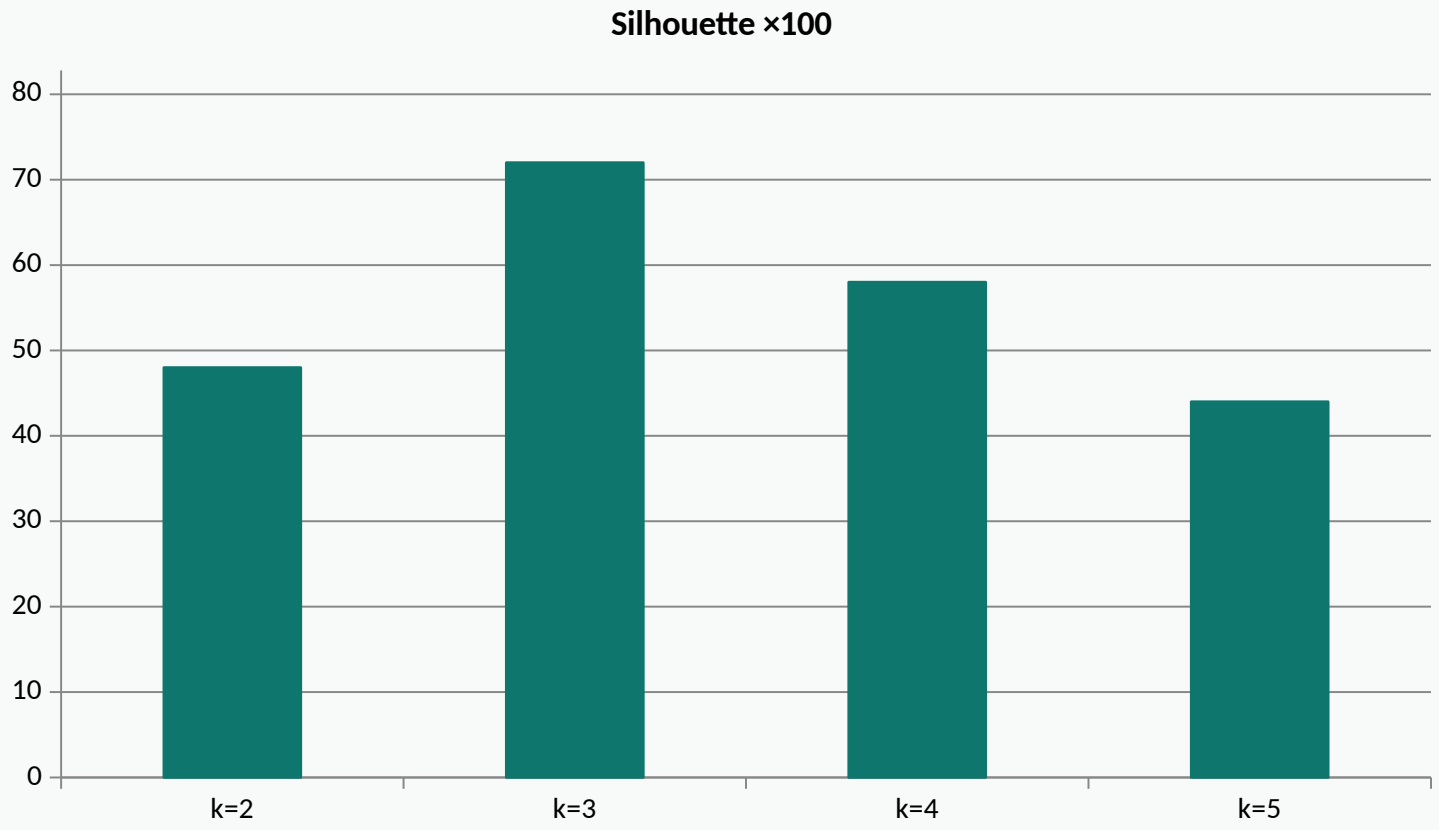
mean and per-cluster silhouette

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Silhouette analysis · evidence



WHAT THE DATA SHOWS

The chosen k has acceptable separation and interpretable cluster sizes.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Silhouette analysis · failure and control

FAILURE MODE

Selecting the highest average while ignoring a tiny or unstable cluster.

CONTROL

Review the distribution, cluster sizes, and business usefulness.

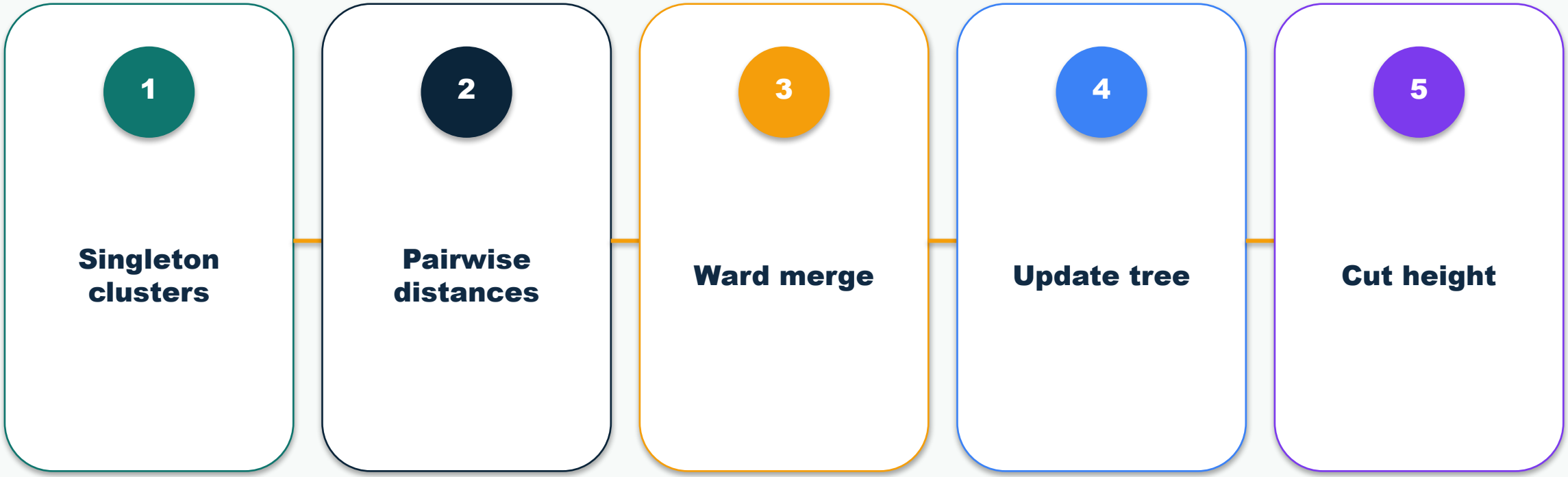
VERIFICATION

The chosen k has acceptable separation and interpretable cluster sizes.

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DATA MODEL METRIC CONTROL

Hierarchical clustering and dendrogram · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Hierarchical clustering and dendrogram · concrete contract

{}	**linkage**	Ward
#	**merge distance**	
#	**dendrogram height**	
{}	**cut**	cluster count

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DATA MODEL METRIC CONTROL

Hierarchical clustering and dendrogram · worked calculation

Ward merges the pair with the smallest increase in within-cluster variance

WORKED EXAMPLE

large jump from 6.2 to 14.8 suggests a cut before the jump

READOUT

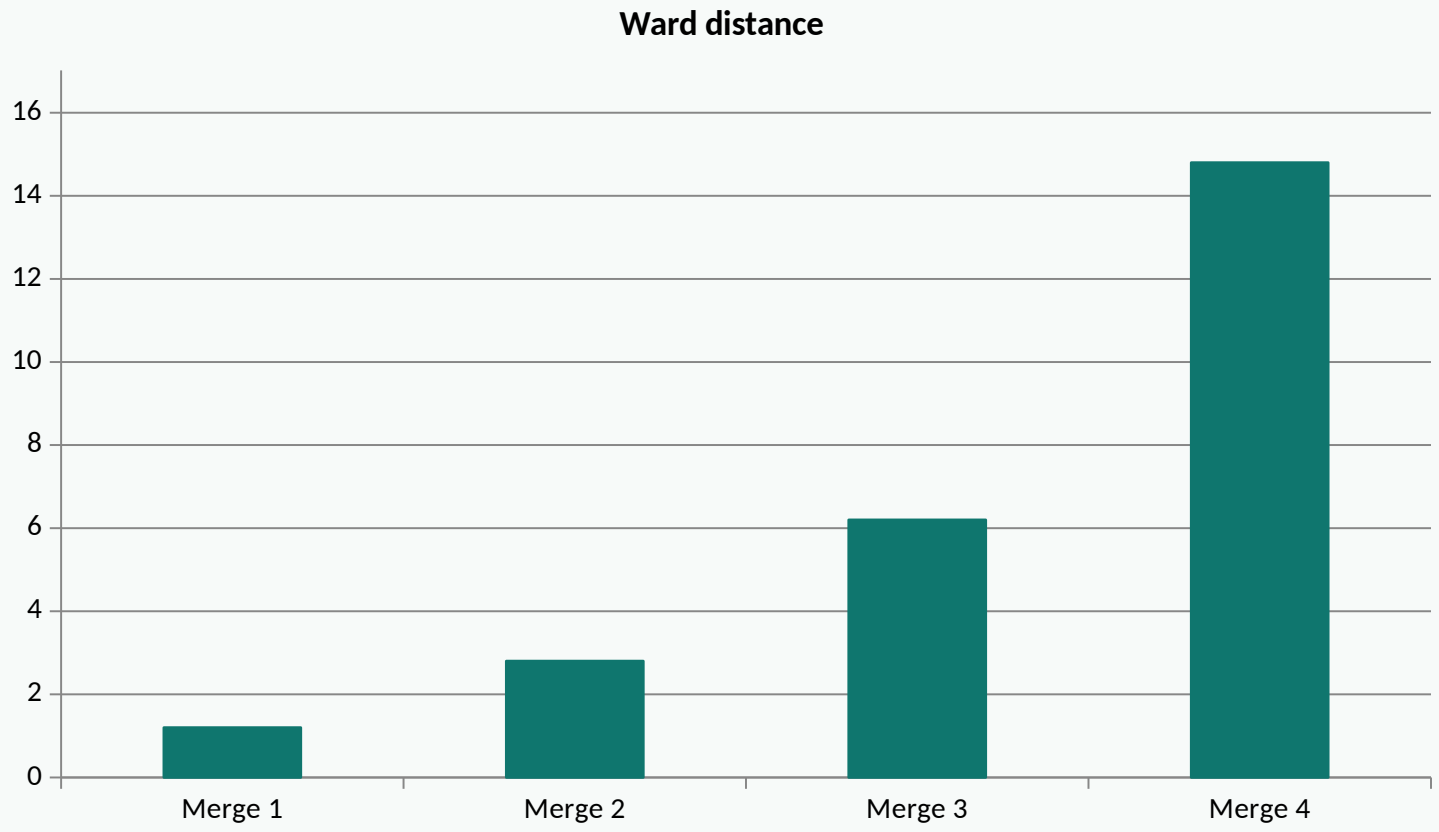
merge-distance jump

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Hierarchical clustering and dendrogram · evidence



WHAT THE DATA SHOWS

The cut produces the stated cluster count and matching scatter colours.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Hierarchical clustering and dendrogram · failure and control

FAILURE MODE

Treating horizontal spacing in a dendrogram as a numeric distance.

CONTROL

Read merge distance from the vertical axis and document the cut height.

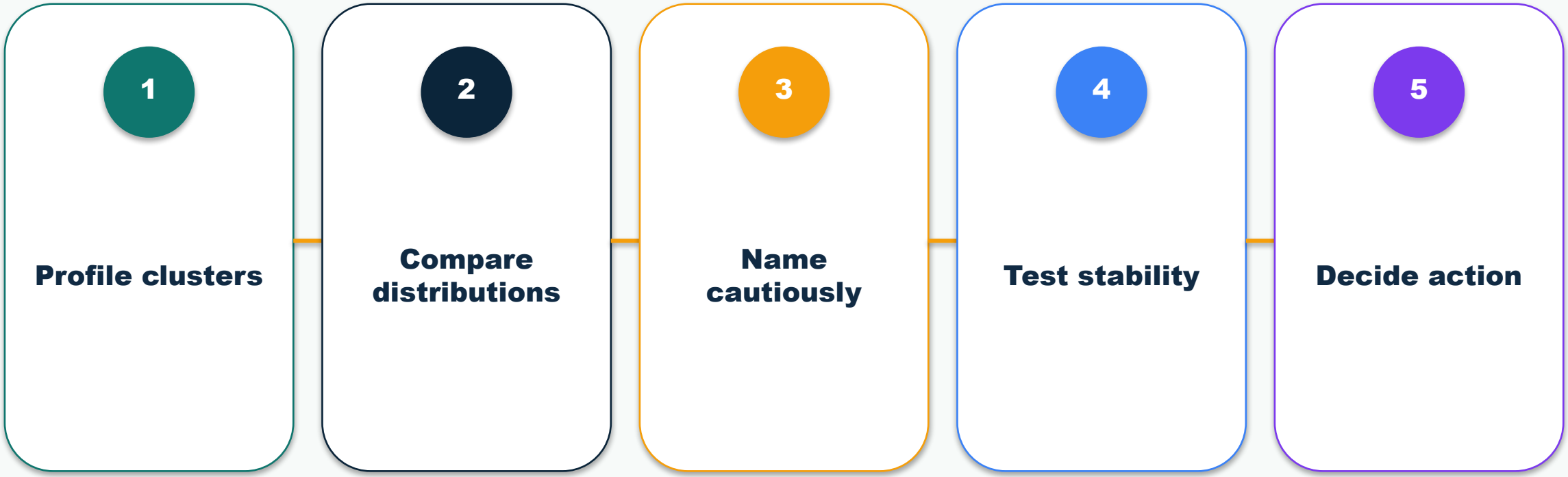
VERIFICATION

The cut produces the stated cluster count and matching scatter colours.

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DATA MODEL METRIC CONTROL

Cluster interpretation and limitations · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Cluster interpretation and limitations · concrete contract

size	n and %
centroid	scaled profile
stability	repeated samples
action	testable intervention

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DATA MODEL METRIC CONTROL

Cluster interpretation and limitations · worked calculation

standardised lift = cluster mean − overall mean

WORKED EXAMPLE

cluster A spend z=+1.2 and visits z=+0.8

READOUT

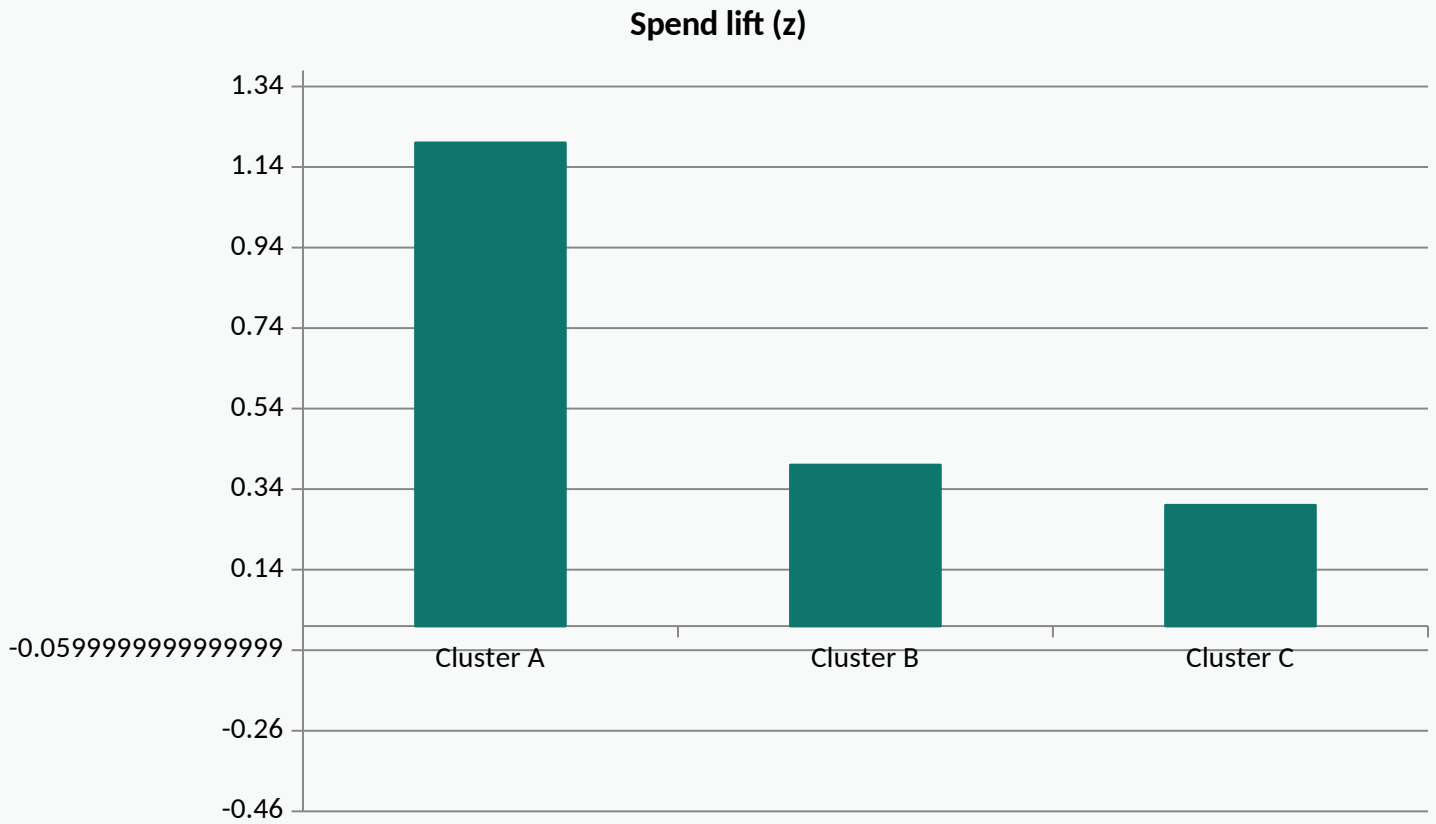
profile lift and stability

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Cluster interpretation and limitations · evidence



WHAT THE DATA SHOWS

Cluster actions are tested and do not encode prohibited discrimination.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Cluster interpretation and limitations · failure and control

FAILURE MODE

Narrative labels exaggerate weak numerical differences.

CONTROL

Publish measured traits, uncertainty, and exclusions before naming segments.

VERIFICATION

Cluster actions are tested and do not encode prohibited discrimination.

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DATA MODEL METRIC CONTROL

Lab 10 · Select K with silhouette evidence

CENTRAL APP

Clustering

OPEN CENTRAL LAB PORTAL

INPUT Lab 10 mock CSV

MECHANISM Clustering

EVIDENCE metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 10 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Select K with silhouette evidence produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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DATA MODEL METRIC CONTROL

Lab 11 · Cut a hierarchical dendrogram

CENTRAL APP

Hierarchical clustering

OPEN CENTRAL LAB PORTAL

INPUT	Lab 11 mock CSV
MECHANISM	Hierarchical clustering
EVIDENCE	metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 11 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Cut a hierarchical dendrogram produces a reproducible artifact and a defensible interpretation —not only a screenshot.

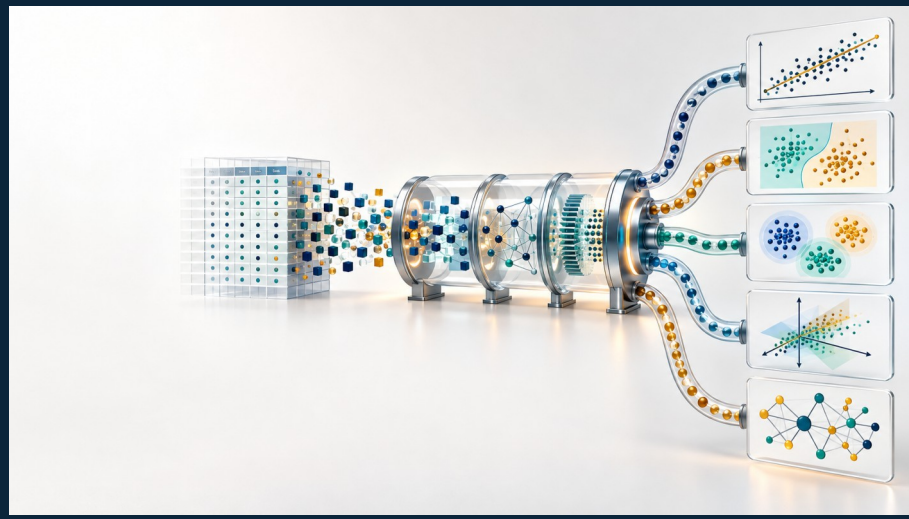
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TOPIC 06

Dimension Reduction

LO6 · A4 · K8 · Compressing variation while retaining evidence

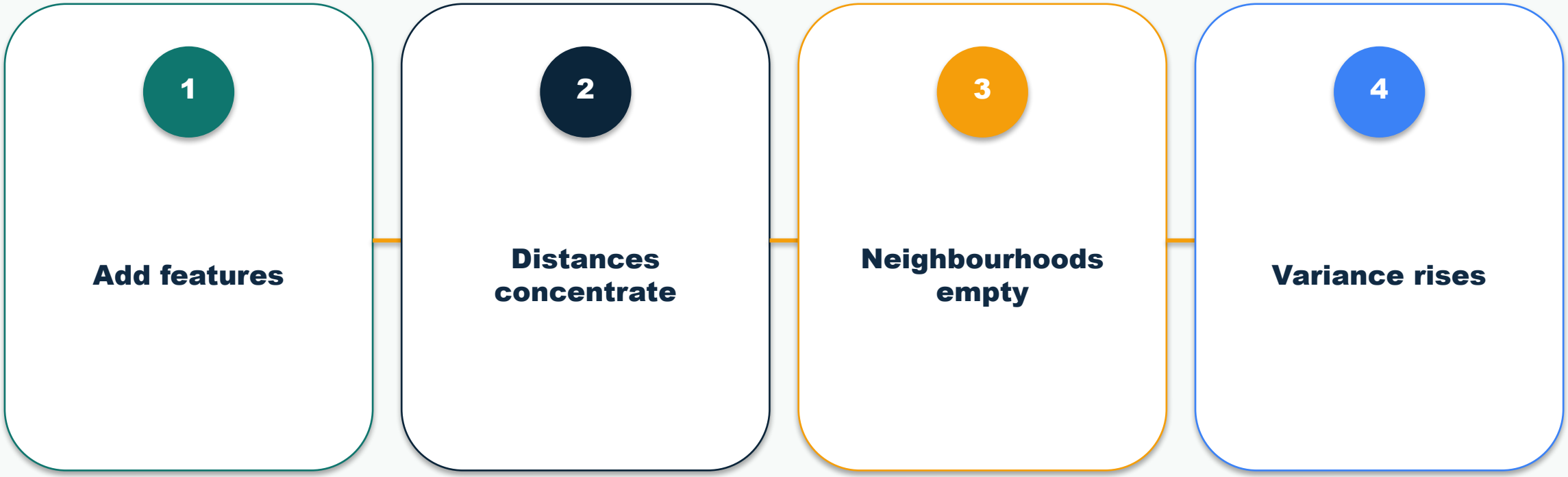
Mechanism → contract → calculation → evidence → control



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DATA MODEL METRIC CONTROL

Curse of dimensionality · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Curse of dimensionality · concrete contract

p	feature count
n	sample count
sparsity	volume effect
redundancy	correlated inputs

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DATA MODEL METRIC CONTROL

Curse of dimensionality · worked calculation

required coverage grows exponentially with p

WORKED EXAMPLE

10 bins per feature → 10^6 cells for 6 features

READOUT

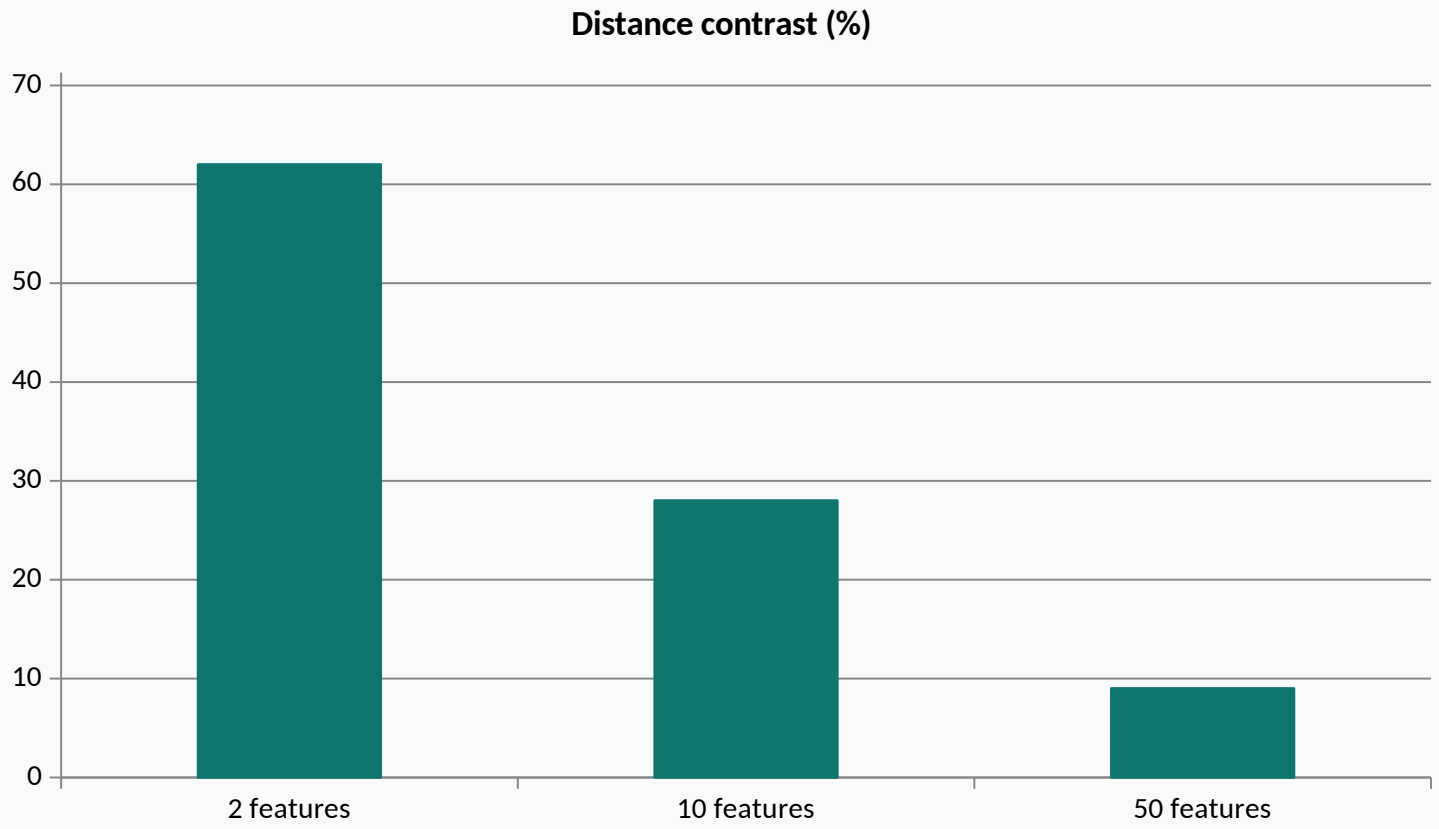
distance contrast

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Curse of dimensionality · evidence



WHAT THE DATA SHOWS

Validation improves or remains stable when a feature is retained.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Curse of dimensionality · failure and control

FAILURE MODE

Adding weak features makes neighbours less distinguishable.

CONTROL

Remove leakage/redundancy and validate feature additions.

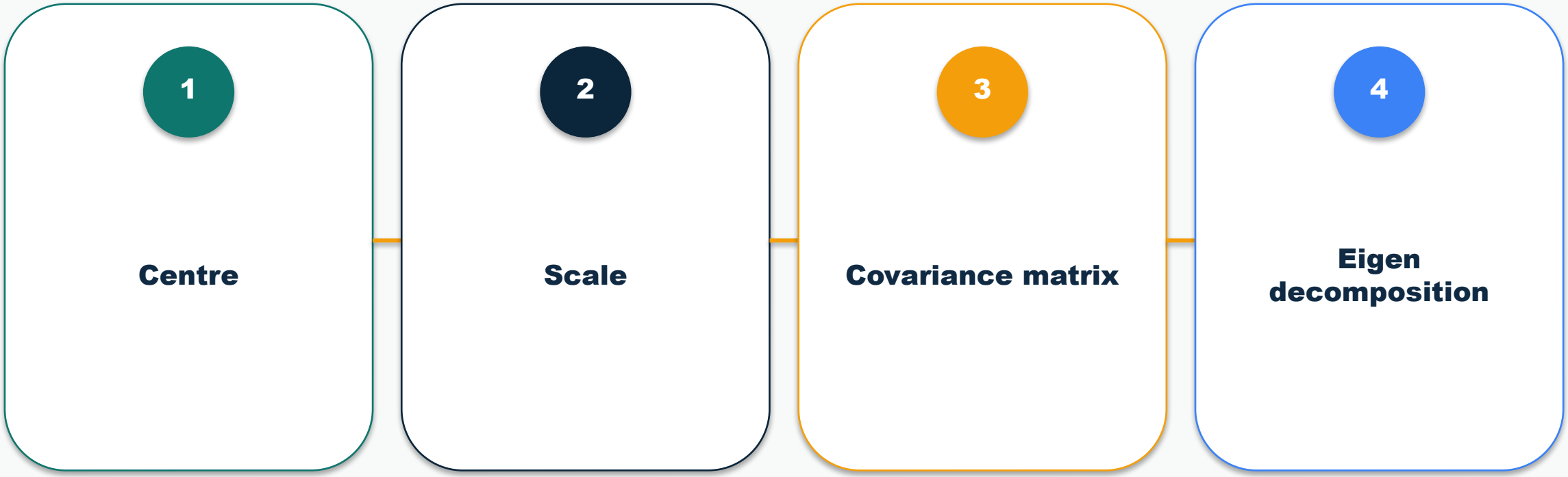
VERIFICATION

Validation improves or remains stable when a feature is retained.

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DATA MODEL METRIC CONTROL

Standardisation and covariance · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Standardisation and covariance · concrete contract

- # **mean vector μ**
- # **standard deviation σ**
- # **covariance Σ**
- # **symmetric matrix**

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DATA MODEL METRIC CONTROL

Standardisation and covariance · worked calculation

$$\mathrm{cov}(x,y)=\Sigma(x-\bar{x})(y-\bar{y})/(n-1)$$

WORKED EXAMPLE

positive covariance means the two centred features move together

READOUT

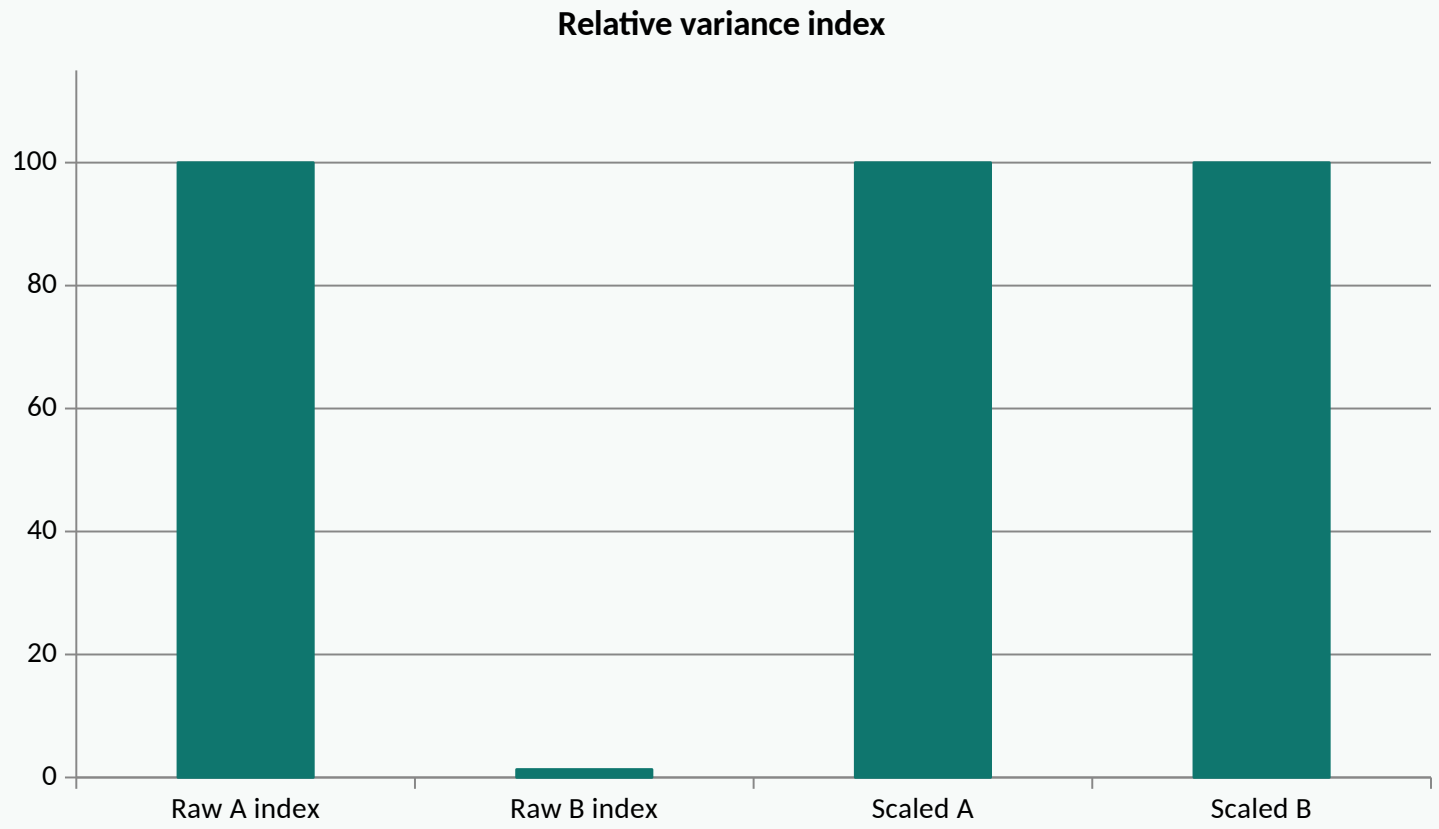
covariance structure

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Standardisation and covariance · evidence



WHAT THE DATA SHOWS

Each scaled feature has near-zero mean and unit variance on training data.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Standardisation and covariance · failure and control

FAILURE MODE

PCA on raw units lets high-variance currency fields dominate ratios.

CONTROL

Standardise features when units and scales differ.

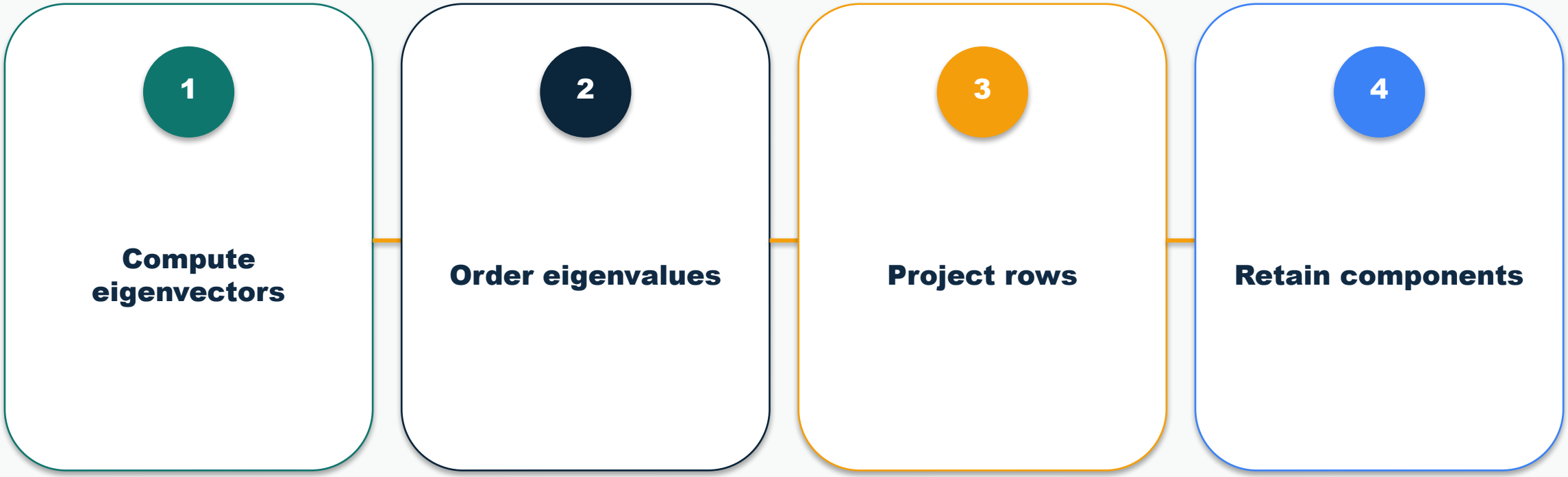
VERIFICATION

Each scaled feature has near-zero mean and unit variance on training data.

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DATA MODEL METRIC CONTROL

Principal components · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Principal components · concrete contract

loading	feature weight
score	projected row
eigenvalue	variance
orthogonal axes	

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DATA MODEL METRIC CONTROL

Principal components · worked calculation

$$Z = XW$$

WORKED EXAMPLE

two correlated measures rotate into PC1 along shared variation

READOUT

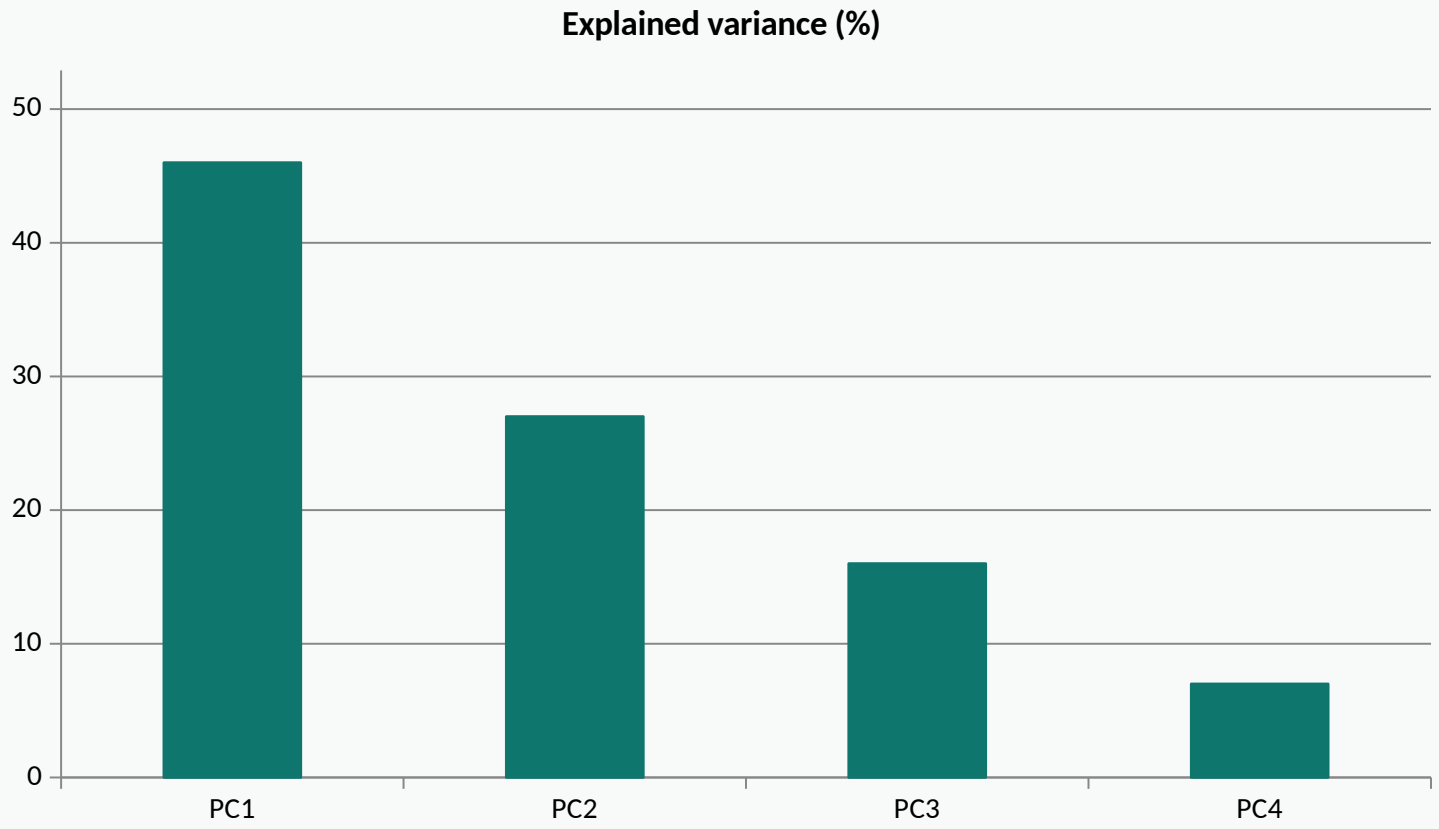
explained variance ratio

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Principal components · evidence



WHAT THE DATA SHOWS

Projected scores reproduce from saved scaler and PCA objects.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Principal components · failure and control

FAILURE MODE

Interpreting component sign as inherently positive or negative; signs may flip.

CONTROL

Interpret relative loading magnitude and relationships, not arbitrary sign alone.

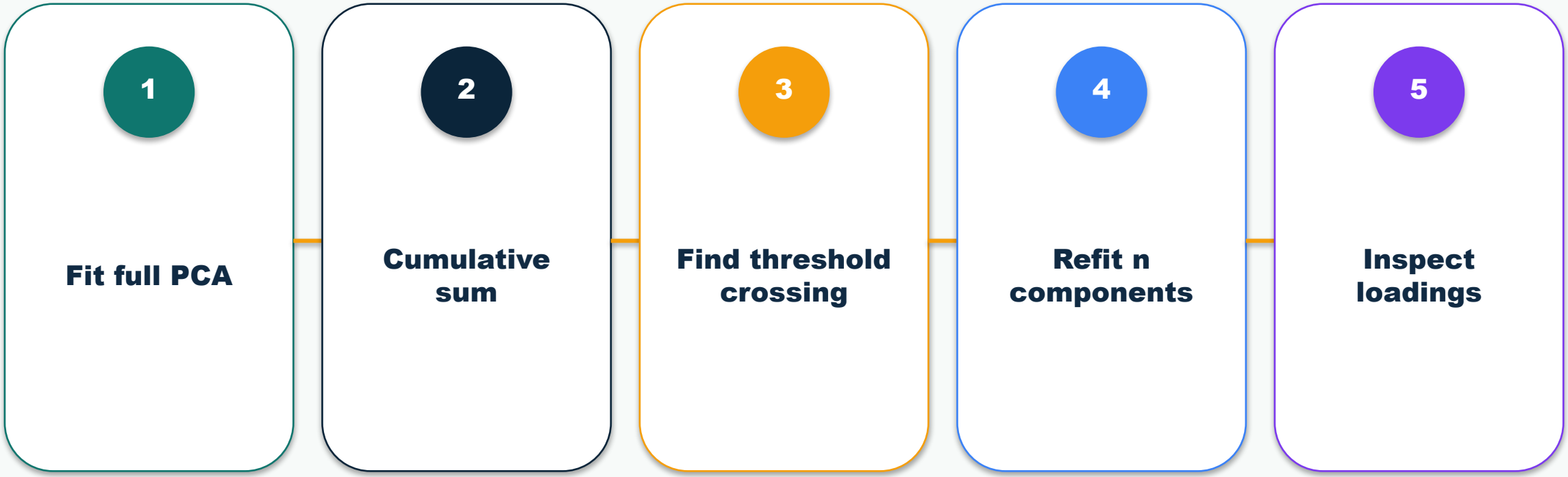
VERIFICATION

Projected scores reproduce from saved scaler and PCA objects.

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DATA MODEL METRIC CONTROL

Retained variance and loadings · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Retained variance and loadings · concrete contract

{}	**variance target**	0.70
{}	**n**	first crossing
#	**loading magnitude**	
#	**reconstruction loss**	

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DATA MODEL METRIC CONTROL

Retained variance and loadings · worked calculation

$$n = \min\{k : \Sigma_{i \le k} \text{variance}_i \ge .70\}$$

WORKED EXAMPLE

46% + 27% = 73% → retain 2 components

READOUT

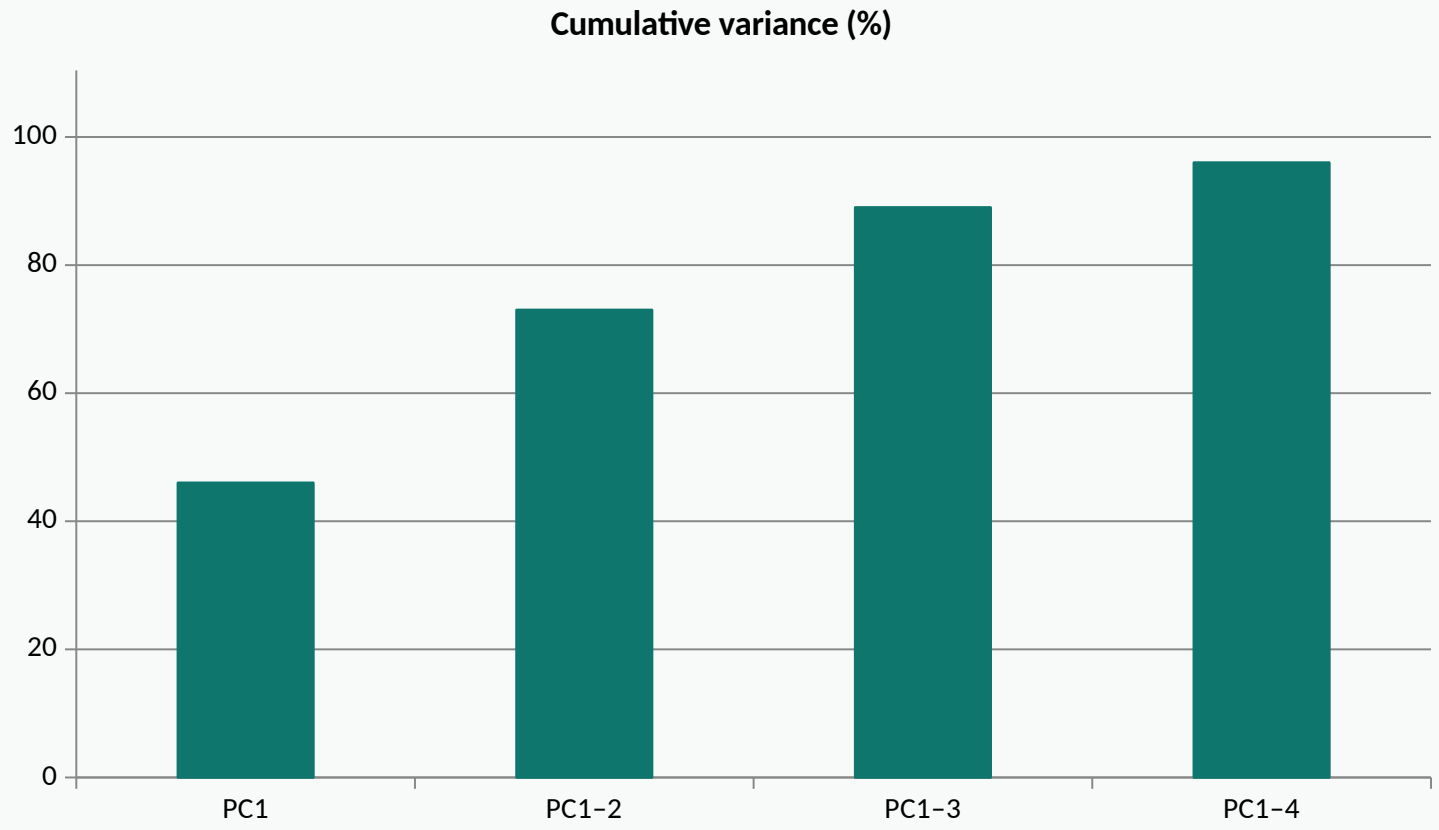
cumulative explained variance

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Retained variance and loadings · evidence



WHAT THE DATA SHOWS

The selected n is the first threshold crossing and is recorded with the data version.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Retained variance and loadings · failure and control

FAILURE MODE

Choosing 70% mechanically when the discarded component contains rare critical behaviour.

CONTROL

Review reconstruction errors and downstream performance.

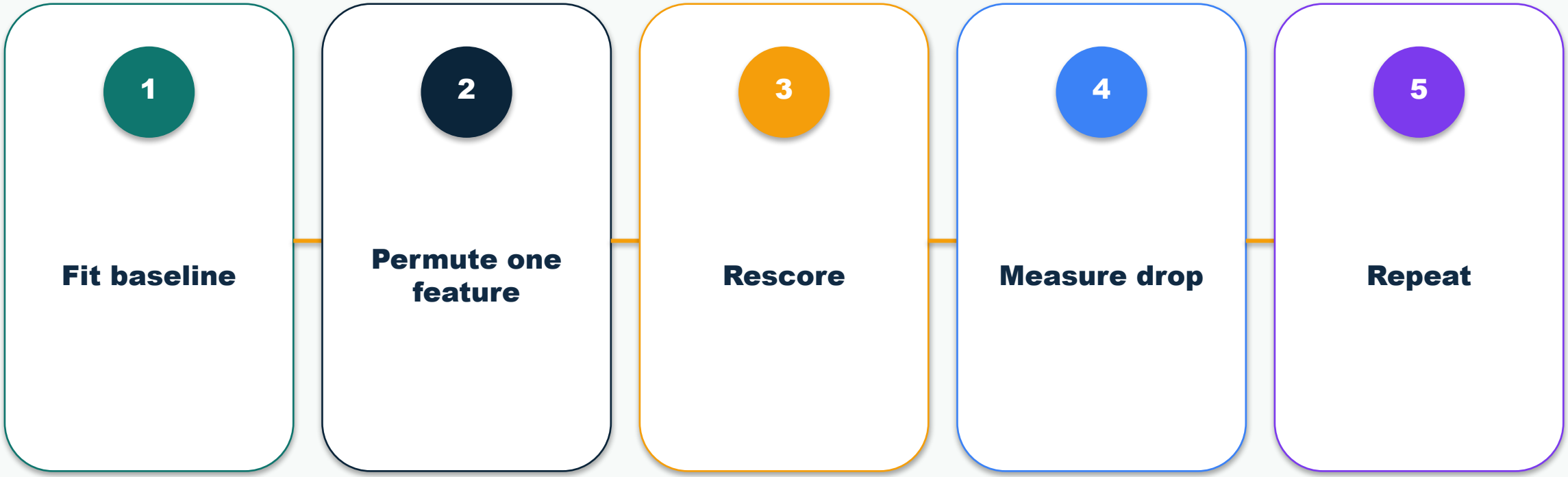
VERIFICATION

The selected n is the first threshold crossing and is recorded with the data version.

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DATA MODEL METRIC CONTROL

Feature ranking · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Feature ranking · concrete contract

importance	score decrease
repeats	uncertainty
correlation	shared credit
target	defined

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DATA MODEL METRIC CONTROL

Feature ranking · worked calculation

$$\text{importance}_j = \text{score}(X) - \text{score}(\text{permute } X_j)$$

WORKED EXAMPLE

.91 baseline − .78 permuted = .13 importance

READOUT

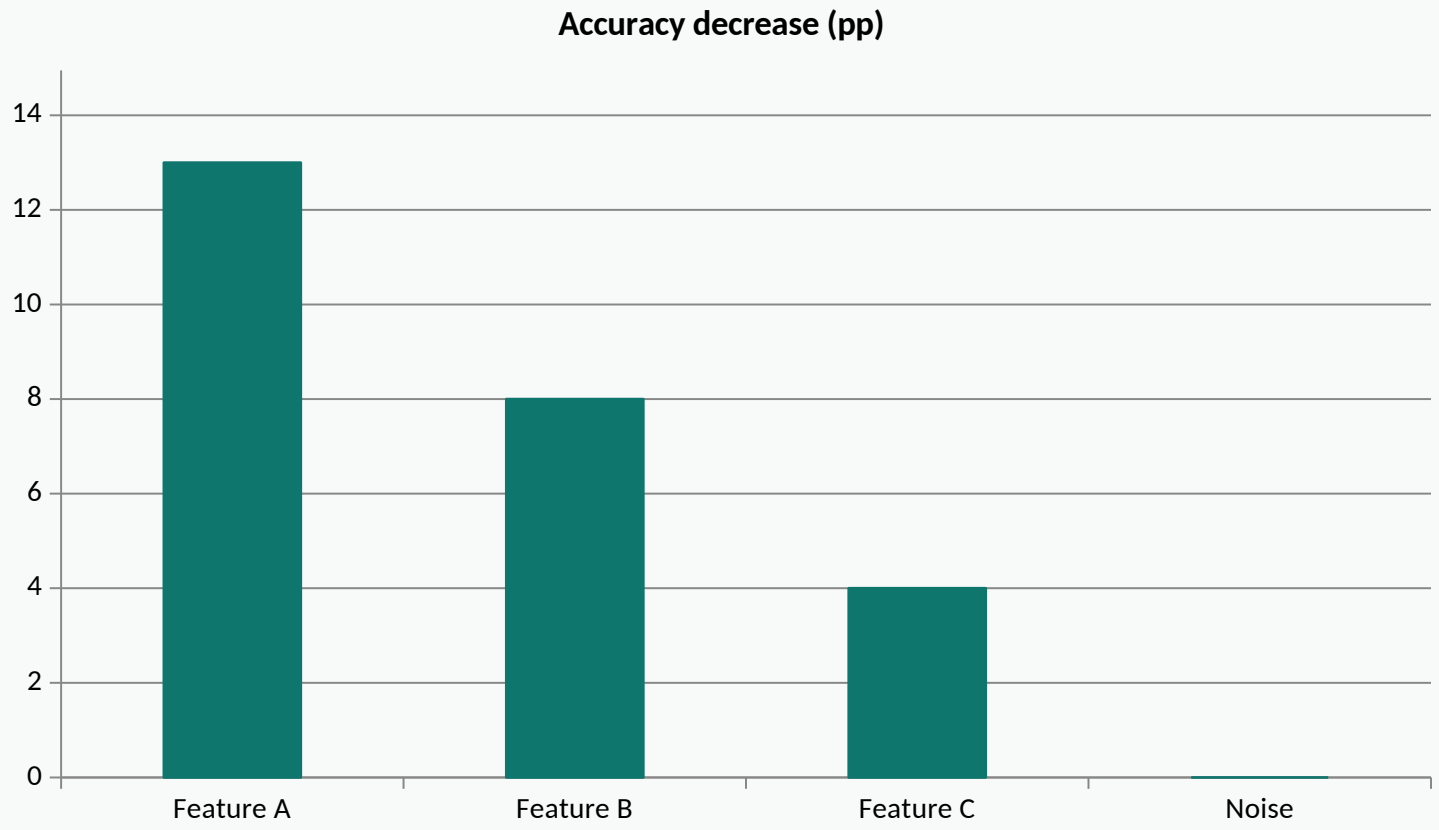
permutation importance

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Feature ranking · evidence



WHAT THE DATA SHOWS

Importance is positive, repeatable, and connected to an auditable model score.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Feature ranking · failure and control

FAILURE MODE

Correlated variables split or mask importance and ranking is mistaken for causality.

CONTROL

Group correlated features and validate rankings on hold-out data.

VERIFICATION

Importance is positive, repeatable, and connected to an auditable model score.

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DATA MODEL METRIC CONTROL

Lab 12 · Retain variance with PCA

CENTRAL APP

PCA & feature ranking

OPEN CENTRAL LAB PORTAL

INPUT	Lab 12 mock CSV
MECHANISM	PCA & feature ranking
EVIDENCE	metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 12 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Retain variance with PCA produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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DATA MODEL METRIC CONTROL

Lab 13 · Rank predictive features

CENTRAL APP

PCA & feature ranking

OPEN CENTRAL LAB PORTAL

INPUT Lab 13 mock CSV

MECHANISM PCA & feature ranking

EVIDENCE metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 13 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Rank predictive features produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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TOPIC 07

Association Analysis

LO7 · A3 · K9 · From baskets to defensible co-occurrence rules

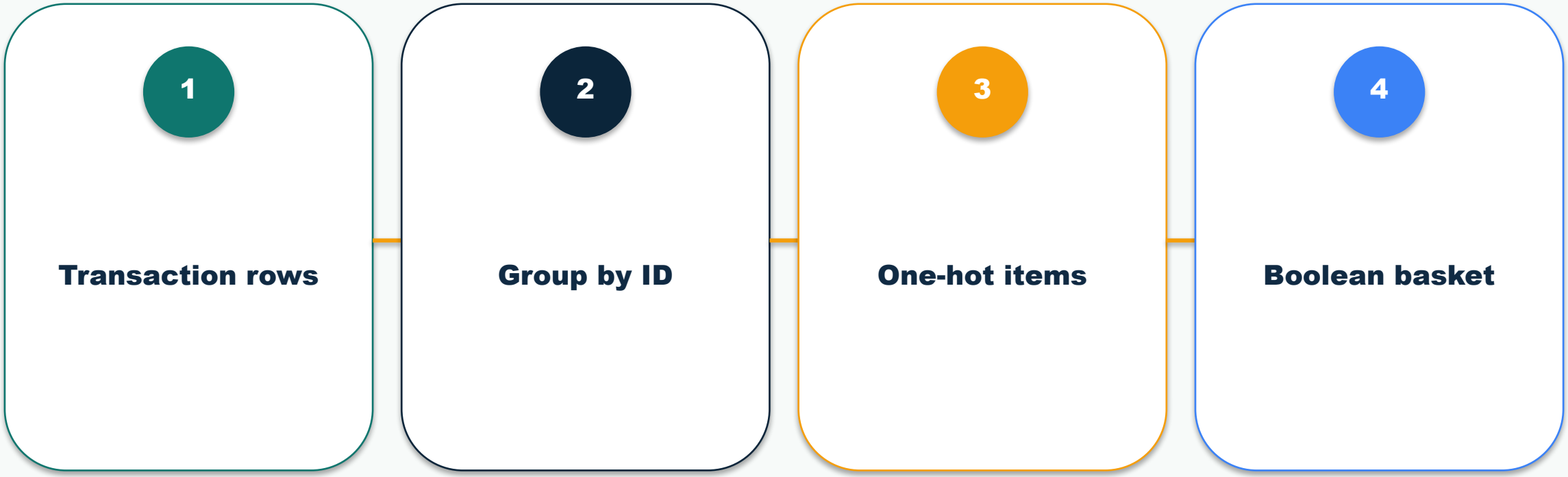
Mechanism → contract → calculation → evidence → control



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DATA MODEL METRIC CONTROL

Transaction-to-basket encoding · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Transaction-to-basket encoding · concrete contract

transaction_id	grouping key
item	canonical label
cell	present/absent
duplicates	collapse

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DATA MODEL METRIC CONTROL

Transaction-to-basket encoding · worked calculation

basket[t,i]=1 if item i occurs in transaction t

WORKED EXAMPLE

T001 milk+diapers+beer → three True cells

READOUT

basket density

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Transaction-to-basket encoding · evidence



WHAT THE DATA SHOWS

One encoded row exists per transaction and item counts reconcile.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Transaction-to-basket encoding · failure and control

FAILURE MODE

Treating purchase quantity as repeated independent transactions.

CONTROL

Define the basket unit and collapse duplicate item lines per transaction.

VERIFICATION

One encoded row exists per transaction and item counts reconcile.

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DATA MODEL METRIC CONTROL

Support · runtime mechanism

1

Count baskets containing itemset

2

Divide by all baskets

3

Apply threshold

Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Support · concrete contract

#	**support(A)**	
#	**support(A∩B)**	
#	**minimum support**	
{}	**denominator**	all transactions

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DATA MODEL METRIC CONTROL

Support · worked calculation

$$\text{support}(A)=\text{count}(A)/N$$

WORKED EXAMPLE

beer appears in 8 of 15 baskets → support=.533

READOUT

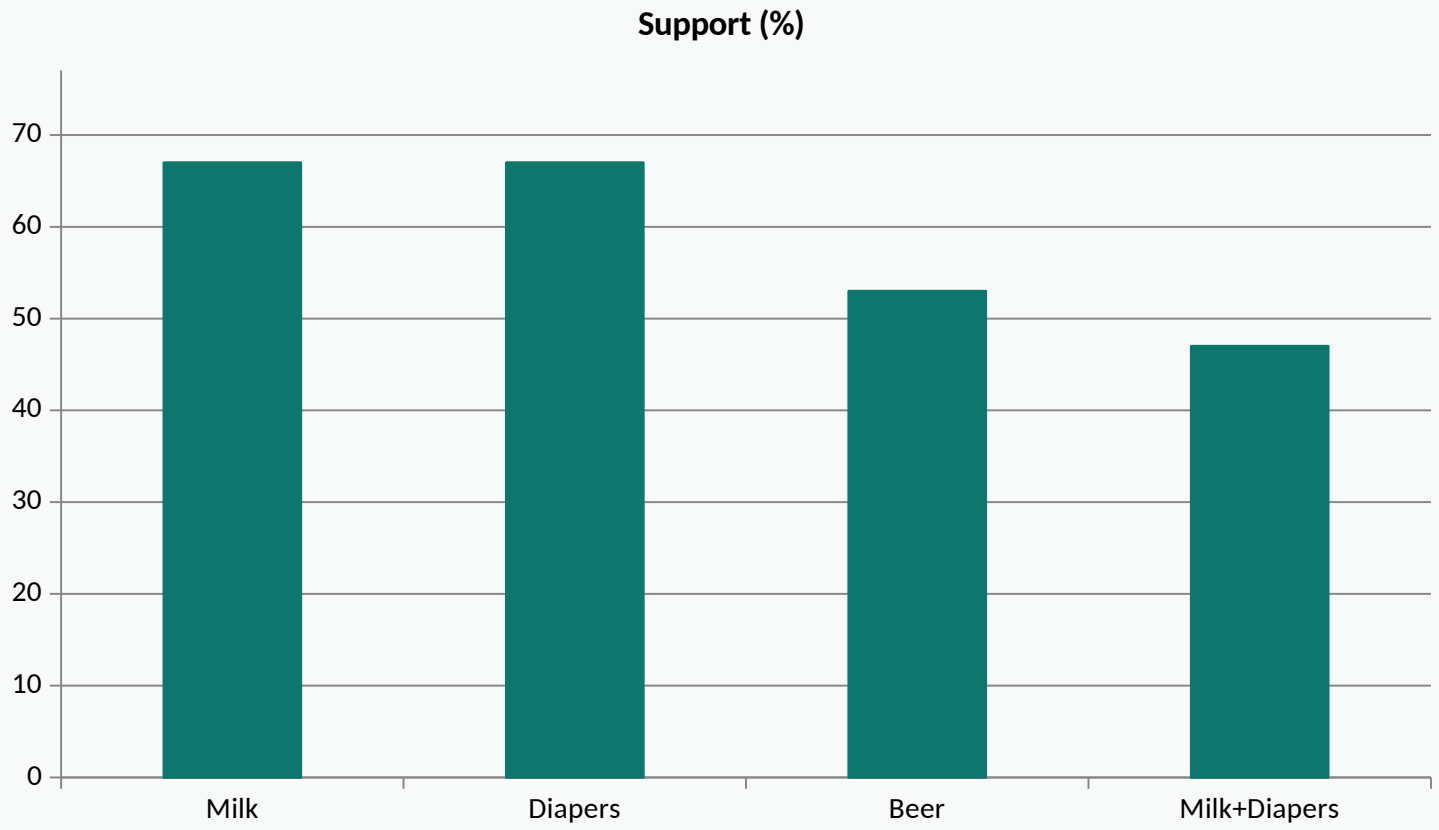
support

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Support · evidence



WHAT THE DATA SHOWS

Manual counts for selected itemsets match the mined result.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Support · failure and control

FAILURE MODE

A high minimum support removes valuable niche patterns.

CONTROL

Tune support to transaction volume and decision value.

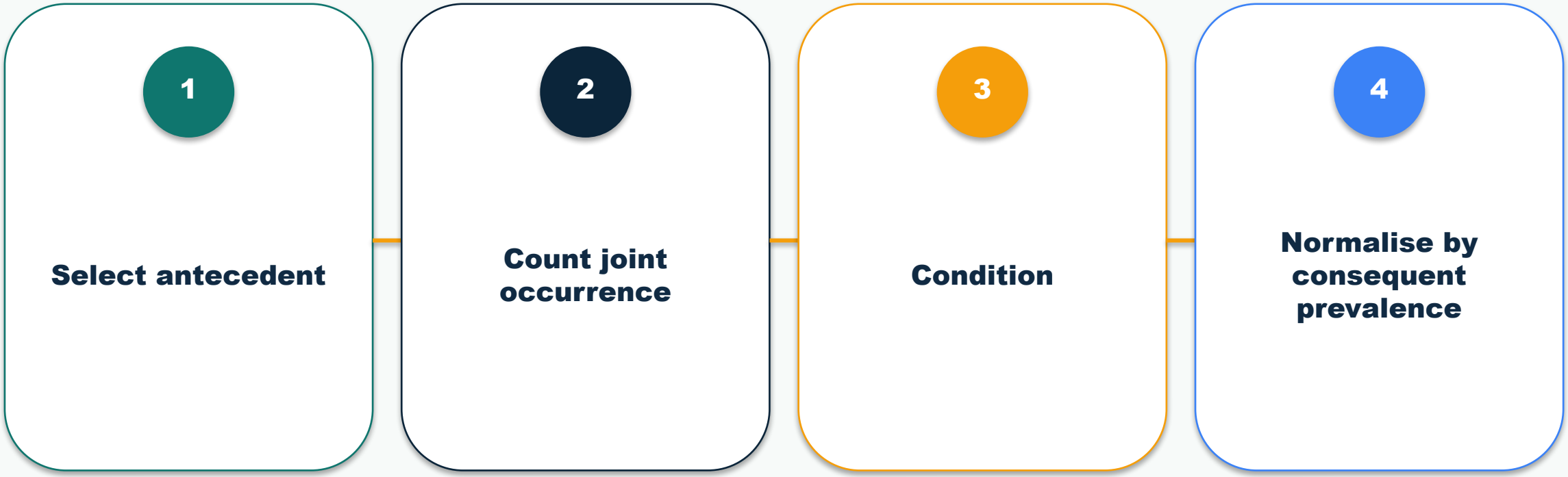
VERIFICATION

Manual counts for selected itemsets match the mined result.

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DATA MODEL METRIC CONTROL

Confidence and lift · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Confidence and lift · concrete contract

- **confidence(A→B)=support(A∩B)/support(A)**
- **lift=confidence/support(B)**
- **lift>1** positive association
- **direction matters**

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DATA MODEL METRIC CONTROL

Confidence and lift · worked calculation

$$\text{lift}(A \rightarrow B) = \text{support}(A \cap B) / (\text{support}(A)\text{support}(B))$$

WORKED EXAMPLE

.40/(.50×.50)=1.60

READOUT

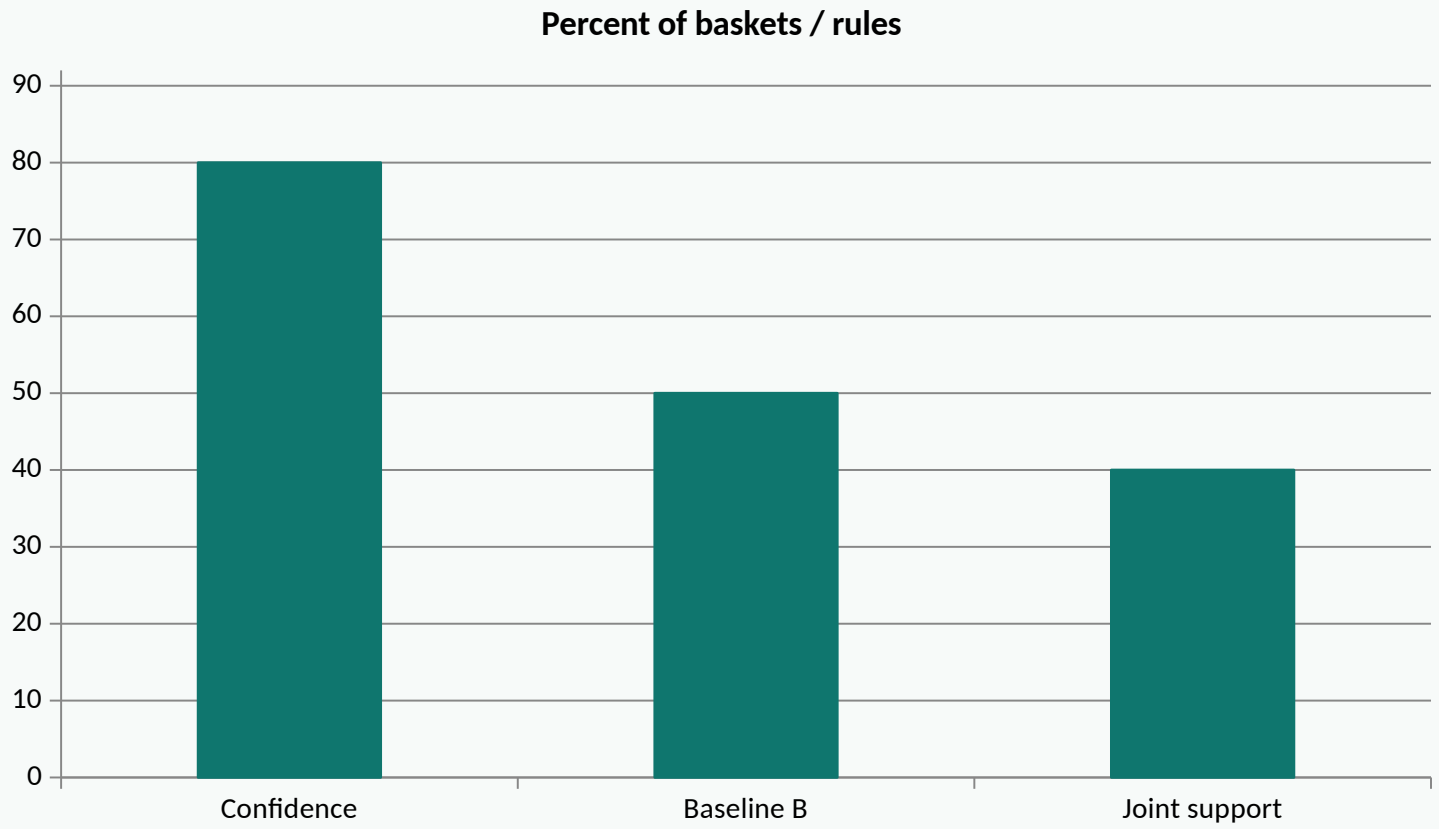
confidence and lift

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Confidence and lift · evidence



WHAT THE DATA SHOWS

The rule includes support, confidence, lift, and basket counts.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Confidence and lift · failure and control

FAILURE MODE

High confidence for a universally popular item is mistaken for a useful rule.

CONTROL

Require lift above 1 and assess absolute support.

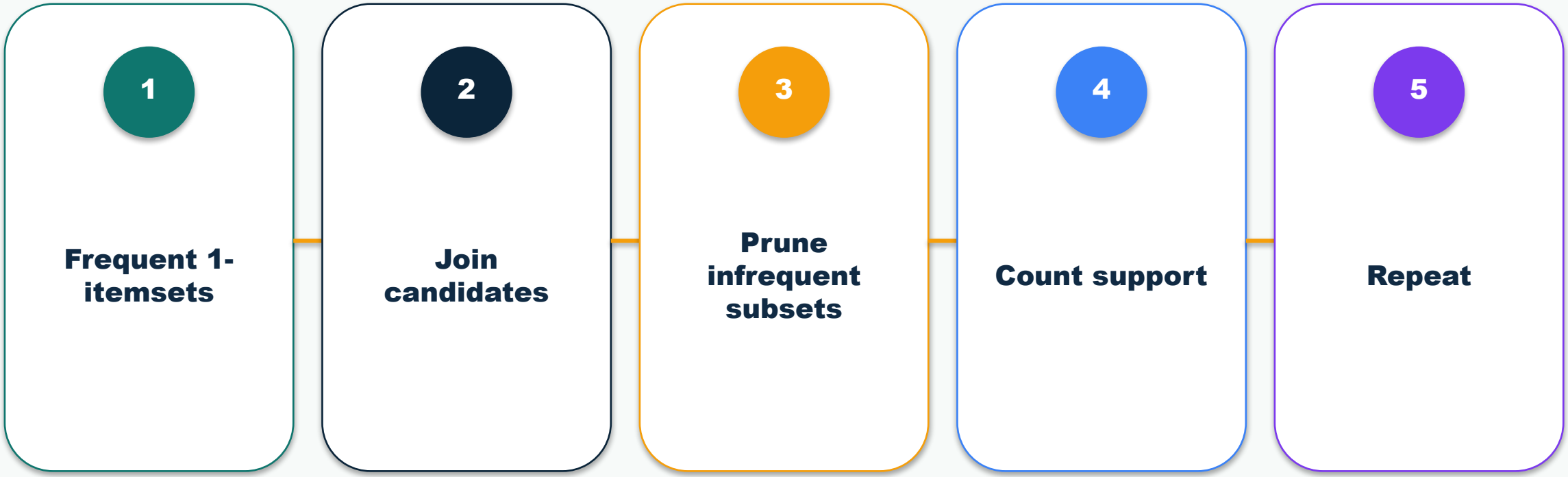
VERIFICATION

The rule includes support, confidence, lift, and basket counts.

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DATA MODEL METRIC CONTROL

Apriori candidate pruning · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Apriori candidate pruning · concrete contract

- # downward closure
- # candidate level k
- # minimum support
- # frequent itemset

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DATA MODEL METRIC CONTROL

Apriori candidate pruning · worked calculation

if itemset S is infrequent, every superset of S is infrequent

WORKED EXAMPLE

{milk,diapers} below threshold → prune all triples containing both

READOUT

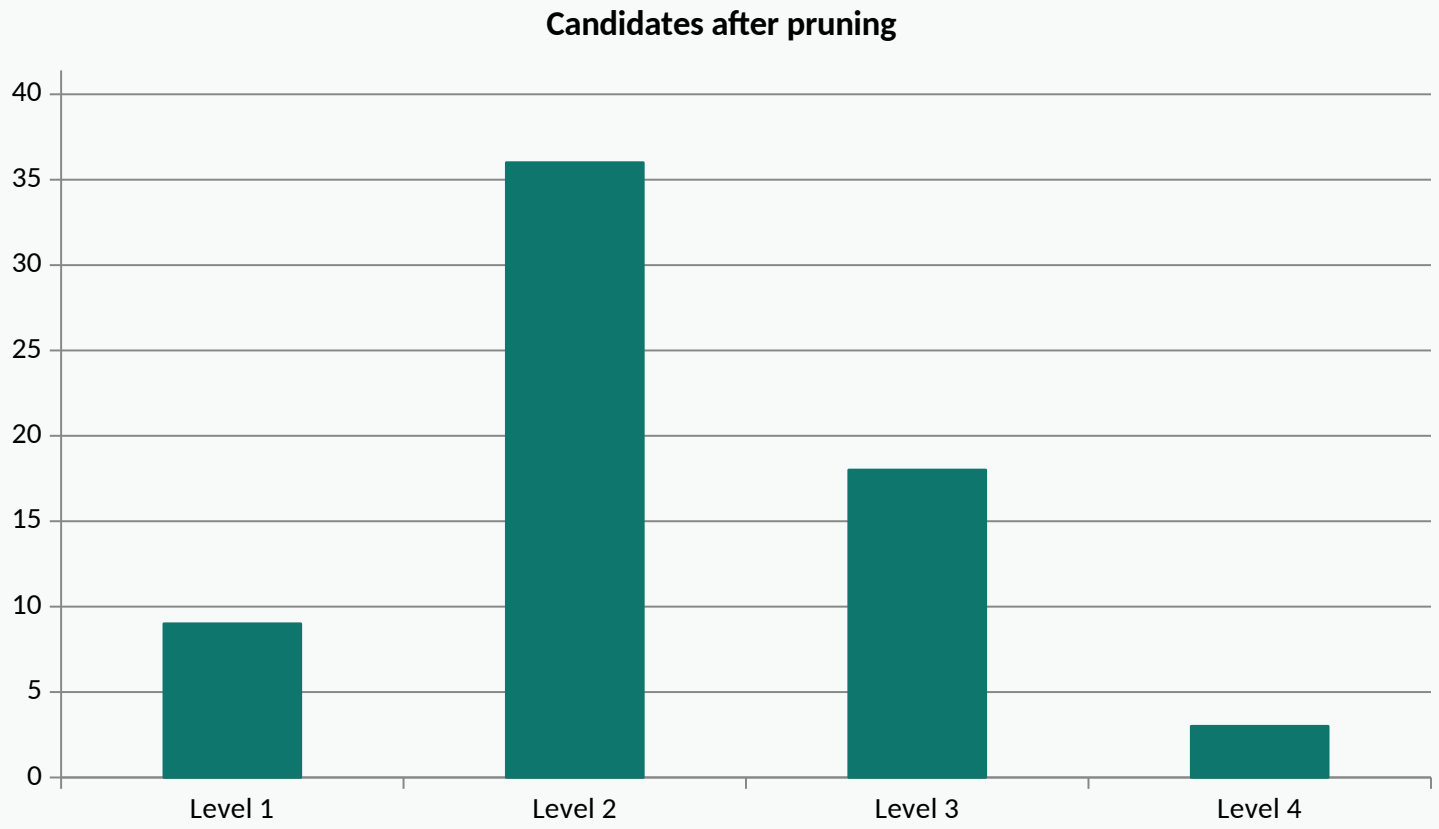
candidate count by level

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Apriori candidate pruning · evidence



WHAT THE DATA SHOWS

Candidate counts shrink after pruning and results reproduce at the same threshold.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Apriori candidate pruning · failure and control

FAILURE MODE

A low threshold creates a combinatorial explosion of candidates.

CONTROL

Start with a defensible support floor and cap itemset length.

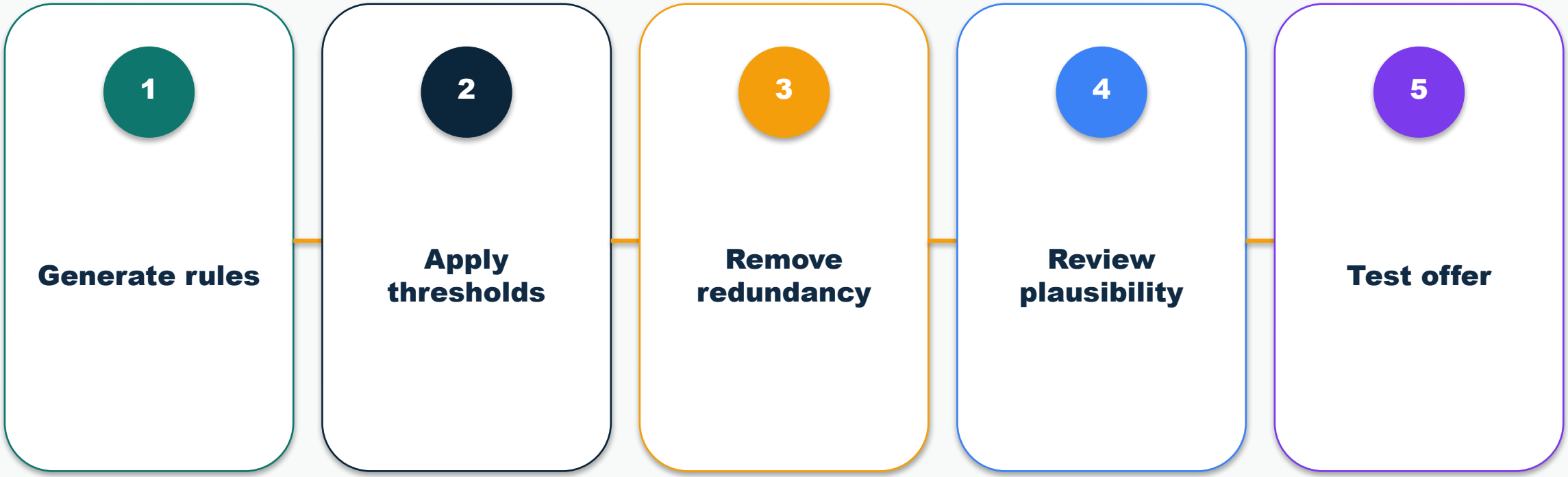
VERIFICATION

Candidate counts shrink after pruning and results reproduce at the same threshold.

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DATA MODEL METRIC CONTROL

Rule filtering and business action · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

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DATA MODEL METRIC CONTROL

Rule filtering and business action · concrete contract

antecedent	trigger basket
consequent	suggested item
leverage	excess joint rate
action	controlled test

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DATA MODEL METRIC CONTROL

Rule filtering and business action · worked calculation

$$\text{leverage} = \text{support}(A \cap B) - \text{support}(A)\,\text{support}(B)$$

WORKED EXAMPLE

$.40 - (.50 \times .50) = .15$

READOUT

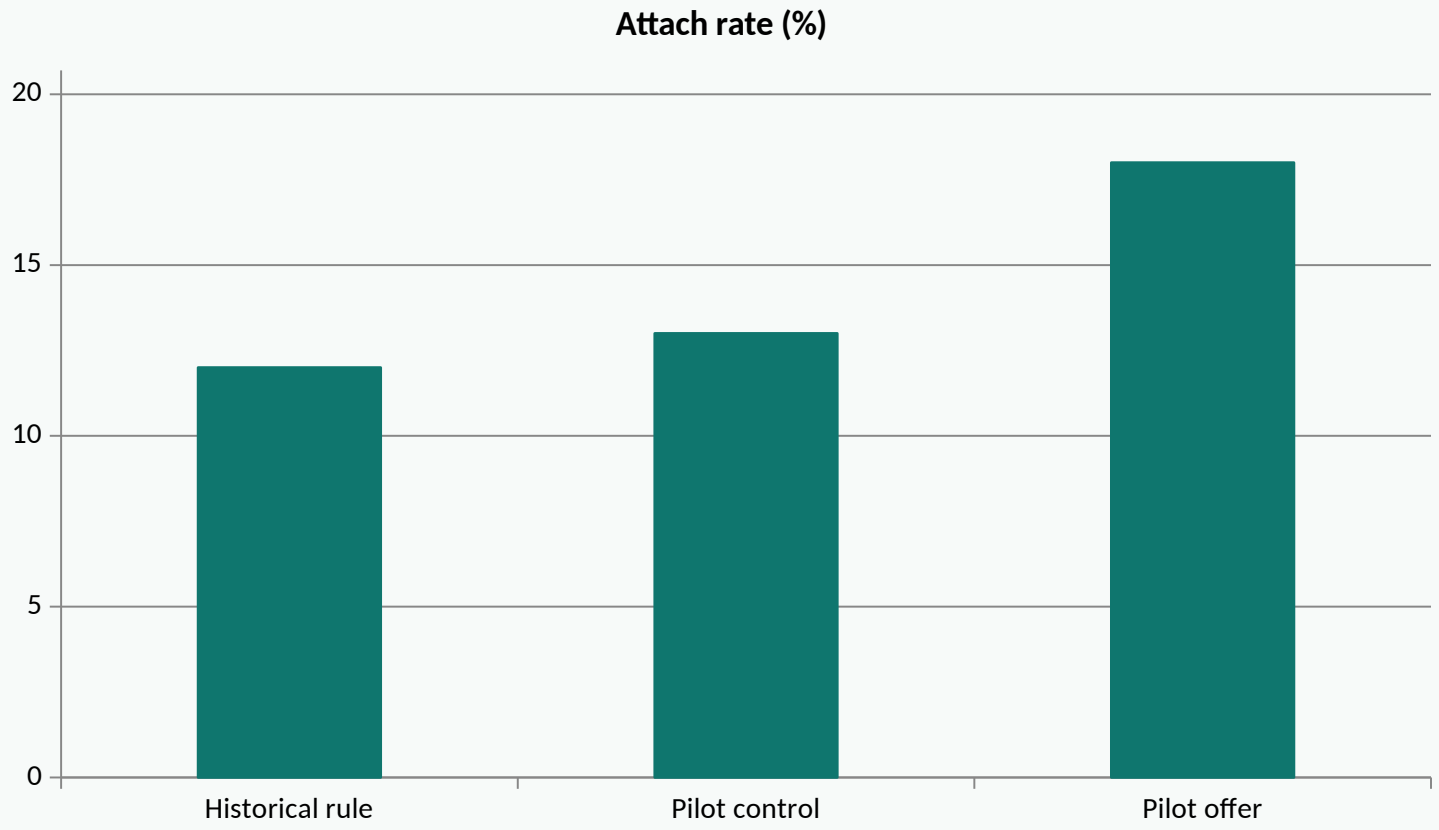
incremental basket outcome

Show units, denominator, and the comparison baseline.

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DATA MODEL METRIC CONTROL

Rule filtering and business action · evidence



WHAT THE DATA SHOWS

The action has an owner, audience, guardrail, and measured outcome.

Evidence becomes a decision only after scope and uncertainty are stated.

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DATA MODEL METRIC CONTROL

Rule filtering and business action · failure and control

FAILURE MODE

Deploying a bundle solely because lift is high on a tiny sample.

CONTROL

Require minimum counts and validate with a controlled promotion.

VERIFICATION

The action has an owner, audience, guardrail, and measured outcome.

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DATA MODEL METRIC CONTROL

Lab 14 · Mine market-basket rules

CENTRAL APP

Association rules

OPEN CENTRAL LAB PORTAL

INPUT Lab 14 mock CSV

MECHANISM Association rules

EVIDENCE metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

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DATA MODEL METRIC CONTROL

Lab 14 · acceptance evidence

1

Record inputs

2

Capture parameters

3

Export results

4

Explain limits

PASS: Mine market-basket rules produces a reproducible artifact and a defensible interpretation—not only a screenshot.

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ORIGINAL CONTENT · CONSOLIDATED

Python mechanism reference

The v16 technical topics are retained below as runnable Python concepts and checks.

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DATA MODEL METRIC CONTROL

CSV import and data table

MECHANISM

Parser assigns types; validation quarantines malformed rows.

VERIFY

Schema + row-count reconciliation

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DATA MODEL METRIC CONTROL

Scatter plots and interactive subsets

INPUT

typed data

PYTHON

Selections must retain stable row identifiers.

PROOF

Selected + excluded = source

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DATA MODEL METRIC CONTROL

Select columns and rows

MECHANISM

Feature role and row predicate are explicit contracts.

VERIFY

No target leakage

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DATA MODEL METRIC CONTROL

Merge and concatenate

INPUT

typed data

PYTHON

Join keys control horizontal integration; concat stacks compatible schemas.

PROOF

Cardinality and totals reconcile

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DATA MODEL METRIC CONTROL

Sampling, transpose, and randomisation

MECHANISM

Sampling estimates; transpose changes unit-of-analysis; shuffling breaks order dependence.

VERIFY

Seed and row identity recorded

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DATA MODEL METRIC CONTROL

Feature statistics and save data

INPUT

typed data

PYTHON

Profile values before export; persist a cleaned dataset with provenance.

PROOF

Hash, schema, and row count

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DATA MODEL METRIC CONTROL

Decision tree

MECHANISM

Recursive impurity reduction forms axis-aligned leaves.

VERIFY

Depth and leaf support

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DATA MODEL METRIC CONTROL

Logistic regression

INPUT

typed data

PYTHON

Linear log-odds become calibrated class probabilities.

PROOF

Calibration plus threshold

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DATA MODEL METRIC CONTROL

K-nearest neighbours

MECHANISM

Scaled distance and neighbourhood size define the vote.

VERIFY

Sensitivity to k and scaling

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DATA MODEL METRIC CONTROL

Support vector machine and kernel

INPUT

typed data

PYTHON

Margin maximisation separates classes; kernels change similarity space.

PROOF

C, gamma, and held-out AUC

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DATA MODEL METRIC CONTROL

Gaussian Naive Bayes

MECHANISM

Class priors multiply conditional Gaussian likelihoods.

VERIFY

Likelihood assumptions checked

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DATA MODEL METRIC CONTROL

Random forest

INPUT

typed data

PYTHON

Bootstrap trees plus random feature subsets reduce variance.

PROOF

Out-of-bag or CV score

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DATA MODEL METRIC CONTROL

Stacking classifier

MECHANISM

Out-of-fold base predictions train a meta-learner without leakage.

VERIFY

Meta-features are out-of-fold

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DATA MODEL METRIC CONTROL

AdaBoost

INPUT

typed data

PYTHON

Successive weak learners reweight previously misclassified cases.

PROOF

Learning rate and error trajectory

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DATA MODEL METRIC CONTROL

Save and load model

MECHANISM

Persist preprocessing and estimator together as one versioned pipeline.

VERIFY

Round-trip prediction equality

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DATA MODEL METRIC CONTROL

Classification threshold

INPUT

typed data

PYTHON

A score becomes a class only after applying an operating threshold.

PROOF

Cost matrix and capacity

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DATA MODEL METRIC CONTROL

Release checklist

FAILURE MODE

A good-looking output is mistaken for a valid model.

CONTROL

Recheck scope, leakage, baseline, metric, error costs, and reproducibility.

VERIFICATION

Every lab artifact has a file, parameter record, chart, and interpretation.

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DATA MODEL METRIC CONTROL

From pattern to responsible action

1. **Validate data**
2. **Fit mechanism**
3. **Measure errors**
4. **Explain pattern**
5. **Test action**

Trace the named state at each boundary; a missing state breaks reproducibility.

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THANK YOU

Keep the evidence reproducible.

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